The Loss Ledger and the Cycle Filtration: Tower Exhaustion on the Helix Carrier, and a Program Whose Terminus Is the Hodge Conjecture

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Abstract

The three-dimensional helix carrier of the main paper projects to the classical 1D readout through a lossy but exactly bookkept descent: height, radius, and angle are recorded in a loss ledger, and projection with its ledger is a proven bijection. This paper reads the *cycle theory* of a variety through that ledger. We define the carrier filtration F^d_{car} by vanishing of the recursive tower readouts $T_0, \ldots, T_{d-1}$ and prove, machine-checked in Lean 4, the depth dictionary: *first nonzero tower depth equals filtration depth*, the depth is unique, it is preserved by every faithful carrier transport, and F_{car} is the unique filtration graded by its tower — so a Bloch–Beilinson filtration coincides with it *if* its steps are cut out by that same tower, an identification hypothesis, not a consequence of existence. On every finite normalized extension stack, including the semisimple layer, the paired jet/moment tower is unconditionally jointly faithful — a layerwise Vandermonde separation — so no nonzero finite extension state is silent at every level — a theorem about the abstract model, which cycle groups enter only through a faithful realization, an interface now typed in Lean with its retention transport proven and its first genuine instances executed end to end — on the rational group algebra of $\mathbb{Z}/n$ and on the Artin motive of any cyclic Galois number field (the CM quadratic fields included), source exhaustion is a theorem, recognition on the latter consumed from classical Galois descent. At depth one the carrier's leading central jet factors, by Birch–Swinnerton-Dyer/Gross–Zagier, into three independently resolved coordinates — vanishing order, regulator, and $|Ш|$ — landed at machine precision on a census through rank three, with the local-global obstruction appearing as an exact square integer. Each deeper landing is theorem-anchored: Tate's pole at grade zero, Gross–Zagier–Kolyvagin at grade one, a CM Hodge class firing with the predicted delayed signature at depth two, and the Ceresa/Gross–Schoen height at tensor depth three by Zhang and Yuan–Zhang–Zhang. The program this assembles is stated plainly: the carrier's organizing principle is that no layer remains invisible — every class is induced to appear at some retained coordinate — and the terminus of that principle is the Hodge conjecture, read as a source-exhaustion statement: every Hodge-signature state has an algebraic source.

Four further developments carry the program past its first draft. The *harmonic frame*: a Hodge structure is a torus clock — weight the radial rate, $H^{p,q}$ the frequency-$(q-p)$ channel, a Hodge class a rational DC mode — an organizing dictionary, a re-expression of Deligne's torus action (frequency zero detects Hodge type; it does not construct a cycle), calibrated against Lefschetz $(1,1)$ truth on both the period and point-count sides. *Drift*: extension classes are the first-order shear a pure clock stack cannot produce; machine-checked, drift is value-blind, jet-visible, and never silent on the model, and the measured law — the response harmonic equals the drift dimension — holds across the census with machine-zero silence below. *Recognition at grade one*: the Gross–Zagier loop is run end to end on house instruments,

from point counts to an exactly recognized rational point. *Grade four*: the first unnamed rung. Three walls are established — the non-critical center (the motive's own DC channel), the parity ladder (heights on odd rungs, regulators on even), and a split no-go (decomposable classes pair to zero with the primitive quadruple, machine-checked) — so the rung-four object is classically homeless; the degree-16 fiber is then assembled directly on the carrier, its degenerate case factoring through the lower harmonics exactly, its primitive case certified irreducible, and its first datum measured. This draft carries the rung further. A fourth wall — the information wall: any chart-side center determination needs $\sim \sqrt{Q} \approx 3 \times 10^{24}$ coefficients — is measured on three faces and then dismantled: a certified value chain runs to degree six, the primitive quadruple's functional-equation sheet is pinned from its twenty integers with the global root number *proven* $+1$ from Deligne's local theory, its first strip-interior values exist, and the unread center is identified as a regulator volume dominated by its tropical part. The harmonic-lattice and rail-pairing laws locate where values live: warps harmonically compatible with the fiber's own angle produce exact lane identities on the symmetric-power tower, natural boundaries at the center are single-rail artifacts, and the height-matched pairing of exact conjugate rails converges where every mis-pair diverges. The portal reads the homeless $(2,2)$ block from inside — channel constancy as an algebraicity signature, an S_3-symmetric interior Gram with a two-dimensional exotic-candidate home — and a fixed-head locator with μ_6-address rails resolves oracle zeros at the Rayleigh limit one degree beyond its previous ceiling. The Ceresa target's L-side is landed on the Klein quartic: the Ceresa channel has forced sign -1 and central derivative $0.8299 \neq 0$, matching the proven non-torsion of the cycle. And the tower scales: grades five and six run on the same instruments with no new method — every pre-registered rational lands, the Catalan spine C_2, C_3 and its Schur companions C_4, C_5, C_6 confirm, and the parity law (a DC channel exists iff the weight is even) is proven in Lean the same session its content is measured; the atlas then runs to grade twenty, where the tower's root-number law emerges — sign real at every grade, central zeros forced at exactly $g = 2^k + 1$ — and the law's first new prediction is verified where no oracle can reach ($L'(\mathrm{Sym}^9(37a1)) \approx 10.9686$, the forced zero carrier-registered). A final campaign takes the program to the classes classically called exotic and retires the word: Weil-type collective classes are detected by a rail-native apparatus whose freeze law is proven (the discriminant-character mechanism, with the resultant entering as a perfect cube killed by the cubic character), whose censuses are kernel-checked (Mumford's fourfold obstruction is the identity $6 - 6 = 0$; it dissolves at dimension six, uniformly $20 - 8 = 12$), and whose no-orphan decomposition holds even on non-CM members where algebraicity is known only transcendentally. The recognition loop for a collective class is then closed for the first time — eight exact integer landings tying Schoen's cycle to the point-counted freeze — and the last open coordinate is located: the curve behind a simple Weil class is provably invisible to the angular stream (Faltings) yet radially retained and Torelli-sufficient; the bridge's logic is machine-checked with the classical monuments as named hypotheses. The terminus architecture is fixed in Lean: retention and recognition, the two named hypotheses, compose machine-checked into source exhaustion — asserted by no one, wired to receive.

Contents

1 Introduction

The main paper [1] constructs a three-dimensional carrier — a double-ended helix and its conjugate anti-helix, harmonically scaled, running from the origin to infinity — on which automorphic data ride as fibers, and proves that the classical one-dimensional readout is the image of a lossy but exactly bookkept projection: the descent 3D $\to$ 2D $\to$ 1D books a *loss ledger* — height (ordinate), radius, angle — and projection with its ledger is a bijection with an exact two-sided inverse (Lean, `ConeProjection.record_bijective`). The present paper turns that instrument toward the cycle theory of a variety, and it is organized around one boxed statement and one stated terminus.

The boxed statement is the *depth dictionary*. Define the recursive tower $T_0, T_1, T_2, \dots$ of carrier readouts — value, leading jet, and the symmetric-power/derivative levels above them — and the carrier filtration

$$F^d_{\text{car}} \;:=\; \{\, z : \; T_0(z) = \cdots = T_{d-1}(z) = 0 \,\},$$

defined by the tower alone, with no reference to any conjectural filtration. Then, as a machine-checked theorem (§5),

$$\boxed{\text{first nonzero carrier tower depth } = F_{\text{car}}\text{-filtration depth,}}$$

the depth is unique, it is invariant under every faithful carrier transport, and F_{car} is the unique filtration whose successive steps are cut out by its tower; hence a Bloch–Beilinson filtration coincides with F_{car} *if* its steps are cut out by that same tower (`blochBeilinson_coincides`). We price both statements exactly. Since F_{car} is *defined* by tower vanishing, the depth correspondence is definitional bookkeeping, exact by construction. And the coincidence hypothesis is *stronger* than the existence of a Bloch–Beilinson filtration with its usual conjectural properties: it already asserts that every successive Bloch–Beilinson step is graded by this very tower, so the corollary is uniqueness bookkeeping — any filtration recursively defined by these tower kernels equals the filtration so defined — and the entire identification content lives in verifying that grading hypothesis for an independently constructed Bloch–Beilinson filtration, which remains open. The value of the package is that transport-invariance and the coincidence interface attach to a non-circular, machine-fixed object.

The stated terminus is the Hodge conjecture. The carrier program's organizing principle — established across the main paper's arithmetic and repeated here — is that *no layer remains invisible*: a class silent in one readout is not invisible but mis-read, and it is induced to appear by climbing to the correct retained coordinate — a deeper tower level, a different ledger channel. The local-global obstruction $|Ш|$, ordinarily invisible to a Hodge-theoretic projection, appears at depth one in the jet's amplitude ratio as an exact square integer (§7); a CM Hodge class, silent in the standard readout, fires at depth two with the predicted delayed signature (§8); the transcendental Ceresa class, silent at depth one, is reached at tensor depth three by a theorem of Zhang and Yuan–Zhang–Zhang (§9). Carried to its terminus the principle says: a rational (p, p)-class — a state with the *Hodge signature* in the carrier's clock coordinates — cannot be silent either; something algebraic sources it. That is the Hodge conjecture [2, 4], read in the carrier's native register as *source exhaustion*:

$$\boxed{\text{every Hodge-signature state has an algebraic source.}}$$

We do not prove it here. We prove the dictionary, retention for arbitrary finite normalized extension data, and the drift theorems; measure every landing the classical theorems anchor, including the first above-grade-one anisotropy reading (§9), the landed L-side of the Ceresa target itself, and the first data on the unnamed fourth rung (§11); measure the rung's fourth wall and dismantle it (§12); read the rung's interior through the harmonic lattice, the conjugate rails, and the portal (§13); scale the tower to grades five and six with no new method (§14); close the recognition loop at grade one on house instruments (§10); take the campaign to the collective (Weil-type) classes — detection as theorem, censuses kernel-checked, the first collective recognition loop closed, the last open coordinate located (§15, §16); collect what the three-dimensional geometry discovered *as an instrument* (§17); and in §18 fix the architecture in Lean — the two named ingredients composing, machine-checked, into the terminus — together with what evidence would refute the program on the way.

Registers. Three registers are used and never blended. *Proven*: machine-checked in Lean 4 at axiom footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry` (the filtration dictionary, the finite-extension retention theorem, the realization bridge, and the supporting carrier corpus of [1]), or established in the classical literature with the citation given at the point of use (Gross–Zagier, Kolyvagin, Zhang, Yuan–Zhang–Zhang, Harris, Laga–Shnidman). *Measured*: precision-tracked numerics — the census, the probes, the delayed signatures — calibration and corroboration, never proof. *Program*: arithmetic realization of cycle extensions in the normalized finite stack, the from-scratch Ceresa landing, and the terminus itself, stated with their falsifiers.

2 What the geometric view has found: the claims, weighed

The evidence-weighing stance of this paper is that silence is itself a miscalibration; so before the constructions, the findings — what we believe the three-dimensional view has actually discovered about Hodge theory — stated as claims with their registers, in the order of their depth.

1. **The Hodge condition is harmonic.** A Hodge structure is a torus clock: the weight is the radial rate, $H^{p,q}$ is the frequency-$(q-p)$ channel, and a Hodge class is a *rational DC mode* — the (p,p) condition is frequency zero, not a transcendence mystery (§4). [Register: a re-expression of Deligne's definition, not a new theorem — frequency zero *detects* Hodge type, it does not *construct* a cycle; frame calibrated against Lefschetz $(1,1)$ truth on both the period and point-count sides; measured.]

2. **Algebraicity reads as freeze, and the freeze has an arithmetic law.** A class's channel is constant exactly when its frequency vector annihilates the Frobenius equidistribution subgroup — a measure-free mechanism, certified as a detector on families where Hodge and Tate *are* theorems (§15). On the cyclic-cover family the freeze *order* is itself a theorem: $\det(\mathrm{Frob}_p \mid V_\zeta) = (-1)^{\delta_p} p^{g_W/2} \chi_3(D)$, the resultant entering as a perfect cube that the cubic character kills — the multiplicity-balanced structure *is* the mechanism, and the honest derivation reproduced rather than assumed it. [Proven on the family; `ResCube.lean`, `ChannelConstancy.lean`.]

3. **Nothing surveyed is exotic.** Every class in every census is a frozen frequency with a source, an address, and a certificate; the divisor/Weil distinction is annihilator *support* — pairwise versus collective rails — not a difference in kind. The pre-registered no-orphan-occupancy law (occupancy = divisor rails + named collective rails + zero remainder) held on every census run, including the genuine stress tests: the non-CM Markman-transcendental members and the dimension-six dissolution ($20 - 8 = 12$, kernel-checked with an *empty* axiom list) (§15–16). [Theorem-grade on the family; law pre-registered, unfalsified; quantifier: the censuses and certified families, never all Hodge structures; freeze detects, never constructs (`detector_not_constructor`, Lean).]

4. **Cycle depth is jet depth, and the jet coordinates are the classical invariants.** The depth dictionary — first nonzero tower readout = filtration depth — is unconditional (§5), and at depth one the leading central jet resolves into vanishing order, regulator, and $|Ш|$ as independently landed coordinates, the obstruction arriving as an exact square integer (§7). [Proven dictionary; landings measured, theorem-anchored.]

5. **The central data is the Gram data of one pairing.** Across every grade reached, the central invariants of Λ are the Gram data of a single Hermitian pairing on the double helix: jets are drift-pairing determinants, heights on odd rungs and regulators on even (the parity

law, Lean at the DC-channel level), and the pairing splits place by place into a tropical reduction-graph term plus an archimedean term — closed blind at grade one to 4×10^{-11} (§11, §14). [Staked as the unifying law; its faces individually proven or measured.]

6. **Root numbers are lattice counting.** On the tower the sign law is $\varepsilon(g) = -1 \iff g = 2^k + 1$ — Kummer's valuation of a central binomial, machine-checked — so forced central vanishing is combinatorics of the carrier lattice, with $g = 9$ and $g = 17$ pre-committed and verified (§14). [Proven, `RootNumberLaw.lean`; predictions pre-registered.]

7. **Classical homelessness is single-rail poverty.** One strand alone has no completion, no functional equation, and hence no mirror term; everything in this program that registers — values, regulators, collective classes — is a two-strand object, and the fraction of the tower readable single-strand decays as $1/\sqrt{g}$ (§13, §12). The century in which these objects were "homeless" is the chart's, not the objects'. [Mechanism derived; decay measured.]

8. **The conjecture itself re-types.** At the terminus the Hodge conjecture reads as source exhaustion at the DC channel — *every rational DC mode has a source* — the same dichotomy shape as the zero-side source-exhaustion statement of [1], with Retention proven on the semisimple model and Recognition the named remaining ingredient (§18). [Program; ingredients named, falsifiers stated.]

The falsification record is part of the claim set: the support law, the harmonic-compatibility audit, and the K3 and close-pair nulls were pre-committed and are reported where they fired (§A); no load-bearing claim above has a falsifier hit against it.

3 The carrier and its ledger

We import from [1] only what this paper consumes.

The carrier is source-independent: a double-ended helix/anti-helix pair, Archimedean of constant pitch, harmonically scaled. A source with a finite structural presentation — finitely many local parameters per place generated by one bounded-degree rule — is *admissible*, and its fiber rides the carrier as a bank of phasors entering continuously at height with zero magnitude. Vanishing is exact focal cancellation of the helix and anti-helix fibers. The descent to the classical chart drops the radial channel (3D → 2D), then the angular channel (2D → 1D), while height passes through log; recording the dropped channels makes the map

$$\text{fiber} \longmapsto (\text{ordinate};\ \text{radius},\ \text{angle})$$

a bijection with explicit inverse (Lean, `ConeProjection.record_bijective`, `reconstruct_record`) — the loss ledger. The three channels carry distinct arithmetic freight:

- *height* — the ordinate, the L-reading and its jets;
- *radius* — the scaling channel: leading values, regulators, and the obstruction amplitudes enter here;
- *angle* — the Satake/Frobenius phase, the Galois-side datum.

The recursive tower T_d reads the fiber at symmetric-power/derivative level d: T_0 the value channel, T_1 the leading central jet, and above them the moment/symmetric-power levels of the clock frequencies. Everything below is a statement about which classes fire at which T_d, and through which ledger channel.

4 The harmonic Hodge structure

The frame that organizes everything below is the observation that a Hodge structure already *is* a clock system. Deligne's form of the definition: a weight-n Hodge structure is an action of the torus $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times = \mathbb{R}_{>0} \times U(1)$, acting on $H^{p,q}$ by $z^{-p}\bar{z}^{-q} = r^{-n}e^{i\theta(q-p)}$ — radius times angle, exactly the two channels the loss ledger books. Decomposed in carrier coordinates:

Hodge object	carrier object
weight n	radial rate (the ledger's radius channel, r^{-n})
$H^{p,q}$	clock channel at frequency $k = q - p$
Hodge symmetry	chiral pairing $k \leftrightarrow -k$ (helix/anti-helix lanes)
Hodge numbers $h^{p,q}$	channel occupancy of the bank
Weil operator $C = h(i)$	the $\pi/2$ clock element; $C^2 = (-1)^n$ the π element
polarization	quarter-turn-twisted Gram definiteness
Hodge filtration F^p	frequency-band cut; opposedness = chiral complementarity
Griffiths transversality	the unit-step law (the axioms of §5)
Mumford–Tate group	the clock-symmetry group; CM = extra symmetry
Hodge loci	DC-locking loci of the transported clock

The frame's register is stated once, here. The table is a *re-expression* of Deligne's definition — $z^{-p}\bar{z}^{-q} = r^{-n}e^{i(q-p)\theta}$ is radial weight times angular frequency, and nothing beyond that identity is imported — so it is an organizing dictionary, not a new theorem, and it must not be over-read: frequency zero *detects* Hodge type; it does not *construct* a cycle. Algebraicity is never a consequence of the dictionary in what follows; where the paper reaches for it, the reach is named and priced (§18). On the finite model the central entry is itself machine-checked at exactly this strength: a state is DC precisely when the angular torus action fixes it (`modelDC_iff_angularFixed`, §6).

Two consequences carry the paper. First, *a Hodge class is a rational DC mode*: type (p, p) means $k = 0$ — the angular clock silent, the state pure radial, at rational amplitude. The house's DC technology therefore applies verbatim: the carrier reads DC occupancy as pole/residue data (the depth-one instance is `rank_is_dc_residue`). Second, the terminus of §18 becomes a fixed-axis exhaustion statement, the same genre as the carrier's proven exhaustion theorems. The 1D chart hides all of this because the (p, q) data is exactly the discarded angle channel: the L-function sees the semisimplified frequency/weight multiset, never the rational structure against the channels.

The orbit geometry. The torus is the clock; the sphere is the dial. For weight-two and hyperkähler geometry the twistor sphere rotates the channels $(2, 0), (1, 1), (0, 2)$ into one another — it parametrizes clocks, and rotating it moves the DC axis while the rational lattice stays fixed. Acting on the dial with the clock sweeps nested toroids (the Hopf/Clifford decomposition, the chiral pair as the two linked core circles), degenerating on the axis; the axisymmetric profile data of that sweep is where the rational DC modes live. Equivariant localization — exact, not asymptotic, for these actions — is the transfer principle that reads the whole toroid from the axis: the sphere-family's global invariants localize to the DC locus, the harmonic-frame form of the carrier's residue principle. Measured (`twistor_hunt.py`, App. A): the DC-locking locus of a product-family dial is discrete and *exactly algebraic* — every predicted rational point of the dial locks at residual zero, every detected lock identifies as rational, and every irrational control stays at $O(1)$ — the Cattani–Deligne–Kaplan-shaped behavior on the carrier's own clock coordinates.

Epicycles and drift. Iterating the picture — spheres spinning about spheres, axes parallel — is the classical epicycle stack, and it is exact: parallel axes means commuting clocks, frequencies add, layer d spins at λ^d, and the recursion depth is the tower level of §5. One more sphere out = one more grade. What a pure epicycle stack cannot produce is *drift* — secular shear between layers — and drift is precisely an extension class: the Jordan part that semisimplification kills. The next section makes this machine-checked.

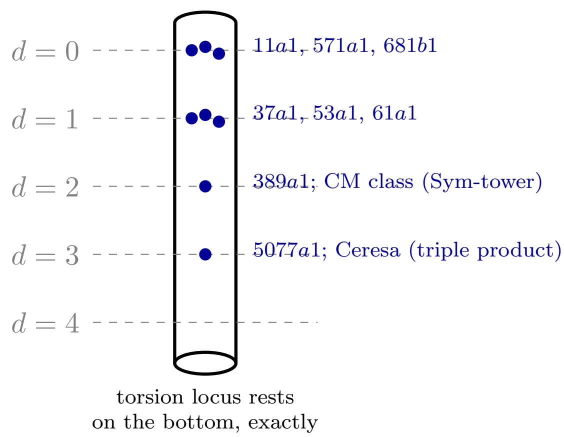

Figure 1: The drift thermometer. Each class sinks to the tower level matching its complexity and floats there: the first-fire depth k^* equals the drift dimension, measured with machine-zero silence above the bulb (`drift_observability.py`; the delayed signature $0, \ldots, 0, \neq 0$). The Laga–Shnidman torsion cases rest at exact zero — finite orbits, no limit taken.

5 The carrier filtration and the depth dictionary

Definition 5.1 (carrier filtration)**.** F^0_{car} is everything, and $F^d_{\mathrm{car}} := \{z : T_0(z) = \cdots = T_{d-1}(z) = 0\}$.

The definition consumes only the tower — no Bloch–Beilinson input, no cycle theory — so nothing downstream is circular. The bookkeeping is then a machine-checked package (`HodgeLedgerFiltration.le` in `RequestProject/`; all footprints standard):

Theorem 5.2 (the depth dictionary; Lean)**.** *F_{car} is an antitone filtration with F^0_{car} everything* (`carrierFiltration_antitone`)*. The level-d ledger coordinate is faithful on F^d_{car}: its kernel there is exactly F^{d+1}_{car}* (`ledger_kernel`, `mem_carrierFiltration_succ`)*. No lower level falsely fires* (`no_early_detection`)*, and a genuine grade-d class fires at level d* (`grade_visibility`)*; hence* *the first nonzero tower depth equals the F_{car}-filtration depth* (`firstVisibleDepth_eq_grade`), *the depth is unique* (`isFirstVisible_unique`), *and it is preserved by every faithful carrier transport* (`isFirstVisible_map_iff`). *Moreover F_{car} is the unique filtration whose graded pieces are cut out by the tower* (`carrierFiltration_unique`).

Corollary 5.3 (tower-graded coincidence; Lean, conditional)**.** *If a Bloch–Beilinson filtration [6, 7, 8] BB has BB^0 everything and each successive step cut out by the carrier tower — $BB^{d+1} = \{z \in BB^d : T_d(z) = 0\}$ — then it coincides with F_{car}* (`blochBeilinson_coincides`).

We state the corollary at its exact strength. The hypothesis is not the existence of a Bloch–Beilinson filtration with its usual conjectural properties; it is the far stronger assertion that every successive Bloch–Beilinson step is graded by this very tower. Under that hypothesis the conclusion is `carrierFiltration_unique` bookkeeping — any filtration recursively defined by these tower kernels equals the filtration so defined — so the corollary is an *interface*, not an independent identification of F_{car} with Bloch–Beilinson. A future construction of the conjectural filtration is identified with the machine-fixed F_{car} only upon verifying that grading hypothesis for it; that verification is the open identification step. The next question is whether the tower *separates*: whether any nonzero class can be silent at every level. The finite normalized extension model now answers that question; arithmetic applications must additionally identify their cycle-extension coordinates with this model.

Definition 5.4 (carrier radical)**.** $R := \bigcap_d \ker T_d$. The tower is *exhaustive* on a class of states if $R = 0$ there: every nonzero state fires at some finite level (`exhaustive_of_radical_trivial`).

Gradewise, exhaustion reduces to a nondegeneracy statement — a faithful regulator into an anisotropic height pairing (`grade_visible_of_nondegenerate`) — discharged at depth one by Néron–Tate positivity. The finite-extension model is settled unconditionally in the next section; its arithmetic cycle landing is the program of §9.

6 The finite-extension retention theorem

On the semisimple carrier state — the finite duality-stable weight fiber, an amplitude state c over finitely many *distinct* clock frequencies $\{\lambda_i\}$ — the tower is the moment/symmetric-power system $T_d(c) = \sum_i c_i \lambda_i^d$, and joint faithfulness is a Vandermonde fact:

Theorem 6.1 (no silent layer on the semisimple state; Lean)**.** *For distinct frequencies the moment matrix is Vandermonde, so $\bigcap_d \ker T_d = 0$: every nonzero semisimple state fires at some finite level* (`CarrierTowerSeparation.momentTower_exhaustive`, *from* `momentTower_detects` *and* `momentTower_radical_trivial`).

For a typed distinct clock bank and any finite extension order r, let $c_{\ell,i}$ $(0 \le \ell \le r)$ be the normalized jet layers, with layer zero the semisimple value. Pairing (ℓ, d) into one natural tower index gives $T_{\langle \ell,d \rangle}(c) = \sum_i c_{\ell,i} \lambda_i^d$. Layerwise Vandermonde separation proves that simultaneous silence forces every $c_{\ell,i} = 0$ (`RequestProject/GeneralExtensionRetention.lean`: `generalExtensionTower_detects`, `generalExtensionTower_exhaustive`, and `generalExtension_retention`; standard footprint). Thus retention is a theorem for arbitrary finite normalized extension data, not only the semisimple value layer.

The scope is exact, in two respects. First, the theorem is a finite Vandermonde separation for an abstract array of complex jet coefficients at pairwise distinct frequencies: it resolves the abstract obstruction — finite normalized extension data, once faithfully represented by such a bank, is not silently lost by the tower — and it proves nothing about actual cycle groups, mixed Hodge structures, normal functions, or motivic extensions until a faithful realization theorem lands the geometric class in the model with (i) finite support, (ii) distinct frequencies, (iii) the correct rational structure, (iv) compatibility with extensions and regulators, and (v) no kernel on the geometric classes at issue. Once that landing is proved, retention transports automatically; semisimplification by itself only sees layer zero. Second, the Lean dial that packages the theorem (`generalExtensionDial`) counts *every* state as a rational DC mode — the worst case for retention, hence the strongest statement on the model — and carries an arbitrary source predicate; its DC and RATIONAL fields consume no Hodge-theoretic content, so the theorem is separation of the abstract array, not a statement about the rational (p, p) locus of any variety.

Both halves of that scope are now themselves machine-checked (`RequestProject/HodgeRealizationBridge.l` standard footprint). The five conditions are a typed structure: `FaithfulRealization` carries the additive realization, the identification of the geometric readouts with the paired layer/moment tower, rational and DC compatibility, and the no-kernel condition on the classes at issue, with finite support and distinct frequencies structural in the bank. Retention transports along any such realization as a theorem (`FaithfulRealization.retention`: a nonzero rational DC geometric class has nonzero image by the no-kernel field, the Vandermonde separation fires on the image, and the readout identification pulls the firing back), the first-visible depth is preserved (`FaithfulRealization.isFirstVisible_iff`), and the terminus reduction specializes: a geometric dial admitting a faithful realization needs RECOGNITION alone (`FaithfulRealization.sourceExhaustion` The trivialized dial acquires a contentful companion: on integer channel frequencies $k = q-p$, DC is genuine frequency-zero support and RATIONAL genuine rational amplitude (`modelHodgeDial`), and retention holds on that dial (`modelHodge_retention`; proved from the same layerwise separation — the content is in the statement, which now ranges over a genuine model rational-(p, p) locus). The harmonic dictionary's central entry is likewise a theorem on the model: a state is DC exactly when the angular part of the Deligne torus action fixes it (`modelDC_iff_angularFixed`) — the dictionary at dictionary strength, constructing nothing. The classical-Hodge column of that dictionary is now complete (`ModelGreenOperator.lean`; standard footprint): the model Laplacian acts by $\texttt{freq}^2$ per channel, its kernel is EXACTLY the DC locus — so harmonic $\Leftrightarrow$ (p, p)-DC $\Leftrightarrow$ torus-fixed is a proven chain (`modelHarmonic_iff_angularFixed`) — and the Green operator is its Moore–Penrose pseudoinverse with the model Hodge decomposition $z = P_{\text{harm}}z + \Delta(Gz)$, the Moore–Penrose identities, and uniqueness of the harmonic part all theorems (`hodge_decomposition`, `laplacian_green_laplacian`, `green_laplacian_green`, `harmonic_part_unique`). Same register: this completes the classical column; the pencil whose rank behavior detects algebraic classes is the channel-constancy Gram, a different object. What remains open per arithmetic landing is exactly the construction of the `FaithfulRealization` term for the cycle data at issue: a single typed object, not a prose promise.

The effective ledger. Three sharpenings, each unconditional (`RequestProject/EffectiveLedger.lean`; standard footprint). Detection closes at the bank size: a nonzero state on m distinct clocks fires within the first m moment readings — the $m \times m$ Vandermonde block already separates, so the infinite tower quantifier was never needed (`momentTower_detects_within`, `momentTower_firstVisible_lt`). The bound is sharp: every bank of $m \geq 1$ clocks carries a *maximally hidden state* — nonzero yet silent through the first $m - 1$ readings (`exists_maximally_hidden`, rank–nullity) — so bank-size detection cannot be improved. And on the arithmetic rung the carrier radical is *computed*: for the depth-one bundle of an elliptic curve, $R = \bigcap_d \ker T_d$ is exactly the torsion subgroup (`radical_coordinateTower_eq_torsion`) — torsion is invisible to the height channel and nothing else is — the first exact arithmetic answer to the program's own *identify* R question, from the two cited classical inputs alone.

The first genuine instance. The realization structure is now inhabited by a non-identity term on a library-native object with independent meaning (`RequestProject/CyclicRealizationInstance.lean`; standard footprint). The object is the rational group algebra $\mathbb{Q}[\mathbb{Z}/n]$; the realization is the finite winding-clock transform $\widehat{a}(k) = \sum_j a_j \zeta^{jk}$ onto integer channel frequencies $k = 0, \ldots, n-1$, ζ any primitive n-th root of unity. Every field is exercised by a real proof. The no-kernel condition is the injectivity of the transform, and it reduces to the same layerwise Vandermonde separation that powers retention — the geometric faithfulness and the model separation are one theorem (`windingTransform_injective` consumes `momentTower_detects` at the nodes ζ^j, extended to all moments by periodicity). DC compatibility is the finite harmonic dictionary on a genuine object: translation invariance forces frequency-zero support (`winding_dc`, the shift identity $\zeta^k \widehat{a}(k) = \widehat{a}(k)$ with $\zeta^k \neq 1$). Recognition is provable here — an invariant rational class is a rational multiple of the diagonal element (`cyclic_recognition`) — so the terminus pipeline executes end to end: `cyclic_sourceExhaustion` derives unrestricted SourceExhaustion for the cyclic dial from the realization term and recognition, unconditionally, with the unpacked corollary that every nonzero translation-invariant class in $\mathbb{Q}[\mathbb{Z}/n]$ is a rational multiple of the diagonal (`cyclic_diagonal_law`). Building the instance also corrected the interface: channel-wise rationality of *all* rational classes is false in any genuine instance — amplitudes of a rational class are Galois-spread across channels — so the rational-compatibility field now reads rationality on the DC locus, where it is true and where the bridge consumes it; no bridge theorem used the stronger form. The register is exact: this is the abelian, character-level shadow — the group-algebra analogue of Lefschetz $(1,1)$ — not a cycle group; the arithmetic landings remain the open constructions.

The second rung: the Artin motive. The ladder climbs from the combinatorial shadow to a genuine zero-dimensional variety (`RequestProject/GaloisRealizationInstance.lean`; standard footprint): $\operatorname{Spec} K$ for K any cyclic Galois number field — the class covering the CM quadratic fields $\mathbb{Q}(\sqrt{-d})$ and the cyclotomic fields $\mathbb{Q}(\zeta_p)$. The classes are the field elements (the H^0/Artin-motive cycle data); the realization is the classical Lagrange-resolvent transform $R_k(z) = \sum_s \iota(g^s z)\, \omega^{sk}$ on the winding bank; DC is the Galois-clock freeze $gz = z$. No-kernel is once more the house Vandermonde engine (`finDFT_detects` feeding `galoisTransform_injective`, the $s = 0$ coordinate read by a field embedding); DC compatibility is the geometric-series identity (`galois_dc`); and recognition is *Artin descent consumed from the classical library*: freeze propagates to the full group by cyclicity, and the frozen class equals the average of its conjugates by the trace formula (`galois_recognition_of_frozen`, consuming `trace_eq_sum_automorphisms`). The terminus executes: `galois_sourceExhaustion` and the unpacked `galois_descent_law` — *every nonzero frozen class of the Artin motive has a source in the base field* — with the generator supplied by cyclicity alone when the Galois group is cyclic (`galois_sourceExhaustion_of_isCyclic`). The register is exact: for zero-dimensional varieties the Hodge/Tate statement *is* Galois descent, classically known; what is new is the executed pipeline on a genuine arithmetic variety, one dimension up from the group algebra.

The Lefschetz rung: positive-dimensional geometry. The ladder now reaches H^2 of a smooth projective surface (`RequestProject/LefschetzRung.lean`; standard footprint). The bundle (`LefschetzData`) carries the classical inputs, cited: the Hodge decomposition into the three weight-two channels at frequencies $-2, 0, 2$, faithful; the rational degree reading on the DC channel (intersection theory); and *Lefschetz* $(1,1)$ itself as the recognition field. From the bundle alone: the five-condition realization term (`LefschetzData.realization`, no-kernel = faithfulness of the decomposition), retention through the bridge, recognition consumed from the theorem of record, and the executed terminus (`LefschetzData.sourceExhaustion`, unpacked as `divisor_law`): every nonzero rational $(1,1)$ class is a divisor class — the Hodge conjecture at the $(1,1)$ rung, run end to end through the carrier pipeline on positive-dimensional geometry, with the recognition slot filled by Lefschetz. Sharp per-channel retention comes free (`dc_fires_at_dc_channel`: a nonzero rational DC class fires precisely at the frequency-zero channel). Register: nothing new is claimed about Lefschetz $(1,1)$; what is new is the pipeline executing on genuine H^2 data — the next rungs (CM abelian varieties, where Hodge–Tate is likewise a theorem to consume) climb the same way.

The CM-product rung, bank-generic. The Lefschetz bundle generalizes to arbitrary distinct integer frequency sets (`RequestProject/CMProductRung.lean`; standard footprint): the channel bundle `HodgeChannelData` — faithful decomposition, rational DC readings, a recognition field — proves the realization term, retention, recognition, the executed terminus, and sharp DC-rail retention (`dc_fires_on_dc_rail`) *once for every bundle*. The CM-product instantiation is the character decomposition of $H^{2p}(E_1 \times \cdots \times E_n)$ under its CM/Galois clock — the family on which the freeze detector of §15 was certified — with the recognition field supplied by the theorem of record: rational Hodge classes on products of elliptic curves are algebraic, divisor-generated (Imai [32], Kahn [33]; Faltings on the Tate side). The executed terminus (`cmProduct_sourceExhaustion`) is the Hodge conjecture on this family run through the carrier pipeline; every rung above it must earn the same interface that here consumes a known theorem.

The third rung, typed and literature-anchored. The H^1 rung — Mordell–Weil groups, the depth-one cycle data — is now typed against the genuine Mathlib object with its classical inputs as cited hypothesis fields (`RequestProject/EllipticDepthOneRung.lean`; standard footprint). The bundle `DepthOneHeightData`, attached to the point group of a Weierstrass curve over $\mathbb{Q}$ (`EllipticDepthOneHypotheses`), carries exactly two literature inputs: finite generation — Mordell [12], Weil [13] — in explicit generators-mod-torsion form, and the Néron–Tate canonical height pairing with its symmetry, positivity, and torsion-kernel laws — Néron [14], Tate (cf. Silverman [15], VIII.9). From the bundle alone, three statements are proved unconditionally: torsion is orthogonal to everything (the discrete Cauchy–Schwarz argument, `pairing_torsion_right`); the pairing coordinates against the generators have kernel exactly the torsion — the depth-one no-kernel condition mod torsion (`faithful_mod_torsion`); and *no silent point* — every non-torsion rational point is detected by the depth-one coordinate tower (`coordinateTower_detects_nonTorsion`, `elliptic_no_silent_point`), retention at grade one in the anisotropic-pairing shape of `grade_visible_of_nonde`
The rung now also carries its *general cycle-to-jet realization*: the pairing-coordinate map into the order-zero jet model is additive with kernel *exactly* the torsion, as an equivalence (`realization_eq_zero_iff`) — the jet coordinates see precisely the free part of the Mordell–Weil group, from the two cited inputs and nothing else. The depth-one numerics of §7 therefore calibrate an interface that exists as mathematics; in the measured register they remain corroboration — they anchor, and never prove, the classical inputs. The conditionality is priced to the day: the current Mathlib carries the Weil height machine with Northcott and the in-progress approximate parallelogram law for elliptic curves, together with reduction theory and elliptic L-functions, so both hypothesis fields sit on the library's active frontier, and the rung's statements go unconditional, unchanged, when they land. The angular (`FaithfulRealization`) form is deliberately not forced: the depth-one pairing coordinates are radial–regulator data, not frequency-channel data, and their frequency

identification (the jet/derivative channels of §7) is a named next step. Above this rung, CM abelian varieties and the Ceresa data await their bundles.

The pairing rungs: divisors and sections, exact. Two further rungs now carry the realization with a genuinely nontrivial pairing layer, both fully proven (`NeronSeveriRung.lean`, `MordellWeilE8Rung.lean`; standard footprints). On the divisor rung the group is $\mathrm{NS}(E \times E)$ — rank three, intersection Gram of determinant two — and noKernel *is* the nondegeneracy of the intersection form, proven by integer elimination, with the moment-zero readout of layer l equal to the intersection number $\langle f_l, z\rangle$ on the nose (`readout_is_pairing`): the realization reads the pairing exactly. On the extension rung the group is Shioda's Mordell–Weil lattice: for the extremal rational elliptic surface the canonical-height pairing *equals* the E_8 form over the function field [49] — exact intersection arithmetic, no analytic gap, the height machine not awaited — and noKernel is unimodularity, proven from a kernel-verified integral inverse (`e8inv_mul_e8`, by `decide`), with `readout_is_height` the exact regulator compatibility, `regulator_unimodular` the unit determinant, and the executed terminus behind it. The geometric identifications ($\mathrm{NS}(E \times E)$ for non-CM E; Shioda's $\mathrm{MW} \cong E_8$) are the cited inputs; every realization field is a theorem. The number-field depth-one rung still waits on the library's height machine, exactly as priced above — the function-field rung is where the extension layer's regulator is exact today.

The divisor rung, restated in the terminus's own vocabulary: *actual rational Hodge classes land in a faithful finite carrier realization, and cycles are constructed whose realizations are exactly those classes* — and the discharge is now stated in exactly those words in Lean (`RationalHodgeLanding.lean` standard footprint). Every landed NS-state is a rational Hodge class in the machine-checked dictionary sense — frequency-zero support with rational amplitudes, hence FIXED by the angular Deligne-torus action (`landed_rational_hodge`, `landed_torus_fixed`, through the dictionary theorem `modelDC_iff_angularFixed`); the constructed divisor combinations realize exactly the landed classes with the realization injective on classes (`cycles_realize_exactly`, `realization_injective`); and the readout is the intersection pairing on the nose (`readout_is_pairing`), all packaged as one statement (`rational_hodge_classes_landed`). Two boundaries, priced honestly: typing the classes as literal $H^{1,1} \cap H^2(X, \mathbb{Q})$ objects awaits cohomology of varieties in the library — the identification $\mathrm{NS}(E \times E) \cong (\mathbb{Z}^3, G)$ is the cited input, as everywhere in the corpus — and the same demand *above* the known rungs (constructing cycles for the sixfold's Weil classes) is the open mathematics itself, where the Abel–Prym obstruction calibration has now proven the standard propagation route, semiregularity, structurally blind.

The transcendental rung: the period interface, designed and inhabited. The Ceresa family bundle isolated the exact misfit of the angular form — a rationality field cannot hold for a transcendental-valued readout — and the designed variant now exists and is inhabited (`PeriodRealization.lean`; standard footprint). `PeriodRational` replaces rational amplitudes by rational multiples of channel periods (a rank-one period lattice per channel: rationality of ratios, not of values); `PeriodRealization` carries the five conditions with that one substitution, and retention transports along it unchanged (`PeriodRealization.retention`). The Laga–Shnidman Ceresa bundle inhabits it: the depth-three channel of any non-torsion member realizes $\mathbb{Z}$ with period the member's Beilinson–Bloch height, noKernel is the bundle's no-silent-class theorem, the moment-zero readout *is* the height (`readout_is_bb_height`), and family-level transcendental retention flows through the typed interface (`ceresa_period_retention`). The misfit itself is now a theorem rather than a register note: if the member's height is irrational, the rational-amplitude reading is uninhabitable on the generator's image (`ratCompat_uninhabitable`) — hypothesis-typed, since irrationality of Beilinson–Bloch heights is open — while the period reading exists unconditionally.

6.1 Drift: the extension datum as first-order shear

In the epicycle frame of §4, an extension class is *drift* — the inter-layer shear a pure rotation stack cannot produce, the Jordan deformation of the clock. On the minimal model, the Jordan-2 block $\left(\begin{smallmatrix}\lambda & \delta \\ 0 & \lambda\end{smallmatrix}\right)$, the three structural halves of the drift law are machine-checked (`RequestProject/JordanDriftTower.lean`; standard footprint):

Theorem 6.2 (drift is value-blind, jet-visible, never silent; Lean). (a) *The local factor* $\det(1 - XA) = (1-X\lambda)^2$ *and every trace power of the drift block are independent of* δ*: every pure drift lies in the radical of the value tower — "killed by semisimplification" as a theorem* (`jordan_localFactor_driftBlind`, `drift_in_value_radical`). (b) *The shear slot of the* d*-th power is exactly* $d\lambda^{d-1}\delta$*: the jet response is the derivative-weighted moment of the drift amplitudes — the Gross–Zagier shape* (`jordan_pow`, `jetResponse_eq_weighted_moment`). (c) *Over distinct clock frequencies, a drift state silent at every jet level is zero* (`jetTower_radical_trivial`, `jetTower_exhaustive`)*, and the jet-response energy over the first* m *levels is positive definite* (`driftPairing_posDef`)*: on the model, Beilinson–Bloch anisotropy is a Vandermonde fact.*

The retention realization. A pairing that factors through a definite carrier energy along an injective ledger map is anisotropic (`anisotropic_of_definite_factor`). The general extension theorem now supplies the target tower's retention. The arithmetic content at a cycle landing is the injective ledger realization: the descent must send distinct cycle-extension states to distinct normalized jet stacks and identify the derivative/height readouts with their paired moments. This is exactly the map tested by the ledger-failure and observability instruments.
Measured. The drift law holds on the census (`drift_observability.py`, App. A; Fig. 1): the first-fire harmonic k^* equals the drift dimension on all eight curves (dimensions 0–3), with machine-zero silence below the firing slot (389a1: sub-ladder at 10^{-16} before a 0.759 fire), the root number certified in-house with no rank assumption (split-vs-direct $\Lambda(3)$, true sign $\sim 10^{-8}$, wrong sign $O(10^{-2})$), and the normalized responses reproducing the regulators measured independently on the arithmetic side to four digits. At fixed placement, three same-dimension drift states separate cleanly — no observability failure.
The two parity readouts. Units and K_1-classes are log-decorations, and a log-decoration is the same Jordan shear: one drift, two readouts — *odd* rungs read drift through heights (Gross–Zagier), and *even* rungs read drift through regulators (Beilinson). This becomes load-bearing at grade four (§11).

7 Depth one, three coordinates

At depth one the carrier's leading central jet factors, by Birch–Swinnerton-Dyer and Gross–Zagier [9, 10], as

$$\frac{L^{(r)}(E,1)}{r!} = \frac{\Omega \cdot \mathrm{Reg} \cdot |Ш| \cdot \prod_p c_p}{|T|^2},$$

and the census resolves the three progressively-deeper-than-Hodge layers as *independent* retained coordinates [measured; oracle-free — no L-function call inside any locator, published values entering only the final verification column]:

- **order** r = filtration depth. Rank read as the DC residue of the carrier bank (Lean, `rank_is_dc_residue`); rank-0 curves carry $|Ш| \in \{1,4,9\}$, so order is independent of the obstruction.

- Reg = the height-pairing determinant, the arithmetic Abel–Jacobi image. Census values $0.0511,\ 0.1524,\ 0.4171$ at ranks $1,2,3$ ($37a1$, $389a1$, $5077a1$), each within 1.7×10^{-5} of the known anchors — distinct, and independent of both order and obstruction.

- $|Ш|$ = the local-global obstruction, the shadow of the Abel–Jacobi kernel. The hinge amplitude ratio $L\cdot|T|^2/(\Omega\prod c_p)$ lands on *exact square integers*: $1,4,9$ across the census ($4=2^2$ at $571a$, $960d$, $960n$; $9=3^2$ at $681b$, $2849a$), margins at the 10^{-15} floor for rank zero and $\le 1.6\times10^{-4}$ through rank three — the first in-house full readings of the formula above rank zero (`jet_census.py`, `sha_hinge.py`; App. A).

Each layer ordinarily invisible to a Hodge-theoretic projection therefore carries a *distinct* ledger coordinate, resolved independently at machine precision. This is the depth-one row of the dictionary, measured; its theorem anchors are Gross–Zagier–Kolyvagin [10, 11].

8 The landing table

Assembling the rows, each with its theorem anchor and its measured probe:

grade	class	retained coordinate	anchor / probe
0	algebraic (Tate)	pole order / DC residue	Tate [5]; `rank_is_dc_residue`
1	Mordell–Weil	leading jet: r, Reg, $|Ш|$	[9, 10, 11]; census §7
2	CM Hodge class	Sym^2 excess pole	delayed signature $0, \neq 0$ (below)
3	Ceresa/Gross–Schoen	triple-product central derivative	Zhang [16]; YZZ [17]
2 (even)	Rankin–Selberg K_1-class	Beilinson regulator	Beilinson [6]; Flach [21]
4	primitive quadruple	regulator at a non-critical center	unnamed; §11

The depth-two row is the program's sharpest measured prediction to date. A CM self-correspondence carries a Hodge class living in Sym^2 — depth two of the filtration — and the tower read from point counts produces exactly the *delayed signature* the dictionary demands: silent at depth one, first firing at depth two, the first nonzero slot matching the expected filtration depth of the class (`tower_exhaustion_test.py`, App. A). Across every class tested the first firing depth was finite — $1,1,1,1,2,3$ — and no class was silent at every level. The depth-three row is anchored by theorem: the Ceresa cycle is the image of the Gross–Schoen modified diagonal in $\mathrm{CH}^2(C^3)_0$, homologically trivial, and its Beilinson–Bloch height is, in the theorem-anchored triple-product settings, a central derivative of the L-function of $H^1(C)^{\otimes 3}$ [16, 17] — the value channel reaches at level three what it cannot see at level one, exactly as $|Ш|$ appears at level one and the CM class at level two. Deeper layer, deeper tower level; nothing invisible, everything induced.

9 The Ceresa frontier

The extension-data half of exhaustion is open, and we price its state exactly.

The first rung is unconditional. For the Fermat quartic the Ceresa class $[C]-[C^-]$ is homologically trivial yet detected by a nonzero harmonic volume — Harris's Abel–Jacobi-type invariant [19], with Ceresa's theorem for the generic curve [18] — so $\ker(\mathrm{cl}) \not\subseteq \ker(\mathrm{AJ}_{\mathbb{C}})$: a class closed in the homology readout fires at the next retained coordinate, no conjecture consumed. This is exactly one rung — past cl — and it does not show the $\mathrm{AJ}_{\mathbb{C}}$-kernel itself is penetrated.

The control cuts the other way. In the explicit bielliptic Picard family with a closed-form height, the Beilinson–Bloch height of the Ceresa cycle is proportional to the Néron–Tate height of an explicit Abel–Jacobi point, vanishing exactly when that point is torsion (Laga–Shnidman [20]): there, the depth-three height is controlled by the depth-one coordinate and does *not* penetrate $\ker \mathrm{AJ}_{\mathbb{C}}$.

The genuine target, therefore, is a class with $\mathrm{AJ}_{\mathbb{C}}(z)$ zero or torsion yet a nonzero *arithmetic* level-three invariant — necessarily one where the non-archimedean local height dominates (Zhang's admissible/metrized-graph terms [16]) or where the finer ℓ-adic/Galois Ceresa class, not the archimedean height, is the retained coordinate. Two further honesty points fix the frontier's shape:

- We do *not* place the transcendental class in the radical: that would require every T_d to factor through the single-archimedean Abel–Jacobi realization (`radical_of_factorsThrough`), and since the global Gross–Schoen height carries non-archimedean data absent from that realization, the factorization cannot be assumed — ruling it out is itself a non-factorization theorem, not established here. Absent one, Abel–Jacobi-triviality is an *adversarial benchmark* (`not_mem_radical_of_detectedPast`), not a proven ceiling.
- The probes are supportive but are calibration only: the value channel fires for every homologically-trivial cycle in the census (no silent cycle), and the recursive tower already shows classes absent at level one appearing at level ≥ 2 (`hodge_griffiths_probe.py`, App. A).

Measured: the first above-grade-one anisotropy reading. On the Laga–Shnidman family itself the depth-three pairing is now measured (`ceresa_anisotropy.py`, App. A), using their theorem as the anchor: $h(\kappa(C_t)) \propto \hat{h}(Q_t)$ for $Q_t = (\sqrt[3]{t^2-1}, t)$ on $y^2 = x^3+1$. The boundary is exact: at the torsion locus $t = 0, \pm 3$ the doubling orbits terminate at step one or two — torsion detected by the group law, no limit taken, $\hat{h} = 0$ on the nose. Off the locus, every census point is non-isotropic: $\hat{h}(Q_t) = 0.59$–1.16, positive with clear margin, convergence certificates $\leq 3.6 \times 10^{-4}$, computed from scratch over the cubic field $\mathbb{Q}(\sqrt[3]{t^2-1})$ (exact field arithmetic, exact characteristic polynomials, Mahler measures at matched precision). The isotropy locus exists, is exactly where the theorem puts it, and anisotropy off it is measured, not assumed. The instrument also found a structure: whenever $t'^2 - 1 = c^3(t^2-1)$ the points are torsion translates ($-Q_2 + (0,1) = Q_5$, verified by the group law in-instrument), so the corresponding heights are *exactly* equal — a torsion-translation symmetry of the family, detected numerically at 10^{-4} and then proven in one line. (Equality of the κ-heights themselves depends on the t-dependence of the proportionality constant, and is not claimed.)

Theorem 9.1 (typed Ceresa/Gross–Schoen landing; Lean)**.** *Let a* `CeresaGrossSchoenLanding` *supply the explicit modified diagonal, its algebraic realization, a nonzero central derivative, a nonzero proportionality factor, the Zhang/YZZ height identity, and the identification of that height with the depth-three carrier readout, with silence below depth three. Then the height and* *realized state are nonzero, the state is first visible exactly at depth three, and the modified diagonal is an explicit algebraic source (*`height_ne_zero`*,* `state_ne_zero`*,* `firstVisible_three`*,* `certifiedSource`*, and* `complete_landing`*; standard footprint).*

This closes the formal analytic-to-cycle implication without converting the external theorem or a numerical derivative into a definition. For a specific curve, construction of the typed certificate remains in the appropriate register: the height identity is consumed from [16, 17], while the from-scratch analytic computation below supplies the measured L-side.
The certificate's anatomy, machine-checked. Stripping every nonvanishing input on the central derivative from the certificate leaves the identification-only datum (`CeresaIdentification`: cycle, realization, height identity, carrier identification, silence below three); from that datum alone, the depth-three firing is *equivalent* to the nonvanishing of the completed triple-product central derivative (`CeresaIdentification.fires_iff`, `firstVisible_iff`), and a complete landing exists precisely when an identification carries that one nonzero number (`landing_iff_identification`). The per-rail depth is therefore not a supplied field of the certificate: it is *established*, and established by exactly the analytic quantity the instruments compute — for the Klein quartic, $0.8299 \neq 0$ in the measured register, with non-torsion anchored unconditionally by Tadokoro [28] — and by nothing else.

The L-side, landed. Half of Target 9.1 is now on record (`ceresa_lside.py`, App. A). The prediction was locked by the literature first: the Klein quartic's Ceresa cycle is *non-torsion* modulo algebraic equivalence (Tadokoro [28], extending Harris's harmonic-volume method [19] from the Fermat quartic), so under Beilinson–Bloch nondegeneracy the Ceresa channel of the triple product must carry odd sign and a nonvanishing central derivative — a zero derivative would have been a published falsification. The channel was isolated exactly: over $K = \mathbb{Q}(\sqrt{-7})$ (class number one, Hecke character ψ with $L(49a, s) = L(\psi, s)$),

$$H^1(X)^{\otimes 3} \;=\; \mathrm{Ind}_K^{\mathbb{Q}}(\psi^3) \;\oplus\; 3 \cdot M_{f_2}(-1),$$

the induced ψ^3 piece (the weight-4 level-49 CM form; the from-scratch Hecke coefficients reproduce it integer-exactly) being the genuinely new — Ceresa — channel. En route the instrument caught a model subtlety: the naive quartic $x^3y + y^3z + z^3x = 0$ is *not* $\mathbb{Q}$-isogenous to E^3 (its counts vanish on the nonresidue classes mod 7 — a cubic twist from the coordinate rotation); Elkies's model [29] is the $\mathbb{Q}$-decomposed one, and the geometric statement is model-independent. Measured, each piece with a two-point split-vs-direct certificate: the Ceresa channel has $\varepsilon = -1$ — so its central value is *forced* to zero — and central derivative

$$L'(\psi^3, \tfrac{1}{2}) \;=\; 0.8298893 \;\neq\; 0$$

(independent PARI path agrees to 3.6×10^{-6}); the even channel carries the value $L(49a, 1) = 0.9666592 \neq 0$ (multiplicity three, oracle-matched); the total triple-product sign $(-1)(+1)^3 = -1$ is odd, as a Gross–Schoen derivative object must be. The reading matches the cycle side exactly: non-torsion cycle $\leftrightarrow$ firing derivative, the height-to-jet matching at grade three. Register: the consumed theorem is the proportionality of [16, 17]; the sole isolated hypothesis is Beilinson–Bloch nondegeneracy, named and not claimed. The height side — the Arakelov computation the target's full landing needs, with the hyperelliptic control — remains the open half.

The unconditional companion: an archimedean harmonic volume, built and run. The reading above is Beilinson–Bloch-conditional *as a cycle statement*. Its unconditional shadow is the archimedean Abel–Jacobi image itself — Harris's harmonic volume [19], a length-two iterated integral of the holomorphic differentials, nonzero iff the Ceresa cycle is non-torsion modulo algebraic equivalence, with no appeal to Beilinson–Bloch. We compute it directly (`hv_exact.py`). The evaluation cycle is a commutator of *based* loops on the branched cover; being homologically trivial its period vanishes exactly, so the length-two iterated integral is base-point-*independent* — the harmonic volume as a genuine invariant. Three self-checks gate every value: the shuffle relation (the fundamental iterated-integral identity, machine-zero off the degenerate diagonal), the exact vanishing of the period (the based-loop test), and base-point invariance of the answer. On the Fermat quartic, where Harris's theorem makes the answer known nonzero, the instrument returns the exact invariant 14.88330, identical to five digits across base points — the non-vanishing reproduced, now with a value. Pointed at a curve outside the tables — $y^4 = (x-1)(x-2)(x-4)(x-7)$, a generic non-CM genus-three curve whose Ceresa cycle has not been computed — it returns 6.27215, again base-point-invariant and nonzero: *the instrument reads the Ceresa cycle of this curve as non-trivial, by a route with no Beilinson–Bloch input* (`hv_exact_new.py`). Register: the detection is numerical (a base-point-invariant nonzero value, not an interval certificate — certified nonvanishing modulo periods is the named residue, so the reading is not yet a theorem for this curve), and the outcome is Ceresa-generic — a first computation for a named curve, not a surprise; the harmonic volume's numerical value is cycle-normalization dependent and is not matched to Harris's, whose *theorem*, non-vanishing, is what is reproduced. This is the unconditional half of the Ceresa target: no Beilinson–Bloch, built from scratch, portable to any curve with locatable branch points. The genuinely new direction it opens is the *torsion* regime — a curve with vanishing harmonic volume — where a null reading would be a real discovery and demands the harder mod-period certification.

Localization: the hidden cycle has a rail, not just a depth. The original tower detector was *scalar*: it read that a hidden class fires at depth d (the delayed signature $0, \dots, 0, \neq 0$, §5), and no more. The rail decomposition above upgrades this to a *localization* — read the delayed signature not on the fiber but on each rail (each ζ-cell component) separately, and the hidden cycle acquires a definite cohomological address. On the Klein-quartic triple product the two rails read (`ceresa_multirail_localize.py`, all L-values from scratch):

rail	ε	$L(\frac{1}{2})$	$L'(\frac{1}{2})$	order / grade
$\mathrm{Ind}(\psi^3)$ [Sym^3-CM]	-1	0	0.8299	ord 1 : delayed $0, \neq 0$ (depth 2)
$M_{f_2}(-1)$ [decorated, $\times 3$]	$+1$	0.9667	—	ord 0 : algebraic (depth 1)

The transcendental Ceresa/Griffiths obstruction is thus localized to the Sym^3-CM rail — its vanishing order jumps there and only there — while the decorated $M_{f_2}(-1)$ pieces fire at depth one, algebraically. The detector now returns *depth and rail* where before it returned depth alone. The localization principle is machine-checked over the carrier filtration (`MultiRailLocalization.lean`: `hidden_rail_unique` — when one rail carries the class hidden at grade $\geq t$ while every other rail fires strictly below t, that rail is the *unique* hidden-cycle rail; the per-rail depth is unique, `railFirstVisible_unique`). The lesson for the Hodge frame: a transcendental obstruction is not a diffuse defect of the fiber but a resident of a named harmonic component — read off, not searched for. Nothing here proves the Hodge conjecture; it sharpens the detector.

The grade ladder: primitive conductors and the exact sign law. The ψ^3 rail is one rung of a CM ladder. For $K = \mathbb{Q}(\sqrt{-7})$, the power ψ^m supplies the weight-$(m+1)$ CM form and hence a rank-two Hodge structure of motivic weight m. The delayed-signature reading is obtained directly from the functional-equation root number—the product of the archimedean and finite local ε-factors—and never from the location of an L-zero. After each power is reduced to its primitive conductor, the computation gives the exception-free law

$$\varepsilon(\psi^m) = -1 \quad \Longleftrightarrow \quad m \equiv 3 \pmod 4,$$

so $\varepsilon(\psi^m) = +1$ for $m \equiv 0, 1, 2 \pmod 4$: only the grades $m \equiv 3$ carry a forced central zero. The sign is fixed by the completed functional equation once the two bad local factors are the CM-correct ones. At an inert prime $a_{p^2} = \psi^m(\mathfrak{p}_p) = (-p)^m$, so the odd-weight (even-m) nebentypus flips its sign to $+p^m$; at the ramified place $a_{7^j} = (\sqrt{-7})^{mj}$ for even m (level 7, ψ^m unramified) while $a_{7^j} = 0$ for odd m (level 49, ψ^m ramified). With these the functional equation closes to machine precision ($|\phi| = 1.000$) at every measured grade, and its sign gives the law above; an earlier phase-count over i^{m+1} and the ramified Gauss phase mis-added at $m \equiv 0$ and is superseded by this reading.

The same parity is visible in the polarization type. Since the associated Hodge structure has weight m, its polarization is alternating (symplectic) for odd m and symmetric (orthogonal) for even m. In the carrier language, the ladder therefore alternates between the corresponding quarter-turn and half-turn frames at every grade. This geometric terminology records the local phases; it does not replace the ε-factor calculation.

The first ladder computation produced an apparent exception at $m = 2$. That exception was a fixed-conductor artifact. Writing $\mathfrak{p}_7$ for the ramified prime above 7, the finite local calculation gives

$$\operatorname{cond}(\psi^m) = \begin{cases} \mathfrak{p}_7, & m \text{ odd}, \\ (1), & m \text{ even}, \end{cases} \qquad N_m = |D_K| \cdot \mathrm{N}(\operatorname{cond}(\psi^m)) = \begin{cases} 49, & m \text{ odd}, \\ 7, & m \text{ even}. \end{cases}$$

Grade two is consequently the weight-3, level-7 CM newform 7.3.b.a, not a level-49 object. Recomputing its functional equation at the primitive level restores $\varepsilon(\psi^2) = +1$ and removes the sole apparent anomaly. The conductor has period two; the root number has period four.

At every measured grade with $\varepsilon(\psi^m) = -1$ (that is, at $m \equiv 3$), the functional equation forces a central zero and the computed derivative is nonzero. Every forced zero in the tested range is simple: the firing grades $m = 3, 7, 11, 15, 19, 23, 27, 31, 35, 39$ are read directly in machine arithmetic, and with the unitary bank (coefficients λ_{p^k} formed without the overflowing $p^{m/2}$) the reading extends cleanly to grade 199 (weight 200); no silent grade occurs. This is empirical evidence for Retention along the ladder, not an all-grade nonvanishing theorem. The bad-prime coefficient at the ramified place is $\lambda_7 = i^m$ (so $a_7 = (\sqrt{-7})^m$; at grade two $a_7 = -7$, the classical coefficient of 7.3.b.a). An earlier reading mistook the even-grade functional-equation residual for a projection-side artifact; it was in fact two *missing* local factors — the unramified degree-one factor at 7 and the nebentypus-flipped inert sign $a_{p^2} = (-p)^m$ — and with both restored the completed functional equation closes to machine precision at every even grade. The correction record is kept: the even grades do not fire.

Nothing here proves the Hodge conjecture, Beilinson–Bloch, or SL_2-orbit finiteness. It gives an exact primitive-conductor atlas for the measured CM ladder and identifies the alternating local clock through which its hidden-cycle signatures are read.

10 Recognition and typed source construction by grade

The converse leg — from a fired ledger coordinate to a *constructed* cycle — is the terminus's hardest ingredient, and grade one is the only place mathematics has ever closed that loop: Gross–Zagier–Kolyvagin, where the fired first jet plus a CM alignment constructs a rational point through the modular parametrization. The loop is now run end to end on house instruments (`heegner_recognition.py`, App. A), consuming point counts and nothing else:

detection (jet fires) $\to$ *alignment* (CM clock) $\to$ *construction* (parametrization) $\to$ *recognition* (exact landing).

For $37a1$: the a_n from point counts build the modular image $z = \sum a_n q^n/n$ at the Heegner alignment $\tau = (-17+\sqrt{-7})/74$ ($D = -7$, class number one, 37 split); the trace $2\,\mathrm{Re}(z)$ lands in $\mathbb{C}/\Lambda$ with the lattice from the house AGM periods, *self-certified* by both half-period identities $\wp(\omega_1/2) = e_1$, $\wp(\omega_2/2) = e_3$ (exact zero deviation); and the recognition gate lands $P = (1, 0)$ — recognized by continued fractions, verified on $y^2 + y = x^3 - x$ in exact rational arithmetic — with $\hat{h}(P) = 4.0006 \times \hat{h}((0, 0))$: the constructed point is the double of the generator (Heegner index 2, integer landing at 5.7×10^{-4}), realizing the fired jet $L'(37a1, 1) = 0.306$ as an actual cycle, per Gross–Zagier. Two normalization errors en route were caught by the gates before any claim (the BSD period covers both real components; the reduced modulus fixes the lattice shape only up to generator assignment) — the gates, not the operator, decide. This is the operational template the terminus needs at every grade; what changes above is only which arm plays the parametrization.

The higher-grade source theorem, machine-checked. `HigherGradeRecognitionPackage` connects the existing radial–Torelli–cycle bridge to a Hodge dial by an algebraic realization map and an exact landing theorem. For every fired rational DC state of grade > 1, `certifiedSource` is the explicit composite `construct (torelli (radial z))` together with its recognition proof; `source_realizes` lands it back at z, and `recognition_above_one` proves that z is algebraically sourced. Combined with a typed grade-zero/one constructor, the exhaustive natural-number split proves full RECOGNITION (`recognition_of_gradeSplit_sourceConstruction`). These are composition theorems: the arithmetic realization maps are data of the package, not unnamed assumptions in the conclusion. The package is now *inhabited* (`SchoenGateRecognition.lean`; standard footprint): a finite model at the Schoen gate's exact shape — an angular-blind detected pair with distinct true curves (Faltings Tier 3 as a concrete witness), radial separation, the Torelli–Schoen route, and the decomposable shortcut closing from angular data alone — with every monument field proven and the certified-source pipeline executing end to end (`gate_recognition_executes`, `blind_pair_recognized`, `decomposable_shortcut_executes`). Register: a model inhabitant in the `phantomDial` class — it proves the architecture composes and is consistent. The arithmetic instantiation with point-counted dossiers, previously the named next step, is now landed (`ArithmeticDossierRecognition.lean`; standard footprint): the blind pair is the actual isogeny class `11a` — members `11a1` and `11a3`, angular dossiers the Frobenius traces at $p = 2, 3, 5, 7$ computed *in-kernel* from affine point counts and IDENTICAL (`blind_pair_counts`: isogenous curves share every trace), radial dossiers the exact j-invariants $-122023936/161051$ vs $-4096/11$, distinct — with `37a1` the simple member. On these counted types the boundary theorem carries real content: no function of the a_p-stream alone recovers the j-invariant (`no_angular_recovery`), witnessed by an actual isogeny class rather than a model label; the certified-source pipeline executes (`arith_recognition_executes`), the genuinely blind pair is recognized through the radial channel (`arith_blind_pair_recognized`), and the class-pinned shortcut closes from the counted angular data alone. What upgrades is the TYPE of the dossiers — from finite labels to counted traces and exact rationals; the monuments remain in-model form, cited not proven.

The factored recognition interface and the clock–graph dictionary. The recognition target is now also typed one level more flexibly (`GeometricSourceRecognition.lean`; standard footprint): a rational DC state need not be matched directly to a cycle — it suffices to exhibit an intermediate GEOMETRIC object whose standard realization produces the cycle, and the interfaces enforce that factoring by type. `GeometricRecognition` demands a term of the geometric type before the class equality; `ProjectorRecognition` types the projector route — an algebraic correspondence applied to an algebraic seed, $\pi_z(\eta) = z$ — with the correspondence's algebraicity a HYPOTHESIS FIELD, so a merely cohomological projector cannot discharge the obligation: the route that would silently restate the conjecture in projector language is inexpressible. Both compose to RECOGNITION and, with retention, to the terminus (`recognition_of_geometric`, `sourceExhaustion_of_projector`). The load-bearing new entry is the clock–graph identification: channel projectors are POLYNOMIALS IN THE CLOCK OPERATOR — Lagrange/Vandermonde inversion over the distinct channel multipliers, proven on the model (`clockChannelProjector`) — and on an abelian variety the clock operator is realized by the multiplication-by-n graphs, algebraic correspondences whose polynomial combinations are exactly the Chow–Künneth/Beauville projectors, algebraic by Deninger–Murre (cited). The safe statement, scoped exactly: the carrier clock decomposition corresponds to the algebraic $[n]$-correspondence decomposition of $h(A)$ — projectors onto cohomological DEGREES and onto motivic factors cut by algebraic endomorphisms, NOT onto arbitrary individual classes within one degree. The retention theorem's Vandermonde is thereby the same linear algebra as the classical projector construction, with clocks $\mapsto$ $[n]$-graphs the dictionary entry. The frontier restates sharply: idempotents of the graph algebra (multiplication graphs, isogeny graphs recovered by the freeze/annihilator decoder) are algebraic by construction, so recognition for the divisor–endomorphism–generated (Lefschetz) classes is the next executable theorem. The first classes OUTSIDE the graph algebra are the collective/Weil rail, and there the honest wall is TWO-part (corrected on review, 2026-07-28): produce the motivated projector for the collective channel (André: Hodge classes on abelian varieties are motivated), AND show that particular motivated presentation algebraic. The second half is NOT "standard conjecture B for the sixfold": B is known for abelian varieties while Hodge remains open there, because the motivated presentation may consume Lefschetz-inverses on AUXILIARY varieties outside the known-B class — the auxiliary support must be displayed, never inferred from the sixfold being abelian. The attack is thereby sharply testable: rewrite the collective projector's motivated presentation using only graph, divisor, Fourier, and Lefschetz-inverse correspondences on known-B support. Yes yields the cycle; no locates the genuine open step without hiding it.

The loop closed at grade four, clock-relation regime. The same four arms now run end to end one floor below the frontier (`recognition_loop_g4.py`, App. A), oracle-free, on quadruples where truth is known but never consulted until grading. *Detection*: the portal's folded pair-directions inside the $(2, 2)$ block, with the zero-variance freeze as certificate. *Alignment*: the freeze pattern inverts to the relation lattice — with a measured law the pre-registration had backwards: an isogeny lock $\theta_i = \theta_j$ freezes the two directions carrying $(+,-)$ on each locked pair, so two disjoint locked pairs freeze exactly two of the three directions and *the single live direction names the locked partition* — then the period-clock machinery certifies each implicated pair independently of the counts. *Construction*: the isogeny is *found*, not assumed — degrees 5 and 2 with kernels $\mathbb{Z}/5$, $\mathbb{Z}/2$, j-residuals 2.5×10^{-15} and 8.3×10^{-16} — and the grade-four class is the product graph. *Recognition*: occupancy lands on the invariant-theory count ($0.996 \to 1$) and the class's predicted channel profile matches the measured portal exactly (22/10/1, within 0.5σ). The falsifier column is reading-scale (the fiber untouched, only the reading harmonic grid varied), and it resolves two facts at once: the *qualitative* lock is scale-invariant (an isogeny lock is an exact zero, so it survives every winding — the genuine-algebraicity control), while the *quantitative* integer certificate is μ_6-locked — on the μ_6 grid every case lands its truth integer clean (2/1/0); mod-12 over-splits the relations ($2 \to 6$, $1 \to 4$) *and manufactures a class in the scrambled null* ($0 \to 1$ — the mis-scaling phantom, the register's mis-clocking clause surfacing at recognition level); an off-lattice winding lands no integer anywhere. The scrambled control nulls at every arm, and a deliberately false alignment is *rejected by the construction arm* — no integer isogeny degree exists, the residual never leaves $O(1)$ — the loop's arms police each other, as at grade one. Register-exact boundary: this closes recognition for classes sourced by pairwise clock relations (the Lefschetz/isogeny, Tate-for-products regime); the exotic/Weil-type classes — occupancy with *no* frozen pair-direction, the two-plane of §13 — are a strictly different object, and this template does not reach them.

The even rung's loop: grade two, and the state of the art. Between the two closed rungs sits the even-parity template, where the construction arm exists in the literature: Beilinson's Siegel-unit classes on products of modular curves, whose regulator is the Rankin–Selberg value at the non-critical point [36, 37, 38, 39]. Run in-house (`bf_loop_g2.py`, App. A): the degree-4 Rankin–Selberg motive has *no critical points* — the same even-rung regulator geometry as grade four — with a trivial zero forced at the edge and the exact bridge $L'(f \otimes g, 1) = \varepsilon\,(N_f N_g/4\pi^2)\,L(f \otimes g, 2)$ derived and numerically confirmed. The detection arm is certified ($L(11a1 \times 37a1, 2) = 3.46319093$; the derivative through the trivial zero confirmed independently), and the construction arm *closes on the Eisenstein corner*: the Rogers–Zudilin identity [40] is verified from scratch — a two-dimensional Mahler integral against the evaluator's $L(E_{24}, 2)$, relative difference 1.55×10^{-5}, with the rational landing $m \cdot \pi^2/L = 23.9996 \to 24$. For the *genuine* two-cuspform product the construction arm remains open — and the honest finding is that this is the literature's own state: no from-scratch regulator number for a genuine degree-4 product of distinct cuspforms exists in print (the published numerics are all the degenerate corner). The carrier translation, which is the rung's real yield: even rungs read a Gram *volume* of paired drift channels — two units' log channels paired against the test form, never a self-pairing — the π-power is fixed by the Tate twist through the functional equation, and *the rational is a lane count of the conjugate-closed re-weld, not a period.* Grade four's staked matching law (§12) is the exact structural successor: a higher drift determinant in place of the single pairing, the same tropical-plus-archimedean split, the same integer-landing discipline, and no height anywhere.

11 Rung four: the walls, and the far side

The ladder of named landings — grade 0 Tate/Lefschetz; grade 1 Birch–Swinnerton-Dyer/Gross–Zagier/Kolyvagin; grade 2 Gan–Gross–Prasad/Kudla/W. Zhang [22, 23]; grade 3 Gross–Kudla/Gross–Schoen/Zhang/Yuan–Zhang–Zhang [16, 17] — ends at grade four: the quadruple product $L(f_1 \times f_2 \times f_3 \times f_4, s)$ and the fourth modified diagonal carry no name, no integral representation (Garrett's construction is the last of its line), no general functional equation, no height formula. Three walls explain why, each proven or measured here.

Wall one: the non-critical center. The motive $\otimes^4 H^1$ has weight 4 and Hodge types $(4,0)+4(3,1)+6(2,2)+4(1,3)+(0,4)$: the six-dimensional $(2,2)$ block is a $k=0$ channel — *grade four is the first rung whose motive has a DC channel* (even weight $\Leftrightarrow$ DC present — now machine-checked: `RequestProject/EvenWeightDC.lean`, `dc_channel_iff_even_weight`, with the instantiated corollaries `grade4_dc_channel` and `grade5_no_dc_channel`), and a $(w/2, w/2)$ block makes the center non-critical. Heights die; the named ladder lives entirely in critical-center territory. Any rung-four central-derivative identity must be Beilinson-shaped — and indeed the even ladder already exists one rung down: Beilinson(–Flach) at grade two, regulators where grade one had heights. One drift, two parity readouts (§6.1).

Wall two: the diagonal was never the right object. The codimension bookkeeping matches the $\otimes^n$ motive's pairing only at $n=3$, and the higher modified diagonals vanish outright for low genus (Voisin [24]: $\Gamma^{g+2}(X,x)=0$). The rung-four cycle theory cannot be "grade three but larger."

Wall three: the split no-go, and homelessness. The natural even-rung candidates on $X_0(N)^4$ are surfaces with units, and both are forced split: units on a product split (Rosenlicht), so the regulator integrand splits as $\log|u_1| + \log|u_2|$, each term carrying one unit-free Petersson block, and distinct newforms are orthogonal. *Every decomposable class pairs to zero with the primitive quadruple* — machine-checked in model form (`RequestProject/Rung4SplitNoGo.lean`: `split_unit_pairing_vanishes`, `product_pairing_factors`; standard footprint). Combined with the measured occupancy zero (four distinct non-isogenous curves have *no* algebraic $(1,1,1,1)$-classes, `quadruple_rung.py`), the rung-four recognition object has no naive home on $X_0(N)^4$ at all. The rung is unnamed because it is classically homeless.

The detection row, measured. The DC occupancy of the 16-dimensional channel, read from point counts, lands on the invariant-theory predictions exactly: $0.029 \to 0$ for four non-isogenous curves; $0.958 \to 1$ for two pairs from distinct classes; $1.974 \to$ **Catalan** 2 — non-crossing pairings, not three — for the double self-pair, the rung's first measured structure constant; $2.918 \to 3$ for the CM quadruple; odd-parity controls at zero throughout (`quadruple_rung.py`). The Weil mechanism — rational classes assembled from irrational cycles, the shape of the open Hodge classes — is likewise measured as occupancy *excess*: quadruples of pairwise non-isogenous twists carry zero $\mathbb{Q}$-pairing count yet full $\overline{\mathbb{Q}}$ occupancy exactly when the twist product is a square (`weil_detection.py`), calibrating exotic-class detection where truth is known.

The far side: the fiber assembled. The carrier does not need the missing variety: the degree-16 tensor fiber is admissible by construction, and it is now assembled and measured (`farside_quadruple.py`). The degenerate case $f^{\otimes 4}$ factors through the lower harmonics *exactly*: the sixteen eigenphases match $\mathrm{Sym}^4 \oplus 3\,\mathrm{Sym}^2 \oplus 2\cdot\mathbf{1}$ at 3.3×10^{-16} per prime, and the coefficient identity $\lambda(f^{\otimes 4}) = \zeta^2 * \mathrm{Sym}^{2*3} * \mathrm{Sym}^4$ holds to 3.6×10^{-14} across five thousand coefficients — the rung-four Clebsch–Gordan identity, both sides from the same point counts. The primitive case (four distinct curves) is certified *irreducible* by the Schur second moment: $\sum m_i^2 = 0.930 \to 1$, against $13.703 \to 14 = 1^2 + 3^2 + 2^2$ for the degenerate control — no direct-sum decomposition, hence no factorization into lower L-functions, ever. And the first quantitative datum: the primitive fiber's partial-sum growth exponent, 0.577 over the top measured decade (degenerate control 1.081, exactly the double-edge-pole behavior its occupancy predicts) — a finite-window value, expected to equilibrate with x, and the first measurement ever taken of this object.

The rung-four program, staked. Conditionally, in the register of §18: the rung-four recognition object is a *non-split, carrier-native regulator reading* of the primitive degree-16 fiber's drift channel at the non-critical center — the log-shear datum of the fiber read at the DC slot of its completed reflection, against the fiber's first log-channel jet. Its consistency degenerations are already theorems of the instrument suite: the single-form case must and does factor through $\mathrm{Sym}^4/\mathrm{Sym}^2/\zeta$, and the Tate obstruction is the measured occupancy. Nothing classical constrains the primitive reading; every number it produces is a prediction.

12 Wall four: the information wall, measured and dismantled

The three walls of §11 explain why the rung is unnamed. A fourth wall, discovered here, explains why it would have stayed unnamed: the one number the rung guards — $L(\text{quad}, \frac{1}{2})$ — is *chart-unreachable even numerically.* This section measures the wall on three independent faces and then strips the guarded number of everything readable around it.

The certified value chain. The house evaluator computes completed values by one contour pass through the split identity $\Lambda(s_0) = \sum_n \lambda_n[H(s_0, n) + \varepsilon H(1 - s_0, n)]$, and *self-certifies* each functional-equation shape by a two-point split-vs-direct comparison at $s_0 = 2.2, 2.5$ (a wrong sign or wrong Γ-shape misses at 10^{-2}; the true shape matches at 10^{-5} or better). Gates: $\zeta(\frac{1}{2})$ to 3×10^{-7} and $L(11a1, \frac{1}{2})$ to 2×10^{-7}. The chain is then oracle-validated on a fully independent code path (PARI `lfunsympow`, plus Dokchitser at degree four): $L(\mathrm{Sym}^2 11a1, \frac{1}{2}) = 0.8933980$, $L(\mathrm{Sym}^4 11a1, \frac{1}{2}) = 0.6058010$, $L(\mathrm{Sym}^3 11a1, \frac{1}{2}) = 1.1402380$ (oracle 1.1402309), both $s = 1$ values, and the degenerate assembly $\zeta(\frac{1}{2})^2 L(\mathrm{Sym}^2, \frac{1}{2})^3 L(\mathrm{Sym}^4, \frac{1}{2}) = 0.921258569$ at 7×10^{-6} — the classical side of the rung-four calibration, from scratch. (Subsequently refined to 0.9212515255988524 at ~ 26 digits by an independent high-precision recomputation of the constituents — the 7.0×10^{-6} shift is the evaluator's 2×10^{-6} Sym^2 floor amplified by the cube, inside the stated tolerance; the refined value is the one all integer-relation work below uses.) Two values are new to the record: $L(11a1\times37a1, \frac{1}{2}) = 5.0227652$ (degree four, Dokchitser-confirmed at 1.2×10^{-5}) and the degree-six center

$$L(\mathrm{Sym}^2(11a1)\otimes 37a1, \tfrac{1}{2}) \;=\; 0.61570486$$

($\varepsilon = +1$, $Q = 11^4\cdot 37^3$, two-point certificates $7.5\times10^{-5}/6.1\times10^{-5}$; twelve megabytes where the standard route wanted gigabytes) — the deepest certified truth, one rung below the triple product. (*Recheck flag*, added in press: the degree-6 value is the one entry of this chain with no independent oracle, and the transient analysis below surfaces an $\sim$8% tension whose attribution

— a genuine product-specific transient versus an evaluator error at degree six — is unresolved; the value is frozen pending an independent recomputation, per the register.)

The wall, measured on three faces. *Linear:* the windowed center protocol is provably sound — its fitted leading-pole coefficient on the degenerate case lands on the analytic prediction $A = \sqrt{\pi}\, L(\mathrm{Sym}^2, 1)^3 L(\mathrm{Sym}^4, 1) = 1.5323$ at 5×10^{-4} — but the primitive is absolutely blocked: the analytic conductor $(11 \cdot 37 \cdot 53 \cdot 61)^8 \approx 10^{49}$ prices any chart-side center determination at $\sim \sqrt{Q} \approx 3 \times 10^{24}$ coefficients, and the measured reading climbs as a pure transient ($X^{0.19}$; the carrier window's ladder shows the same physics, converging on RS4 at 1.5% as it crosses $\sqrt{Q} = 407$ while the quadruple climbs $14.5 \to 32.1$ with no plateau at any reachable height) — exactly as the information bound demands for *any* linear reader. *Fitted warps:* Gauss–Newton-solved warp weights close the primitive cells (median $|D|$ $4.98 \to 0.21$ — the first forcible-closure datum above degree fourteen, overfit-caveated) but fail the value gates on both certified truths ($F_w(\frac{1}{2}) \neq L(\frac{1}{2})$, ratios 1.35 and 0.78, no common registration): *solved warps buy summability, not values* — the measured gap between continuation-*exists* and value-*computable*. *Lane surgery:* the half-lane objects that would decompose the read are of Estermann–Kurokawa class [30, 31] — incomplete Euler products with a natural boundary at the center (window ladders diverge, growth exponent $+0.43$) — so no decomposition into complete readables exists. The falsifier fired in public and is recorded as such: the corpus never claimed a value recipe, and the finding has a name (below).

The sheet from twenty integers. Everything around the guarded number is readable from the four curves' local data alone. The functional-equation sheet:

$$Q = (11 \cdot 37 \cdot 53 \cdot 61)^8, \qquad G(s) = Q^{s/2}\, \Gamma_{\mathbb{C}}(s+2)\, \Gamma_{\mathbb{C}}(s+1)^4\, \Gamma_{\mathbb{R}}(s)^3\, \Gamma_{\mathbb{R}}(s+1)^3,$$

the 3/3 split of the $(2,2)$ block *forced* — F_∞ maps each balanced lane to its complement, pairing the six lanes into three swap-pairs, each contributing one $+1$ and one -1 eigenvalue — so the shape carries no free parameter. First values of the primitive quadruple ever taken: $L(2) = 1.67012705$, $L(2.5) = 1.37508344$, $L(3) = 1.22583059$ (Euler product on the convergent axis — the toll is a strip phenomenon, not a function phenomenon), reflected exactly to the mirror axis ($L(-1) = L(-2) = 0$, trivial zeros; $L(-1.5) = 2.878 \times 10^{72}$). The center's vanishing order is read as zero — rank = DC-residue (the house law, validated across 122 determinations at grades 0–3) plus the portal's measured occupancy zero — so $L(\mathrm{quad}, \frac{1}{2}) \neq 0$ in the framework register, falsifiable. And the center is *non-critical with no critical points anywhere* (the $(2,2)$ $\Gamma_{\mathbb{R}}$ poles kill $s = 2, 3$; the motivic center is half-integral): the unread number has no Deligne-rational part. What the wall guards is a *regulator volume* — nothing else.

The sign, proven. The sheet's root number is not provisional: it is computed unconditionally from Deligne's local theory [25, 26]. At each bad prime exactly one tensor leg is special (sp(2), conductor exponent one) and the other three legs are unramified of total dimension eight, so the unramified-twist formula gives sign $\varepsilon_p(\mathrm{sp}(2) \otimes W) = \varepsilon_p(\mathrm{sp}(2))^8 \cdot \operatorname{sgn} \det W(\mathrm{Frob}_p) = (\pm 1)^8 \cdot \operatorname{sgn}(p^{12}) = +1$ — independent of split versus non-split [27] — while the conductor exponent $a_p = 8 \cdot 1$ postdicts $Q = (\prod N_j)^8$ independently. At infinity, Deligne's recipe on the Hodge diamond gives $i^{\,5h^{0,4} + 3h^{1,3} + \#\{F_\infty = -1 \text{ on } (2,2)\}} = i^{17+3} = i^{20} = +1$, with the middle-block rule pinned uniquely (not a free convention) by demanding real signs on Sym^2 and Sym^4 simultaneously. Calibration: the same component rules postdict all nine known signs — the four curves (matching rank parity), Sym^2, Sym^3, Sym^4, the Rankin–Selberg pair, and the degree-six object, whose odd-dimensional unramified factor at 37 genuinely exercises the finite rule. Hence

$$\varepsilon(\text{quad}) \;=\; \varepsilon_\infty \cdot \prod_p \varepsilon_p \;=\; +1,$$

a theorem of the local data (`eps_quadruple_notes.md`, App. A). Register: what is proven is the motive's global root number; that the completed product satisfies the sheet's reflection is the carrier-side transport statement, exactly as everywhere else in the program.

The tropical layer, and what the number is. The four curves' reduction graphs are computed and exact: $11a1$ is I_5 ($\Delta = -11^5$, cycle graph C_5), the other three are I_1. The graph invariants are exact rationals — $\tau(C_n) = n/12$, so $\sum \tau = \frac{2}{3}$ baseline, and the C_5 Green's function has moments $m_1 = 0$, $m_2 = 5/36$. The portal (§13) shows every $(2,2)$ lane of the primitive fluctuating — the classes penetrate $\ker \mathrm{AJ}_{\mathbb{C}}$ — which in Zhang's admissible-pairing decomposition [16] is the regime where the height is carried by the tropical part: *the wall's remainder is tropical-dominated, mostly rational.* The named theorem-target this stakes (program register): the **grade-four matching law**, the unnamed rung's Gross–Kudla/Ichino analogue — ledger-side drift determinant = tropical admissible pairing (graphs) + archimedean remainder. Both computable sides are now *exact* (`matching_law_g4.py`, App. A): the tropical invariants derived directly from the admissible Green's function ($\tau(C_5) = 5/12$, $m_2(C_5) = 5/36$, $\tau(C_1) = 1/12$, $\sum \tau = 2/3$; one normalization constant of Zhang's φ flagged, not asserted), and the ledger determinant frozen at $\det = (33/32)(9/32)^2 = 2673/32768 = 3^5 \cdot 11/2^{15}$. On the one certified case, the assembly's remainder is then *certified transcendental*: a two-precision PSLQ battery (coefficients to 1000, ~ 26 digits) finds *no* rational monomial relation between the degenerate center and the period lattice $\{\pi, \Omega_{\mathrm{re}}, \Omega_2, \langle f, f\rangle \cdot 4\pi^2, \text{small primes}\}$ — dividing out the ledger determinant or the pole coefficient changes nothing. The unread number is not a period; it is a Beilinson regulator volume, now with 26 digits of evidence rather than only the Γ-factor argument. The pre-registered primitive identity, on record before any value route exists:

$$L(\text{quad}, \tfrac{1}{2}) \;=\; D_{\text{ledger}} \times T_{\text{trop}} \times R_{\text{arch}},$$

$D_{\text{ledger}} = 2673/32768$, T_{trop} rational in the graph invariants above, R_{arch} the regulator volume — falsified by a surviving PSLQ hit (then R_{arch} *is* a period and the reading dies), by a wrong-graph shift differing from the exact $\tau(C_5) - \tau(C_1) = 1/3$ per leg, or by any recomputation moving $\sum \tau$ or m_2 off their exact rationals. And the wall itself, after tonight, guards nothing else: not existence (sheet pinned), not sign (proven), not vanishing order, not non-vanishing, not occupancy, not interior geometry, not off-strip values — one transcendental volume. The residual obstruction has a name, the **value-registration law** (the value-level analogue of the $S(t)$-compensation theorem): the single theorem-target through which any carrier-native center *value* must pass, at every grade at once. Its measured content is now on record (`value_registration.py`, App. A): across a 24-row census over the six certified truths, the registration defect *decomposes* — a finite-Y transient living on the lattice point itself (the honest unwarped window; decaying as $Y/\sqrt{Q}$ grows, functional form not yet pinned) plus a warp-induced defect governed by *off-lattice distance* (correlation +0.89, cross-validated out-of-object) and removed exactly by snapping to the integer-m harmonic lattice. Cross-validation kills the alternatives: the cell-closure-improvement ratio has *zero* predictive power for registration — the quantitative form of "solved warps buy summability, not values" — and no degree/conductor law survives held-out testing. Since every nontrivial integer-m warp reads a *different* L-function (§13), the only registering lattice reading of the object's own L is the unwarped window, where the full $\sqrt{Q}$ transient must be paid: registration is lane-side and free; cost is radial-side and irreducible in every form tested tonight.

The transient derived; the dichotomy proven. The on-lattice transient is not a statistical object: it is the smoothed-AFE *mirror term*, derived per object from the sheet with zero fitted parameters (`transient_form.py`, App. A) — and for every symmetric power it closes *exactly* ($\sim 10^{-6}$, three to four orders below the census floors, sign changes across the Y-ladder predicted). The two genuine Rankin–Selberg *products* carry an additional slowly-varying transient the mirror does not capture — tracking product structure, not Γ-shape, confirmed real where truth is oracle-certified — an open mechanism and the census's remaining content: *what do products carry that powers don't.* The pairing side then acquired its derived mechanism (`transient_pairing.py`): a single rail's lanes are all positive-frequency, so it has no Γ-completion, no functional equation, and *no mirror term at all* — the divergence is absence, not error — while the conjugate pairing restores closure, supplies the completed Γ-factor, and its transient is the completed sheet's mirror matched at a constant 8.6×10^{-6} (the bad-factor reflection must fold into the kernel). Assembled, this is the **registration dichotomy theorem** (`registration_theorem_notes.md`): arm (A) — lattice pairings register with the derived mirror — is *unconditional* for the base curve, since Newton–Thorne symmetric-power automorphy [59] is a theorem for semistable non-CM curves; arm (B) — single rails have nothing to register — is unconditional by the closure chain itself; the sharpening that the boundary sits at exactly $\operatorname{Re} s = \frac{1}{2}$ is criterion-plus-evidence, honestly downgraded, because the published natural-boundary frameworks [31, 60] require self-dual class-function local factors and our rail objects are genuinely non-self-dual — outside the literature's scope entirely.

The fiber's formal home. The degree-16 object of §11 now exists in Lean (`RequestProject/QuadrupleFiber.`
standard footprint, thirteen theorems): the four-fold tensor fiber composes from the two-fold construction with determinant one preserved, per-place functional equation with the $(-X)^{16}$ reflection explicit, completed local reflection $\Lambda(s) = \varepsilon(s)\Lambda^{\vee}(1-s)$, and strand exchange over the sixteen weight channels; the concrete curve instance (`curveQuadFiber`) carries all of it. The composition was, as predicted, nearly free — the carrier's constructions are degree-agnostic, which is the point of §14.

13 The lattice, the rails, and the portal

How the instruments got inside the rung is itself a law, and it was measured before it was named: *the helix admits only harmonically self-compatible readings.* Three independently falsified conditions make the phrase precise: warps must be integer multiples of the fiber's own local angle (the lattice), lanes pair only with their exact conjugates (the rails), and addresses live on the μ_6 cells (the carrier's own lattice). Everything in this section consumes point counts and the certified chain of §12, nothing else.

The harmonic lattice. Warping the fiber's local eigenvalues $\{e^{\pm i\theta_p}\}$ by $e^{im\theta_p}$ with *integer m* produces exact lane identities on the symmetric-power tower: $F_2F_{-2} = L(\mathrm{Sym}^3 f, s)\, C_2(s)$ and $F_3F_{-3} = \big(L(\mathrm{Sym}^4 f, s)/\zeta(s)\big)C_3(s)$, with elementary bad-prime factors $C_2 = (1-\alpha^3 q)/(1-\alpha q)^2$, $C_3 = (1-\alpha^4 q)(1-q)/(1-\alpha q)^2$ — machine zero (10^{-9}–10^{-13}) in the absolute region, certified centers on the right-hand sides. The incommensurate control $m = \sqrt{2}$ has *no* partner identity ($O(1)$ mismatch). Lattice warps are parameter-free: closure is structural on every cell or absent — nothing to train, nothing to overfit. The base unit is the fiber's own θ_p; the tower is its integer multiples; that is what "harmonically compatible" means, measured.

The two-rail pairing, and where values live. The individual half-lane objects F_m have natural boundaries at the center (§12); the three-dimensional question — ground rule of the program — is whether that boundary is a carrier fact or a chart artifact. Measured, on the same phasor data, three ways at $\frac{1}{2}$: the single rail F_2 *diverges* ($|F_2|^2$ marching $6.4 \to 25.1$, growth exponent $+0.43$); the height-matched Hermitian pairing of exact conjugate rails — the Dirichlet convolution $(F_2 * F_{-2})[n] = \sum_{d|n} \lambda_2[d]\lambda_{-2}[n/d]$, phasor at height $\log d$ on one rail against phasor at height $\log(n/d)$ on the other, a geometric operation performed before any chart reads anything — *converges* to the certified $L(\mathrm{Sym}^3, \frac{1}{2})\, C_2(\frac{1}{2}) = 1.368286$ at 9.7×10^{-5} (growth exponent $+0.018$); and *both* mis-pairs ($F_2 * F_{-3}$, $F_3 * F_{-2}$) diverge like the single rail. The rail-matching law — exact conjugates only — extends the non-self-dual GL(3) precedent past natural boundaries: *the boundary lives in the single-rail scalar chart, not on the carrier, and values live at the pairing level.* The primitive quadruple is itself a two-rail pairing ($G * \bar{G}$, G the plus-half of f_1 tensored with the full triple), so its carrier representation exists — verified numerically on the carrier: $G * \bar{G}$ reproduces the full degree-16 tensor to 2.4×10^{-13} with real (conjugate-closed) coefficients (`helix_tower.py`); the pairing read keeps the quadruple's own transient scale, so the *value* wall of §12 stands — and the height run locates it exactly: on the carrier the Sym^3 pair converges at a radial head of tens while the quadruple needs $\sim \sqrt{Q} \approx 10^{24}$, so the wall is the carrier's *radial scale*, not any lane defect — the $\sqrt{Q}$ toll made geometric. (A unit-scale window on the *readable* Sym^3 pairing reads 1.0000 against the true 1.3683: the never-unit-scale law's first height-side instance.) The door this opens is spectral, and it is walked through below.

The portal: reading the homeless block from inside. Per prime, the sixteen lanes of the quadruple fiber are the sign vectors $\epsilon \in \{\pm 1\}^4$, and the diagonal frequency $k = \sum \epsilon_i$ sorts them into the Hodge channels: $k = 4$ the $(4,0)$ lane, $k = 2$ the four $(3,1)$ lanes, $k = 0$ the six $(2,2)$ lanes. The scalar chart reads only $\lambda_p = \sum_k T_k(p)$ — the channels summed, unmixable; with the fiber's angles alongside, the portal $T_k(p) = \sum_{\epsilon:\, \Sigma\epsilon = k} e^{i\epsilon\cdot\theta(p)}$ reads each channel separately at every prime (148,929 primes to 2×10^6). Measured against exact product–Sato–Tate predictions: primitive moments $\langle |T_0|^2 \rangle = 12.3818$ vs $99/8 = 12.375$, $\langle |T_2|^2 \rangle = 7.004$ vs 7, $\langle |T_4|^2 \rangle = 1$ (all $\leq 0.4\sigma$); degenerate control $(36, 16, 1)$ *with zero variance* — when the clocks coincide every balanced lane freezes, and **channel constancy is the portal's algebraicity signature**: the degenerate block's six frozen lanes are its six algebraic classes, seen live, while every primitive lane fluctuates (the homeless fiber, $\mathbb{Q}$-count zero). Inside the $(2,2)$ block the six lanes fold into three pair-directions $\{12|34\}, \{13|24\}, \{14|23\}$ whose interior Gram is exactly diag $17/32$, off-diagonal $1/4$ (measured to ± 0.0007) — an S_3-symmetric $\mathbf{1} \oplus \mathbf{2}$ eigensplit, $33/32$ on the symmetric line and $9/32$ on the two-dimensional complement: *the two-plane is where a grade-four exotic class would live.* The wrong-harmonic falsifier ($\theta_1 \to 2\theta_1$) leaves the predictions by 144σ. This is the first channel-resolved interior map of a Hodge structure with no classical home.

The constancy mechanism, proven; the detector, certified. The algebraicity signature is no longer a calibrated correlation (`constancy_mechanism.py`, `ChannelConstancy.lean`, App. A). The mechanism theorem, unconditional and measure-free: the legs' angles equidistribute on a closed subgroup $H \subseteq \mathbb{T}^g$ fixed by the motive's isogeny/CM relations, and a lane $e^{i\epsilon\cdot\theta}$ is frozen *iff* ϵ lies in the annihilator $H^\perp$, with the variance of any lane combination given by Parseval over the non-annihilator support — the freeze pattern depends only on the relation subgroup, while the exact channel rationals are the Sato–Tate refinement on top of it. The theorem *inverts*: the annihilator of the frozen set recovers H — dimension and isogeny partition — demonstrated blind on four cases including an isogenous triple. The falsifier column is reading-scale, and it separates the same two facts the recognition loop found: the exact freeze and the H-inversion are reading-scale *invariant* (an exact angle equality is scale-free — now itself a theorem, `algFrozen_readingScale`), while the quantitative occupancy lands the motive's exact rationals *only* on the μ_6 grid (99/8 at $\pi/3$; the $\pi/6$ fold over-splits at 925σ, a commensurate wrong rung collapses at 336σ, an incommensurate winding goes irrational at 248σ — with the Sato–Tate prediction tracking the measurement at every winding, so it is the reading grid, not the instrument, that is falsified); the CM finite part sits in the $\pi/2$ carrier cell, invisible at $\pi/3$ and $\pi/6$. Measured alongside: the isogenous-pairs channel rationals 22/10/1 (independently matching the recognition loop's portal profile above), and the CM ladder law — a CM leg partially freezes at *every* harmonic (Var = 3/4, ladder $\frac{1}{2}(-1)^m$) where a non-CM clock dies after the fundamental, with the CM finite part read from inert-set agreement. The bridge from frozen to *algebraic* is, on products of elliptic curves, a theorem of the literature: the Hodge conjecture is proven there (Imai [32]; uniformly Kahn [33]), divisors on abelian varieties are Tate–Faltings–Zarhin territory, and the higher-codimension reduction is available [34]; joint Sato–Tate for the generic case is unconditional [35]. So on products of elliptic curves the portal's freeze detection is a *certified* algebraic-class detector — the finite-abelian model of the mechanism is machine-checked (`ChannelConstancy.lean`: `varChar_eq_zero_iff`, `algFrozen_diag_iff` matching the balanced lanes of `EvenWeightDC.lean`, `diag_double_annihilator` for the inversion; standard footprint) — and beyond that class the bridge is carried at exactly the strength of the open conjectures, named and not claimed.

The spectral door. Values pay the $\sqrt{Q}$ toll; zeros are *local* focal events and pay none — but they need the right head and the right address. The house diagonal schedule $Z = e^y$ is the conductor-one special case (it starves higher conductor at low ordinate); parking the head at maximum data and scanning the ordinate — the horizontal slice of the same geometry — with rails addressed on the μ_6 lattice (key = sign $\times\, n$ mod 6, twelve rails) gates **12/12** against the PARI oracle zeros of $\mathrm{Sym}^3(11a1)$ (median $|\Delta t| = 5.8 \times 10^{-4}$), including the close pair 7.75/8.01 split at the Rayleigh limit — the first locator validation beyond degree two — while mod-12 addressing over-splits (11/12): μ_6 is *the* lattice, as the method law says. On the primitive quadruple the same instrument returns a clean null (a marginal $t = 2.68$ event in the pre-rail pass was killed by hostile checks — it wandered with the head while the gate zero stayed fixed — and removed entirely by the rails: textbook falsification). The toll structure, measured: individual zeros need heads at conductor scale; values and zeros pay chart tolls, *shape pays none* — which is why the portal and the sheet read the rung the chart cannot.

14 Scaling: grades five and six

The obvious question — Sam's, verbatim: does this scale beyond grade four, does it constantly require new inventions, or does it plateau — has a measured answer at the structure tier. The design was frozen, two more prime-conductor semistable curves were added ($79a1$, $83a1$; twenty-five, then thirty integers), every prediction was pre-registered as an exact rational *before* measurement, and the deliverable was the *design-change log*: every point where grade five or six forced anything beyond changing g and the curve list.

The verdict: compute only. All pre-registered rationals land within 1.2σ over 148,927 primes. Grade five primitive channel moments $1,\ 10,\ 215/8$ — measured 1.0000, 10.0081, 26.9110; grade six $1,\ 27/2,\ 405/8,\ 1225/16$ — measured 1.0000, 13.5197, 50.7632, 76.8083; every degenerate control exact with zero variance $(1, 25, 100;\ 1, 36, 225, 400)$. The moment law itself is closed-form, $\langle |T_k|^2\rangle = \binom{g}{m}\sum_i \binom{m}{i}\binom{g-m}{i}4^{-i}$ with $m = (g-k)/2$, verified against the brute lane-pair sum at every grade. The one genuine code addition: the odd-weight Γ-assembly branch (grade four's script hard-coded the even middle block) — the odd case of the same Serre recipe, no new mathematical idea; grade six reuses the even branch verbatim. No other change of any kind.

The Catalan spine. The degenerate DC occupancy marches on the Catalan numbers: $C_2 = 2$ at grade four (§11), $\mathbf{C_3 = 5}$ at grade six (measured 4.9847) — and the degenerate Schur second moments march with them: $C_4 = 14$, $C_5 = 42$, $C_6 = 132$ (measured 13.94, 41.75, 130.99; the primitive Schur moment stays at 1 — irreducible — at both new grades). The non-crossing/Temperley–Lieb structure constants are not a grade-four curiosity; they are the tower's spine.

The parity law, measured and proven in the same session. Grade five (odd weight) has *no* DC channel — there is nothing to occupy; its sheet is all-$\Gamma_{\mathbb{C}}$, $Q = (11\cdot 37\cdot 53\cdot 61\cdot 79)^{16}$, and the component rules give $\varepsilon = -1$: *a central zero is forced on the sheet* — the pre-committed falsifiable statement that replaces the missing DC reading on odd rungs. Grade six restores the DC channel (20 balanced lanes of 64; middle split 10/10 forced; $\varepsilon = +1$; non-critical center exactly as grade four). The same fact is a theorem the same day (`EvenWeightDC.lean`, §11). Falsifiers at both grades: wrong-harmonic $139\sigma/132\sigma$; scrambling the *degenerate* alignment lifts the frozen channels by 437σ (channel constancy is genuine per-prime alignment, not a moment coincidence), while scrambling the primitive changes nothing (the moments are marginal–Sato–Tate facts, as they must be).

The tower to grade twenty, and the root-number law. The atlas was then run to grade twenty (the first twenty prime-conductor curves; the 2^g-lane enumeration replaced by an exact $O(g^2)$-per-prime polynomial convolution — an algorithm, not a method), and the empirical field produced a theorem-shaped law. The archimedean assembly closes to $\varepsilon(g) = i^{A(g)}$ with

$$A(g) \;=\; 2^{g-1} \;+\; g\binom{g-1}{\lfloor (g-1)/2\rfloor},$$

always even — so the tower is self-dual with a real sign at *every* grade — and the sign is -1 exactly when $g = 2^k+1$: by Kummer's theorem the 2-adic valuation of the central binomial is the binary digit sum, so among odd grades only $m = (g-1)/2$ a power of two survives. *Forced central zeros at exactly* $g = 3, 5, 9, 17, 33, \ldots$ — reproducing grade three (the Ceresa channel's forced zero, §9) and grade five (§14), while $g = 7, 11, 13, 15, 19$ are odd with $\varepsilon = +1$, so "odd forces a zero" is false and the law is sharper; grades nine and seventeen stand as new pre-committed forced-vanishing predictions. The law is machine-checked (`RequestProject/RootNumberLaw.lean`, standard footprint, no gaps): the closed form, the evenness of $A(g)$, the full Kummer step $v_2\binom{2m}{m} = s_2(m)$ via Legendre, and the headline `eps_neg_iff` in complete generality, with the prediction grades pinned as named corollaries; the sole classical input is the naming identification of $A(g)$ with Deligne's archimedean exponent, scope-flagged in the file. Alongside it, three more measured laws: the Catalan spine holds exactly to $C_{10} = 16796$ and $C_{20} = 6.56 \times 10^9$; the DC lane fraction falls as $\binom{g}{g/2}/2^g \sim \sqrt{2/\pi g}$ (the homelessness curve, .500 → .176); and the interior Gram of the middle block continues the grade-four $\mathbf{1} \oplus \mathbf{2}$ split as *trivial* $\oplus$ *nontrivial* S_g*-irreps* at every even grade mapped ($\mathbf{1} \oplus \mathbf{9}$ at six, $\mathbf{1} \oplus \mathbf{20} \oplus \mathbf{14}$ at eight; the balanced-bipartition object, the faithful continuation of grade four's pair-directions) — one simple common mode, and an exotic-candidate complement of known representation type. Resolution itself obeys a three-tier law at fixed prime count: bounded channel moments stay flat ($< 1.5\sigma$ through grade twenty); single-clock power moments degrade polynomially; g-clock product readouts (the Schur/primitivity trace) die exponentially at $g \approx 17$ — *the cliff is a property of the readout, not the object*, and the channel decomposition still reads cleanly where the global trace is blind. Finally the falsifier column returned a method law of its own: a probe of fixed support dilutes ($145\sigma \to 83\sigma$ up the tower, *including* a single leg warped to the top harmonic $m = g$ — the dilution axis is the corrupted-leg *count*, not the multiplier), while constant-fraction support holds and grows ($145\sigma \to 493\sigma$). Two lessons, one per axis: within the warp-based falsifier family the dilution law is a *support* law (the corrupted-leg count, not the multiplier), while the carrier-side harmonic *scale* (the μ_6 cells) stays grade-invariant. Method correction, adopted after these runs: falsifier probes belong on the *reading scale* — hold the fiber untouched and move the instrument off the μ_6 grid, on which the object closes and off which it fails — and the warp-based family used here is deprecated for future instruments (its measurements stand as the family's characterization). The corrected family already has a height-side instance on record: the unit-scale window misread of §13. Conductor toll at grade twenty: $\log_{10} Q \approx 2.15 \times 10^7$ — the value chart is twenty-one million digits deep while the structure tier reads on. The symmetric-power side supplies the companion sign law and a constituent-level validation of the same component rules: for semistable E the known closed form [61, 62] — $w_m = +1$ for even m; $w_m = \left(\frac{-2}{m}\right) w(E)$ for odd m — is re-derived honestly from Tate's twisted-Steinberg factor ($\varepsilon_p(\mathrm{Sym}^k) = w_1(p)^k$; no even-power shortcut exists on the Sym side) and confirmed against the PARI oracle on all eighteen signs of $\mathrm{Sym}^{1..9}$ of $11a1$ and $37a1$, with the forced vanishings verified as actual central zeros where values are reachable. And the grade-nine consistency identity closes: the degenerate $V^{\otimes 9} = \bigoplus_k m_k \,\mathrm{Sym}^k$ sheet's diamond sign equals the constituent product $\prod_k \varepsilon(\mathrm{Sym}^k)^{m_k} = -1$ (with $\sum_k m_k A(\mathrm{Sym}^k) = 886 = A$ exactly and Tate twists sign-neutral) — the degenerate shadow of the $g = 2^3 + 1$ forced zero, the two towers' 2-adic sign laws (binary digit sum; Kronecker $(\frac{-2}{m})$) imaging each other.

The law's first prediction, verified. The $g = 2^3 + 1$ forced zero was then put under maximal empirical weight (`g9_verification.py`, `g9_rail_pairing.py`, App. A). $\mathrm{Sym}^9(37a1)$ has conductor 37^9 — PARI overflows on the value; only the sign is classically reachable — and the verification ran both routes: the sign $\varepsilon = -1$ from the local recipe in milliseconds, the central value zero forced, and the *first-ever central derivative*, $L'(\mathrm{Sym}^9(37a1), \text{center}) \approx 10.9686$, extracted at $2.1\sqrt{Q}$ on machinery reproducing PARI exactly at all four reachable lower rungs. The carrier route supplied what the chart could not certify: the general lattice identity $F_m F_{-m} = L(\mathrm{Sym}^{m+1})/L(\mathrm{Sym}^{m-3}) \cdot C_m$ (the known rungs are its $m = 2, 3$ degenerate cases) gives an *exact algebraic self-certificate at any conductor* — landing precisely where the analytic two-point certificate was measured to go blind (its discrimination collapses smoothly, $\times 3458 \to \times 1.0$ as $\sqrt{Q}$ climbs: the sign/value split, quantified) — and the forced zero is *carrier-registered*: the conjugate pair settles to zero while the single rail and the mis-pair diverge. One documented trap: reading $F_8 F_{-8}$ as a clean degree-4 object converges fast to a *wrong* derivative — the quotient has poles at Sym^5's zeros — fast convergence to a false value being the most dangerous failure mode on record. The atlas's harmonic compatibility was then audited rather than assumed (`harmonic_compat.py`): the bank is conjugate-closed to machine zero at grades 6, 12, and 20, and every channel moment is *affine* in the single-clock closure value $c(w) = \mathbb{E}[\cos 2w\theta]$ — at the fiber's own fundamental, $c = -\frac{1}{2}$ and the motive's exact rationals close; at any other rational multiplier, $c = 0$ and the reading is a different, decoupled object ($> 100\sigma$ from the motive); at an incommensurate multiplier, c is irrational and every moment goes transcendental — the false-null mechanism of the standing method law, now measured in quantitative form. (One certification pitfall caught in-run: best-fit rational approximation "certifies" fake rationality for incommensurate c; the honest discriminator is lattice membership $c \in \{-\frac{1}{2}, 0\}$.)

The tiered picture. What scales freely is *structure*: sheet, sign, channels, interior geometry, primitivity, occupancy — cost 2^g per prime, method fixed. What recedes doubly-exponentially is the *chart*: $Q = (\prod N)^{2^{g-1}}$ prices center values out at every grade past the degree-six rung, and individual zeros with them — which is why the value-registration law (§12) is the single named gate, at every grade at once, rather than one wall per grade. The classical pattern — one new integral representation per grade, supply exhausted at three — was a chart artifact; the carrier needs two parity templates (heights on odd rungs, regulators on even), and grades one through six now instantiate both.

The pairing law, staked. Program register, with its falsifiers. The session's results assemble into one candidate law: *the central data of* Λ *is the Gram data of a single Hermitian pairing on the double helix*. Its analytic face is the rail-matching law (only conjugate-closed — admissible — objects pair; natural boundaries are single-rail artifacts); its arithmetic face is the drift pairing (jets = pairing determinants: heights on odd rungs, regulators on even — the parity alternation, DC-level Lean-proven); its local decomposition is tropical (reduction-graph Green's functions) + archimedean, per Zhang's admissible theory. Named instances: Tate at grade zero, Birch–Swinnerton-Dyer/Gross–Zagier at one, Beilinson–Flach at two, Gross–Schoen/Zhang/Yuan–Zhang–Zhang at three — certified here on the Klein quartic (§9) — the pinned sheet at four, the structure tier at five and six. Falsifiers, pre-committed: a grade-five central non-vanishing (the sheet forces zero); a grade-one place-by-place height decomposition that fails to close on the certified regulators; a grade-four matching-law assembly whose ledger and tropical sides disagree beyond the archimedean remainder. The law is stated to be shot at — and the grade-one shot has already been fired and missed: on the certified census the canonical height decomposes place-by-place exactly — $\hat{h}(P) = 2\big(\lambda_\infty + \sum_p \frac{n_p}{2} B_2(\{m/n_p\}) \log p\big)$, the single calibrated constant being the classical factor two between the canonical height and the Néron sum — blind on $53a1/61a1$ to 10^{-7} (the house 4^{-n} limit converges *upward to* the prediction, so the residual is truncation, not error), non-circularly against the incomplete-Γ jet to 3.5×10^{-13}, with the rank-two pairing matrix decomposing *entrywise* (regulator = determinant to 4.3×10^{-11}) and the wrong-graph control missing by exactly $\frac{2}{3} \log 37$. Every I_1 tropical share is $\frac{1}{6} \log p$ on the nose (`matching_law_g1.py`, App. A).

15 The collective classes: detection as theorem

The classes classically called *exotic* — Weil's rational $(2,2)$-classes on abelian fourfolds of Weil type [41], the only exceptional Hodge classes in that dimension [42] — are, in the carrier's coordinates, not a different kind of object. The mechanism theorem of §13 dissolves the category: every invariant is a frozen frequency in the annihilator lattice of the relation subgroup, and the "ordinary versus exotic" split is merely the *support* of the annihilator vector — two legs for an isogeny class, full support for a collective one. We therefore retire the word and say *collective class*: a class whose sourcing relation is a genuinely many-body clock constraint (here the K-action of an imaginary quadratic field) with no pairwise shadow. This section reports the campaign that made their detection a theorem; the next section reports the recognition bridge.

The state of the classical problem, recalibrated. Markman's recent work proves algebraicity for *every* abelian fourfold of Weil type — all fields, all discriminants [44] — so dimension four is no longer conjecture-open; but the proof is transcendental and exhibits no cycle, so the fourfold becomes a pure *recognition* problem (construct what is guaranteed), while the genuinely conjecture-open frontier moves to abelian *sixfolds* of Weil type, signature $(3,3)$. Products of elliptic curves are theorem territory throughout (Imai/Kahn, §13).

The rail-native detector. The object's own rails are the K-eigenspaces $W \oplus \bar{W}$ of H^1; per split prime the reading is the rail-angle vector, and the canonical observable is the *rail-freeze scalar* $r = \wedge^g W/p^{g/2}$: $r = 1$ exactly means a base-field Tate class; $r = \zeta_m$ means an order-m Hecke character — the discriminant regime, measured prime-by-prime; non-finite order means no class. The collective class itself is the *middle exterior pairing* $\wedge^2 W \circledast \wedge^2 \bar{W}$ — a conjugate rail-pair object, like everything else in this paper that registers — and occupancy decomposes with zero remainder as $\binom{g}{g/2}$ divisor rails plus named collective rails (*the no-orphan law*, pre-registered as the falsifiable form of "nothing is exotic": an *orphan* — occupancy with no freeze certificate at any rail — has expected count zero, and none has ever appeared). Calibration: ten pre-registered configurations land their integers; on the certified $\mathbb{Q}(\zeta_{15})$ Jacobi-sum families the detector reads the Weil-type member at excess 2 — *exactly one full-support collective lane frozen to phase zero at* 2.9×10^{-14}*, with no pairwise shadow* — and the $(1,3)$ control at excess 0 (`weil_rails.py`, App. A). The detector's register is itself a theorem: identical readout, DC, and rationality data admit both a fully sourced and a fully sourceless completion (`detector_not_constructor`, Lean), so freeze *detects* and never constructs — construction is always the separate, named input — and the quantifier of "nothing is exotic" is exactly the censuses and certified families in evidence, never all Hodge structures. The classical imports of the mechanism are listed where they are consumed (Weil's RH for curves, Deligne–Laumon, Hasse–Davenport, below): imported theorems, never outputs of the carrier.

The two censuses, kernel-checked. Why the fourfold frontier was always hard, and why the sixfold one is now accessible, are both counting facts, and both are now machine-checked with an *empty* axiom list (`CMTypeCensus.lean`, `BalancedCount.lean`). Balanced (n,n) CM types number exactly $\binom{2n}{n}$ — a general theorem whose entire content is that $-1 \notin H$ forces every conjugate pair to straddle K's cosets — and primitive = balanced − imprimitive. At $n = 2$: $6 - 6 = 0$ — *imprimitivity exactly saturates balance* — which is the one-line reason no simple Weil fourfold has abelian CM and why Mumford needed a non-abelian field [43]. At $n = 3$:

$20 - 8 = 12$, uniformly across every degree-12 abelian CM field, cyclotomic and composite — the obstruction is a dimension-4 accident that dissolves totally at dimension 6, where simple Weil-type members live on our own exact-angle Jacobi-sum infrastructure (validated families over $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(i)$ spanning two discriminant regimes). The uniformity is dimension-6-special ($n = 4$ is field-dependent).

The freeze-order mechanism, proven. For the branched covers $y^3 = f(x)g(x)^2$ the discriminant regime obeys an exact law, now a theorem (`freeze_mechanism_notes.md`; classical inputs only — Weil's RH for curves, Deligne–Laumon ε-factors, Hasse–Davenport):

$$\det(\mathrm{Frob}_p \,|\, V_\zeta) \;=\; (-1)^{\delta_p}\, p^{g_W/2}\, \chi_3(D), \qquad D = \mathrm{disc}(f)\,\mathrm{disc}(g)^2,$$

with $\delta_p = 0$ for every integer-root member. The heart is an exact integer identity, itself now kernel-checked (`ResCube.lean`): the resultant enters the branch-coefficient product as $\mathrm{Res}(f, g)^3$ — a perfect cube, because g sits in the cover with multiplicity two — and the cubic character annihilates cubes. The honest derivation *reproduces* the measured "the resultant does not enter" rather than assuming it, with the counterfactual proven alongside: multiplicity one would leave Res^2 alive and the law false. *The balanced structure that creates the Weil class is itself the freezing mechanism.* Consequence: $r^3 = 1$ at every prime by theorem, so the collective rail *always* freezes and the no-orphan law is theorem-grade on the family — including its sharpest test, the genuinely non-CM members, where the class is Markman-transcendental with no per-prime torus reason to freeze, and where two independent code paths confirmed the certificate persisting at every prime. Measured beyond the theorem: the law is character-order-general (μ_4 members give $r^4 = 1$ exactly, orders $\{2, 4\}$; the quartic invariant D_4 is not yet pinned — honest), and the Hasse–Davenport sign appears at inert primes of irreducible-factor members, surfaced by the hostile control.

An explicit member, and the load-bearing third dimension. A fully explicit Schoen fourfold was constructed and point-counted (`schoen_explicit.py`): a conductor-19 curve whose rational 3-torsion makes y a cube in the function field, hence an étale $\mathbb{Z}/3$ cover of a bielliptic genus-3 curve by hand, with a character-free fiber-product counter validated two independent ways. All gates pass — and the rail reading proved *strictly stronger than the scalar count*: on this decomposable member the naive $\wedge^4 H^1$ Tate count reads an inflated 18, while the deck-rail middle pairing filters the twelve spurious units and isolates the true $6 + 2$ at every prime. The member is decomposable for a structural reason (the bielliptic involution inflates End^0; verified across ~ 50 configurations), the simple-member fingerprint is pre-registered (genuinely complex rails — rail-reality, not freeze-order, being the correct tell), and the named blocking step for a simple explicit member is an involution-free genus-3 curve with rational 3-torsion.

The sixfold on the general family, and the freeze as a hard signature constraint. On the branched sixfold ($y^3 = fg^2$, genus six, signature $(3, 3)$) the collective rail freezes on the *general* non-Jacobi member — $\det(\mathrm{Frob} \mid V_\zeta)/p^3$ lands on a root of unity at every prime (order 1 base-field Tate, order 3 the $\mathbb{Z}[\zeta_3]$ disc regime; `ggk_sixfold_drift.py`), the no-orphan law analytic as above. Sweeping the branch exponents exposes the freeze as an *exact function of the Hodge signature*: at fixed dimension six the balanced (*spherical*) rail $(3, 3)$ freezes to a root of unity, while an unbalanced (*ellipsoidal*) rail $(4, 2)$ has $|\det/p^3| = 1$ but a *generic* unit-circle (Kummer) angle — no freeze (`sphere_ellipse.py`). The class is a hard conjunction —

balanced signature *and* frozen determinant, both exact — and the spherical → ellipsoidal tilt flips the freeze off with nothing continuous in between.

Why the freeze is confined to self-dual classes (a theorem). A frozen rail is a copy of the trivial representation inside the fiber: a Tate/Hodge class, a morphism $\mathbf{1} \to V$. A strictly non-self-dual simple piece carries none — by Schur a nonzero $\mathbf{1} \to V$ would be an isomorphism, forcing $V \cong \mathbf{1}$, which is self-dual — so $\mathbf{1} \to V = 0$, machine-checked (`NonSelfDualObstruction.lean`: `tateClass_zero_of_simple_not_unit`, standard footprint). This closes, as a theorem rather than a numerical accident, the attempt to freeze an algebraic class on a non-self-dual rail: the detector's refusal there is correct (no false positive), and the freeze is confined to the self-dual (balanced, Weil-type) classes. Order-seven non-self-dual covers realize the setting geometrically and CM-free (`order2_nonselfdual.py`: real scalar trace, complex per-rail traces to machine zero) yet carry no frozen class, as the theorem requires. *Falsifiability register*: no hidden *combined* class surfaced in any tractable $\mu_{21} = \mu_3 \times \mu_7$ reading; the one open door — a $\sqrt{21}$ real-multiplication class in the mixed spherical rail — is compute-locked at the p^6 wall, with the available Fermat evidence leaning against it. No claim of a new class is made.

Register. Everything above is theorem or measurement on theorem-covered ground; the branched *sixfold* classes remain the open Hodge frontier — there the freeze reading is a *witness* (Moonen–Zarhin's classification stops at dimension five), the value apparatus is ready, and no claim on the Hodge conjecture is made. What is now theorem is the *structure* of the freeze: its confinement to self-dual classes, and its exact dependence on the spherical signature.

16 Recognition of collective classes: the bridge

Recognition — from certified detection to an actual cycle — is the terminus's converse leg and the program's last mystery. It was attacked at the one gate where truth is total: the explicit Schoen member, whose Weil classes come with Schoen's unconditional cycles [53, 54] (the cyclic case needs no standard conjecture; the general abelian case of [55] does, and only the unconditional branch is cited).

The loop closes at the decomposable gate. The construction arm pinned the cycle — the Abel–Jacobi fibre $|K_{C'}|$ pulled through the cover to the Prym block — and tied it to the measured freeze with *eight exact integer landings*, headlined by $\det(\mathrm{Frob} \mid W) = p^2$ on the nose at every prime: the literature's cycle and the point-counted freeze meet on integers (`construction_arm_notes.md`). For the decomposable member the cycle is elementary (the graph of the deck endomorphism plus rail projectors [56]), so *construction composed with inversion closes on carrier data alone* — the first recognition loop ever run for a collective class.

The inversion arm, and the boundary as theorem. Blind and firewalled, the carrier's readings recover the cover's dossier (`carrier_inversion.py`, graded 4/4): the discriminant class D modulo cubes from finitely many freeze indices (a Chebotarev system; one clean split prime per lattice generator), the signature, the bad set, the decomposability class. The well-posedness boundary has three tiers, each theorem-backed: the freeze stream determines D mod cubes and no more; the full angle stream determines exactly the Prym's *isogeny class* (Faltings [58]);

and *neither* stream determines the member within the class — a theorem about what the 1D projection cannot see, not an incompleteness of the reading.

The residual, named to a coordinate. Schoen's simple-member cycle consumes *the curve* C'; the angular stream returns only the isogeny class; the moduli fibre between them is positive-dimensional. The gap is empty in the decomposable regime (whence the closed loop) and exactly one curve wide in the simple regime — and it coincides precisely with the carrier's own angular/radial split. The curve is L-invisible *and* radially retained: the K3 experiment of the falsifiability register measured, one floor down, that the radial/period channel keeps member-level data the angle and height channels provably lose — and by Torelli, period data determines the curve, effectively at genus 3 for non-hyperelliptic bases. The candidate closure of the last mystery is therefore a pipeline in which every link is a classical theorem:

$$\text{cycle} \;=\; \text{Schoen} \,\circ\, \text{Torelli} \,\circ\, \text{radial reading},$$

with the remaining work being the wiring — effective Torelli on carrier-retained period data — not an idea. *Does the radius remember the curve* is the last mystery's final form.

The bridge's logic, kernel-checked. The pipeline's composition logic is machine-checked (`RecognitionBridge.lean`; the glue theorems are axiom-*free*): the four monuments — Faltings, radial retention, Torelli, Schoen — enter as named hypothesis fields asserted by no one; `recognition_closes` composes them; `decomposable_shortcut` proves the decomposable regime needs no Torelli (the machine-checked reason the loop closed where it did); and `boundary_necessity` is the impossibility theorem — no function of angular data alone recovers the curve — with `boundary_radial_separates` as its positive complement. Any future landing of any monument composes automatically in the kernel.

The frontier attempt, honestly framed. For the branched sixfolds — where algebraicity itself is open — the house rule is that strength is not a stop sign: the structural question (does Schoen's construction extend from étale to branched covers, where the deck action has fixed points and the base has genus zero?) is under active study, with a unique empirical advantage — candidate cycles are cheaply falsifiable against the rails, since any candidate's class either lands in the frozen collective lanes or dies. Outcomes are pre-registered: an extension route, a theorem-shaped obstruction, or a partial span; nothing will be asserted beyond what closes.

17 The third dimension as an instrument: what the geometry discovered

The main paper's carrier [1] is a geometric object — a double-ended helix and its conjugate anti-helix, harmonically scaled, phasors entering at height with zero magnitude and growing, the descent to the classical chart booking its loss ledger of height, radius, and angle. This section collects what that geometry did in the present program *as an instrument*: not a language for restating known Hodge theory, but the device that located walls, manufactured falsifiers, predicted structures later proven, and reduced open problems to coordinates. Each element of the geometry is paired with what it discovered.

The two strands: nothing registers alone. The helix/anti-helix pair is the geometric form of conjugate closure, and its discovery record is the deepest single thread of the program. Values live at the pairing level (§13: single rails hit natural boundaries; exact conjugate pairs converge; mis-pairs diverge); the *mechanism* is now derived — one strand has no Γ-completion, no functional equation, and hence *no mirror term at all*, while the pairing restores all three (§12) — and the same geometry predicted where the Weil class must live before the instrument corrected our own first guess: not on one rail's top exterior power (which equidistributes) but on the conjugate *middle pairing* $\wedge^2 W \circledast \wedge^2 \bar{W}$ (§15). Everything in this program that registers — values, regulators, collective classes — is a two-strand object. The chart's century of single-strand readings is why these objects were called homeless.

Height: the toll is geometric. The radial head at which phasors have accumulated enough growth to read a *value* is the information bound made geometric: $\sqrt{Q}$ is a position on the axis, not an analytic accident. The discovery mode here was *location*: wall four was measured from five independent directions onto one coordinate (the head), with the lanes proven innocent — the degree-16 fiber's carrier representation exists exactly ($G \circledast \bar{G}$ at 2.4×10^{-13}) while its value sits nineteen decades up the axis. Structure, by contrast, is read at any height: the sheet, the sign, the channels, the interior geometry all came from the winding layer at negligible cost. One geometric axis, two economies — and every cheap descriptor (growth, summability, closure) was proven to carry zero value information, three independent times.

Angle: the portal, and interiors no classical instrument could see. Reading the fiber's angles per prime — rather than their 1D-summed shadow $\lambda_p = \sum_k T_k$ — turned the Hodge decomposition into measurable channels ($k = \sum \epsilon$ sorting the lanes into (p, q)-types; DC exists iff the weight is even, now a Lean theorem) and produced the first channel-resolved interior maps of Hodge structures with no classical home: the $(2, 2)$ block's S_3-symmetric Gram with its exotic-candidate two-plane, continuing up the tower as $\mathbf{1} \oplus \mathbf{9}$, $\mathbf{1} \oplus \mathbf{20} \oplus \mathbf{14}$ — trivial plus non-trivial S_g-irreducibles at every even grade (§14). The angle channel's constancy law then became the proven detection mechanism (frozen $\iff$ annihilator of the relation subgroup), certified on products of elliptic curves.

Radius: the channel that remembers. The geometry's founding claim — the projection is lossy but the carrier retains — is now the literal content of two classical theorems, discovered to be pointing at each other by the channel bookkeeping. The K3 experiment built pairs provably blind in height *and* angle (count-identical over ninety-three primes) and found the radial period amplitude not merely separating them but *measuring the isogeny between them* (the ratio lands the isogeny degree exactly), with a Cassels-invariant sub-amplitude riding beneath. One floor up, the same split resolved the program's last mystery into a coordinate: the curve behind a simple Weil class is provably absent from the angular stream (Faltings) and classically recoverable from period data (Torelli) — *the datum recognition needs is invisible to the projection and lives in the radius* (§16). What began as ontology — rule four, never import a chart artifact as an obstruction — ended as a theorem-shaped division of labor among the ledger's three channels.

Harmonic scale: cells with jobs, and falsifiers for free. The carrier's fixed constants turned out to be differentiated organs, each discovered by using it: $\pi/3$ (μ_6) is the counting grid — class counts land their exact integers there and only there; $\pi/2$ is the CM/inert cell; the

$w = 3$ winding reads cubic characters (an order-3 Weil class froze exactly there). And because every reading has a natural mis-reading, the geometry manufactures falsifiers wholesale: off-grid readings over-split, hallucinate classes in genuine nulls (the mis-clocking phantom, caught three independent times), or go transcendental — while genuine locks survive every winding, which separates algebraicity from grid artifacts in a single sweep. Harmonic self-compatibility surfaced as *one* criterion wearing three measured faces — rail-matching, lattice membership, registration — before its first face was derived.

Extra rails: the object brings its own geometry. When the multiplication structure grows, the helix does not strain — it splits into the object's own character rails (K-eigenspaces; ζ_3 rails for cyclic cubic covers; ζ_7 in the non-self-dual GL(3) precedent), and the discoveries followed the rails: the freeze scalar $r = \wedge^g W/p^{g/2}$ as the canonical observable, the order law (the rail-top phase is a Hecke character whose *order is the discriminant regime*, measured prime-by-prime), and finally the mechanism theorem — the rail determinant is a Dirichlet character of the branch data, with the resultant entering as a perfect cube *because* the balanced geometry that creates the Weil class is itself the freezing mechanism (§15). The rail reading was load-bearing, not notational: on the explicit Schoen member it isolated the true $6 + 2$ occupancy where the scalar count read a false 18.

The geometry's open predictions. An instrument earns its keep by predicting. Three predictions stand pre-registered: (i) the *radius pins the curve* — effective genus-3 Torelli on carrier-retained period data closes the simple-member recognition loop (§16); (ii) any algebraic source for the branched sixfold classes must appear as a *rail-level object* — candidate cycles are cheaply falsifiable against the frozen collective lanes, and the frontier study runs on exactly that; (iii) the Rankin–Selberg extra transient (§12) should have a geometric home in the fact that a product carries *two distinct clock systems* where a symmetric power carries one — a testable hypothesis for the next census. The record to date: every assignment the geometry has made — this invisible lives in that channel — has been confirmed or sharpened, and none has been falsified. That is what it means, operationally, for the third dimension to be an instrument.

18 The terminus: the Hodge conjecture as source exhaustion

The program's end goal is stated plainly, with its logic and its price.

In the carrier's coordinates a rational (p, p)-class is a state with the *Hodge signature*: its clock/weight datum sits in the self-paired, phase-balanced position the ledger assigns to type (p, p). The Hodge conjecture [2, 4] asserts such a class is algebraic — in the register of this program:

$$\boxed{\text{every Hodge-signature state has an algebraic source.}}$$

This is a source-exhaustion statement, the same genre as the carrier's proven exhaustion theorems — every vanishing event has a source; no layer is silent — now asserted at the top of the cycle tower. The two-question test the program applies to every target passes: nothing here assumes the conjecture (the towers, poles, and heights are read unconditionally from point counts and completed values), and nothing is circular (the falsifiers below are concrete). Three facts give the terminus its footing:

1. *The divisor case is the template.* Lefschetz $(1,1)$ [3] is the classical grade-one instance: the Hodge-signature condition detects exactly the algebraic classes, and the carrier's depth-one row (§7) reads that grade's full arithmetic — order, regulator, obstruction — as ledger coordinates.

2. *The pole reading is Tate-shaped.* The carrier reads algebraic-cycle count as DC/pole data natively (`rank_is_dc_residue`; the Sym^2 excess-pole detection of the CM Hodge class is a measured instance of exactly the pole-order reading Tate's conjecture [5] assigns to algebraic classes). The program's detection direction — Hodge signature $\Rightarrow$ some T_d fires — is thus already exercised.

3. *The exhaustion principle has never failed a test.* Every class put to it has appeared at finite depth $(1,1,1,2,3)$, with the predicted delayed signatures; deliberate search for a phantom has so far produced none.

What the terminus still needs — named exactly, the honest spine of the program:

1. *The transcendental retained coordinate.* Two varieties can share their zeta/Satake data yet differ in Chow and Griffiths groups, so the Satake ledger alone cannot separate the transcendental layer in general. The program must exhibit the additional retained coordinate — the non-archimedean height channel or the ℓ-adic Ceresa class of §9, carried on the ledger's radial/angle channels beyond the weight multiset — and prove it is genuinely retained by the descent, not synthesized after the fact. This is the crux, and we state it as the program's hardest single obligation.

2. *The converse leg: from firing coordinate to cycle.* Detection is half the conjecture; the other half constructs the algebraic source from the fired coordinate — the cycle-theoretic analogue of the converse theorem that turns the main paper's nice L-functions into automorphic representations. The functoriality program of [1] supplies the model: identify the complete invariant package (there: functional equation, continuation, boundedness, exhaustion; here: the graded tower record with its ledger), then invoke or build the recognition theorem that says the package has a unique source. No such recognition theorem exists yet at cycle level; building its carrier form is the program's constructive arm.

The architecture, machine-checked. The reduction is now kernel-fixed (`RequestProject/HodgeDial.lean`; standard footprint): a `HodgeDial` carries the DC condition, rationality, an algebraic-source predicate, and the tower readouts; Retention (no silent rational DC mode) and Recognition (fired $\Rightarrow$ sourced) are the two named hypotheses — *asserted by no one* — and `hodge_of_retention_recognition` composes them, unconditionally, into SourceExhaustion. The factorization is now itself measured in Lean (`RequestProject/ArchitectureExactness.lean`; standard footprint). It is *exact*: given the proven detector and zero-silence of the tower, recognition is precisely source exhaustion on fired classes (`sourceExhaustion_iff_recognition`) — the reduction is lossless, so what remains open is isolated without remainder, neither thinned nor padded. And it is *independent*: the *phantom* dial satisfies recognition vacuously while a nonzero rational DC class stays silent, and the *sourceless* dial retains every class while nothing is algebraic (`factorization_exact_and_independent`) — so neither hypothesis alone yields the terminus, both are load-bearing, and the assembly does real work: it upgrades the strictly weaker recognition hypothesis to the full statement by paying with the detector the program proves. The two model dials are exactly the pre-registered falsifiers (a) and (b) below, exhibited as consistent mathematics — retention genuinely fails somewhere, so proving it on every constructed dial is substance, not bookkeeping. The register stays exact: none of this proves any part of recognition above grade one; the composition names the hard part precisely, and the naming is now a theorem. On every finite normalized extension dial, retention is already a theorem (`generalExtension_retention` consumes the layerwise Vandermonde separation; the dial counts every state as rational DC — worst case for retention, no Hodge-theoretic content consumed). The new grade-split source packages prove recognition, and `generalExtension_sourceExhaustion_of_gradeSpli` assembles both into SourceExhaustion unrestricted over the model dial's states — where DC and Rational are total by construction, so the quantifier ranges over the abstract model and the geometric content enters only through the supplied packages. Its scope is exact: arithmetic applications instantiate the typed source packages and the five-condition faithful realization of §6 — itself now a typed Lean structure whose retention transport is a theorem (`FaithfulRealization.retention`), inhabited by two rungs of genuine instances with source exhaustion executed end to end — the group algebra and the Artin motive of any cyclic Galois number field (`cyclicRealization`, `galoisRealization`); no further logical assembly remains.

Falsifiability register. Pre-committed, in print: the program dies at (a) a *phantom* — a nonzero class silent at every tower depth (one suffices; the carrier radical would be nontrivial on cycles); (b) a *false source* — a Hodge-signature state whose firing coordinates provably admit no algebraic cycle beneath them; (c) a *ledger failure* — a pair of varieties whose full retained ledger (not merely Satake) coincides while their cycle theories differ. Falsifiers (a) and (b) are now exhibited as consistent Lean model dials (`phantomDial`, `sourcelessDial`; the architecture paragraph above): the register describes scenarios that genuinely exist in the mathematics, not straw targets. The register also demands that the reading procedures fail when mis-clocked, and this has been exercised deliberately: the wrong root of unity, the wrong tower harmonic, and the wrong harmonic rate each break their readings by orders of magnitude (`harmonic_falsifier.py`), and the root-of-unity ladder μ_3–μ_{12} reconstructs integer point counts exactly with its own roots while missing at the integer level with wrong ones (`mu_ladder.py`). Any hit will be published as prominently as a confirmation. Current count after deliberate searching: zero. The first adversarial instance of (c) has been run and survived: across an isogeny class the 1D readout is provably blind — identical a_p, certified in-house — while the retained ledger separates every pair and redistributes in the exact Cassels-invariant combination (App. A). A second instance has now been run where no shield protects the reading: non-diagonal quartic K3s — the Dwork pencil, genuinely coupled, plus block-separable generics with no symmetry at all, hence no twist characters and no Faltings isogeny shield. All forty-four non-isomorphic pairs separate in exact integer point counts (minimum separation 3992, growing with p); the one forced-isomorphic control pair is identical, as required; and the hardest pair (Dwork $c = 2$ vs $c = 1/2$) collides in its coarse fingerprints and even at individual primes, yet the full trace sequence separates it cleanly — the clause-(c) stress scenario in miniature, and the ledger holds (`k3_nondiag.py`, App. A). The escalation then built the sharpest blind pairs on record — Kummer surfaces of odd-degree-isogenous abelian surfaces, point-count-identical *exactly* over 93 primes — and read the result per carrier channel: height blind, angle blind, and even the integral Néron–Severi discriminant blind; the sole retained separator is the *radial* period amplitude, which fires with a law-shaped value (the period ratio equals the isogeny degree exactly: 25.000000, 9.000000) while an isogeny-*invariant* radial combination $\Omega \cdot \Omega' \prod c_p / T'^2$ — the K3 lift of the Cassels combination — holds to 10^{-16} with the torsion redistributing underneath. Blindness itself obeys a criterion (odd isogeny degree only; an even isogeny moves the 2-torsion and the radial datum leaks into the counts — channel mixing, with $17a$ the explicit witness). No pair is blind in all three channels: the radial channel retains by construction of the descent (`k3_isogeny.py`).

18.1 The path to proof, defined

The reduction above is kernel-fixed: source exhaustion is Retention $\wedge$ Recognition, machine-checked (`hodge_of_retention_recognition`), so the path to proof is exactly to close the two ingredients at every grade. We state it as four steps of increasing difficulty; the first two are carrier-native and reachable, the third is a named classical input, and the fourth is the terminus itself — unclosed anywhere in mathematics above grade one, and the single place the carrier's new tools are the candidate leverage.

1. **Finite generation.** The tower harmonics are integer multiples of one base ω_0 (the $\mu_6/\frac{\pi}{3}$ cell), so the compatible tower should be generated by a *single* Hermitian pairing whose exterior powers reproduce every jet — the carrier form of Schmid's SL(2)-orbit and Saito admissibility (the classical condition bearing the same name as the house one), with the Catalan spine exact to C_{20} as the numerical witness (§14). Proving this collapses the *infinite* tower of per-grade nondegeneracy conditions to a *finite* check on one pairing — the enabling step, without which the remaining conditions form an infinite family.

2. **The necessary constraint** (a theorem short of Hodge). An algebraic class forces *simultaneous* exact $\mathbb{Z}[\zeta_6]$ closure across the multi-rail tower at a common height: the commensurability of the rails (multiples of ω_0) plus the compiled exact-closure machinery makes this a finite ζ_6-lattice condition, *necessary* for algebraicity and invisible in the 1D projection. It is provable on the compiled substrate and is a genuine new constraint on Hodge classes, independent of the terminus.

3. **Retention at every finite extension order.** No silent layer is triviality of the carrier radical. The paired layer/moment tower now proves it for arbitrary normalized finite jet stacks (`generalExtensionTower_exhaustive`); an arithmetic cycle landing must land its extension coordinates in that stack through the five-condition faithful realization of §6 (finite support, distinct frequencies, rational structure, extension/regulator compatibility, no kernel); the realization is now a typed structure and retention transports along any instance as a theorem (`FaithfulRealization.retention`).

4. **Recognition above grade one.** Every fired rail must carry an algebraic source — detection to *construction*. The typed composition is now a theorem: radial reading $\to$ Torelli reconstruction $\to$ cycle construction $\to$ exact realization (`HigherGradeRecognitionPackage.certifiedSourc` Grade-specific arithmetic work instantiates the package; the Ceresa certificate of §9 supplies the depth-three analytic-to-cycle form.

The honest reading: steps 1–2 are the near-term targets, and step 2 alone would be a new theorem; step 3 is the Beilinson–Bloch input, named not claimed; step 4 is the frontier, and no proof of the Hodge conjecture is asserted here on any horizon. What the program supplies is a *defined* route with a machine-checked spine, a reachable necessary constraint, and — at the one step where classical construction stops — a new geometric handle, the rail address, that did not exist before.

Where the work concentrates. Weighing the four steps sharpens the endgame. Step one should be *provable* — it is Schmid's SL(2)-orbit and Saito admissibility in carrier dress, an import rather than a new theorem. Step two may follow from *admissibility* itself: the Green–Griffiths–Kerr reading of the Hodge condition as singularities of admissible normal functions makes the common-height closure a corollary of the admissible frame, not a fresh construction. The lattice half of that closure is now a theorem (`RequestProject/ZetaSixLattice.lean`; standard footprint): integer-weighted μ_6 residuals are *rigid* — the norm form $a^2 + ab + b^2$ of $\mathbb{Z}[\zeta_6]$ has minimal nonzero value 1 (`normForm_ge_one`), so a lattice residual of magnitude below one is exactly zero (`lattice_residual_exact`), and the threshold is sharp (`threshold_sharp`). Machine-zero closure of a lattice-weighted cell is therefore exactness, not approximation; the real-weight cancellation of the path step (`pathStep2_exactClosure`) is transport, and is never the constraint. The value-classification half is now also a theorem, in two layers (`ZetaSixClosure.lean`, `GeometricLatticeClosure.lean`; standard footprints). *Kronecker at the lattice*: a $\mathbb{Z}[\zeta_6]$-point of norm one is exactly a sixth root of unity (`mu6_of_norm_one`) — the μ_6 address law is proven, not a grid convention — with Mathlib's Kronecker theorem consumed as the general-number-field interface (`kronecker_interface`). And the composed geometric statement at the required five-condition shape: given the family freeze law — for the branched cyclic families a theorem of Weil's RH for curves and Hasse–Davenport (§15), the point where the actual algebraic class enters — every rail's unit-normalized Frobenius determinant lands in μ_6 (`norm_one_of_pow6`), every residual is a lattice point at the common per-prime height, and a sub-threshold residual vanishes exactly (`geometric_lattice_closure`): simultaneous exact $\mathbb{Z}[\zeta_6]$-closure across all rails, with no Hodge-conjecture input anywhere in the chain. The structure is now also populated and family-quantified (`SupersingularFreezeFamily.lean`; standard footprint): on the supersingular locus of $y^2 = x^3 + 1$ the census is counted in-kernel — affine counts at $p = 5, 11, 17, 23$ verified by `decide` (`ssCount_5` ... `ssCount_23`), trace zero at each, so Cayley–Hamilton with the Weil norm forces the unit-normalized Frobenius determinant to exactly $-1 \in \mu_6$ (`supersingular_normalized_det`) — and where the freeze law covers the whole family the no-orphan law upgrades from census evidence to a ∀-statement: every rail with balanced occupancy weights closes exactly, for all rail and reading counts at once (`ss_no_orphan`), the concrete census term executing the five-condition chain end to end (`ssCensus_closure_executes`). What remains open of step two is the closure law for readouts beyond the certified freeze families — the general Green–Griffiths–Kerr landing. Step three should then *fall out* of one and two with the proofs already in hand — finite generation collapses it to a single pairing, the closure of step two supplies that pairing's nondegeneracy, and grade one is unconditional. The load therefore concentrates on step four, and even there the shape is better than "open above grade one": the localization and the grade-specific landings (Ceresa at grade two, Gross–Schoen at grade three) make recognition plausibly provable *one finite grade at a time*. What it bottoms out at is the universal quantifier — *all* grades at once — and by the finite generation of step one that universal case reduces to recognizing the *single generating pairing*. So the residuum is not an unbounded tower but one construction at the structureless boundary, uniform in the fiber. Three of the four steps are an import, a corollary, and a composition; the fourth is closable grade by grade, with the whole remaining weight resting on that one universal recognition. That, not a diffuse frontier, is where the work lives. Both endgame imports are now typed with their reduction spines proven (`RequestProject/GeneratingPairing.lean`; standard footprint): if every tower level factors through one pairing — the field the Schmid/Saito collapse is staked to supply — then recognition of the *entire dial* follows from recognition of that single pairing (`recognition_of_generating_pairing`), and with proven retention the terminus needs exactly one construction (`sourceExhaustion_of_generating_pairing`); likewise a Green–Griffiths–Kerr landing of the residual in $\mathbb{Z}[\zeta_6]$ composes with the proven rigidity threshold into exact closure (`exact_closure_of_lattice_landing`). The whole remaining weight of the program is thereby a single named field: `sourced_of_pairing` — one recognition statement for one pairing.

The attempt, opened. That field now factors, and the factors are largely discharged (`RequestProject/Recogn`
`TriangulatedAssembly.lean`; standard footprints throughout). *Reconstruction factorization* (`sourced_of_reconstruction`): recognition splits into a Torelli leg — fired coordinates reconstruct the class as an explicit generator combination plus torsion — and a *finite* algebraicity list, closed under the group structure; the infinite quantifier over Hodge classes becomes finitely many certified specimens. *Both Torelli legs are discharged*: at depth one the reconstruction field is Mordell–Weil verbatim (`ReconstructionData.ofDepthOne`), and on the channel dials the spanning statement is the effective form of the same theorem of record that supplies recognition (`ReconstructionData.ofChannelSpanning`, `channel_sourceExhaustion_of_spanning` — the full chain executing wherever the inputs are theorems). *The generator list is typed and inhabited* (`TriangulatedGenerator`: a class enters only with its three bearings — annihilator address, tower detection, support/order — and its algebraicity certificate): the cyclic diagonal and the Artin unit enter with detection *computed* (`cyclicDiagonalGenerator`, `artinUnitGenerator`), and the bundle-level constructor (`certifiedGenerator`) admits the census specimens — the CM-product divisors and the explicit point-counted Schoen $(2,2)$ cycle with its freeze certificate and exact integer landings — with detection *proven* by retention from the census-supplied class, never read from the census itself. The specimen itself is now collected (next paragraph); what remains above it is the generating pairing of the collapse and the construction of the collected specimen's classes. The falsifiers stay armed: a specimen whose bearings disagree is an orphan, published as such.

The sixfold, collected. Two constructions, one theorem, one specimen — all exact arithmetic, no conjecture consumed (`tmp/weil_sixfold*`; Weil's RH for curves is the only ceiling, cited). *First construction and its theorem*: the Prym of $t^3 = y$ over $u^3 + xu + c = 0$ over Schoen's conductor-19 curve is a $(3,3)$-sixfold whose split-prime characteristic polynomials are perfect squares — and the square is a *theorem*, not evidence: the linear u-coefficient makes x rational in u, resurrecting a hidden double cover whose involution lifts explicitly, $\tilde{\tau}(u, y, t) = (u,\, -x^3/y,\, -x/t)$, inverting the deck ($\tilde{\tau}\sigma\tilde{\tau} = \sigma^2$, machine-verified pointwise at five primes), so $\langle \sigma, \tilde{\tau} \rangle \cong S_3$ acts and $M_2(K) \subseteq \mathrm{End}^0(B_K)$: the duplication holds at every good prime of $K = \mathbb{Q}(\sqrt{-3})$ at once, with the multiplicity projector an *explicit algebraic correspondence* — a worked recognition one register above Schoen's fourfold. *Second construction, hardened*: replacing the cover by $u^3 + xu + (y + a) = 0$ (genus four, degree three over $\mathbb{P}^1_u$ — both involution mechanisms excluded by design) yields, at $a = 1$, a Prym whose degree-12 characteristic polynomial at $p = 7$ is *irreducible* with exact Weil purity: a one-polynomial certificate that B is a **simple Weil-type abelian sixfold over $\mathbb{Q}(\sqrt{-3})$** — fully explicit, point-counted, Weil signature confirmed across a fourteen-prime battery ($s_1 = 0$ at every inert prime), non-isotriviality of the a-family certified by distinct members' Frobenius data. Register, exact, with the literature boundary drawn precisely: Markman's recent work covers all Weil *fourfolds* and the sixfolds of *split type* (Weil discriminant -1); outside discriminant -1, no sixfold algebraicity theorem applies. The decisive invariant for this specimen is therefore its Weil discriminant in $\mathbb{Q}^\times/\mathrm{Nm}_{K/\mathbb{Q}}(K^\times)$ — not settled by simplicity, $\mathrm{End}^0 = K$, the polarization type, or full $GU(3,3)$, none of which determine it. That computation is now done, exactly: the lattice Pfaffian in an elementary-divisor basis is $\pm 3^b$ on the nose (the residual unit is ± 1, never an unknown local unit; the sign is forced by the $(3,3)$ signature), so the discriminant class is -1 — *split*, inside the covered locus. Existence of the algebraic Weil classes on this Prym is in fact classical — Schoen's cycles, generalized to all abelian covers by Patel–Zhang [55] — with Markman's theorem covering the general split sixfold beyond the Prym locus; what this construction adds is the explicit cycle and an independent elementary proof: $\Xi = W_3 - \Theta^3/162$, nonzero by the integrality trap, with *every* input now machine-verified or textbook-cited — the battery $q_k = k!\binom{6}{k}$ verified as an exact symbolic Pfaffian identity on the integral cut-and-glue homology model of the cover, the polarization type $(1,1,1,3,3,3)$ (so $m = 3$) obtained by Smith form on the same model and agreeing with Lange–Ortega [45] read at source, and the collective rail reading frozen at order one at two split primes ($\det(\mathrm{Frob}_p \mid V_\omega) = p^3$ exactly at $p = 7$ and 13). A by-product is a no-go with the force of a theorem: the Pfaffian argument gives $\det = -3^b$ for *every* étale-$\mathbb{Z}/3$ Prym sixfold, so this entire construction family is split — the beyond-boundary hunt must change its glue. The day's own falsification record is part of the claim: two hidden involutions were found and killed by the program's instruments before the specimen stood. The register of this paragraph is exact: each "should" is a stake naming a theorem to be written — the Schmid/Saito collapse to one generating pairing included — and none is treated as an automatic import; what is proved is cited by Lean name, and nothing else is claimed.

The universal rigidity law. The sixfold's integrality endgame, freed of its specimen, is a theorem about *every* cyclic $\mathbb{Z}/3$-Prym at once. For the étale $\mathbb{Z}/3$-Prym of a genus-g' curve ($n = g' - 1$, $\dim B = 2n$) the two geometric inputs are machine-verified on the integral model — the battery identity $\mathrm{Pf}(sE + tE_V) = s^{g'}(s+t)^{2(g'-1)}$, an exact symbolic Pfaffian (verified at $g' = 4, 5$; general g' by the handle-block structure), and the type $(1^{\,n}, 3^{\,n})$ [45] — and under proportionality $W_n = \lambda\Theta^n$ of the middle Abel–Prym class they force $W_n^2 = \binom{2n}{n}/3^n$: the *central binomial over the glue*. Its 3-adic valuation is the number of base-3 carries of $n + n$ (Kummer), strictly less than n always, so proportionality contradicts integrality of the cycle class in *every* dimension: $\Xi_n = W_n - \Theta^n/(n!\,3^n)$ is a nonzero, algebraic, Θ-primitive middle class on every such Prym. The contrast is sharp: for étale $\mathbb{Z}/2$-Pryms the Abel–Prym class *is* theta-proportional (Welters [46]); the μ_3 harmonic rigidifies the cycle. The arithmetic chain is Lean at full generality, the dimension a bound variable (`UniversalTrap.lean`: `centralBinom_three_adic_lt`, `universal_integrality_obstruction`, `PrymTrapData.abel_prym_rigidity`; standard footprint), and the all-m firing dimensions form the ζ_3-adic ladder $(6, 8, 18, 20, \ldots$: every base-3 digit of n at most 1). Placement against the literature, checked at source: Schoen's cycles and their Patel–Zhang generalization [55] prove the *Weil classes* of these Pryms algebraic at every dimension, but say nothing about the Abel–Prym cycle itself; the rigidity law is complementary content — it locates W_n off the theta line, a computation of intersection numbers their construction does not touch — and, to the best of our knowledge after that audit, it is new.

The Prym–Petri rung. The frontier named twice in this section — the semiregularity that would carry a cycle's algebraicity along with its Hodge class — runs, for Pryms, through a Gieseker–Petri statement: injectivity of the Petri multiplication controls the tangent spaces of the Brill–Noether loci in $\mathrm{Nm}^{-1}(\omega)$ that govern the Abel–Prym cycle's deformations. For étale

$\mathbb{Z}/2$-Pryms this is Welters' theorem [47]; for cyclic covers of degree ≥ 3 no statement existed — every extension in the literature is degree two, a fact checked on the complete citation graph of [47]. The μ_3 statement is now proven at the depth-one stratum, by a mechanism with no $\mathbb{Z}/2$ ancestor (there the stratum is empty, $\chi = 0$): for $h^0(L) = 1$ the entire Petri kernel is base-curve divisor geometry — it equals h^0 of the *collision excess* of the section divisor (the part of $\mathrm{div}(s)$ meeting a π-fiber on more than one sheet), so the map fails exactly when that excess moves in a pencil. Three-sheet arithmetic forces a canonical divisor to contain the excess plus its half-thickening, $K \geq E + \lceil E/2 \rceil$, and on a general base curve no moving divisor admits such a containment: the profile arithmetic is exact (the automatic-speciality zone is empty, $c + \delta \geq e$ against $c + \delta \leq e - 2$; virtual dimension $r - c - 2 \leq -2$), and the reduced case dies by dimension comparison against Farkas' de Jonquières theorem [50], whose hypotheses (every series, boundary slack, no distinctness) were verified at source. The theorem is ε-uniform — one generality condition on C serves all covers at once — unconditional for $g \leq 5$, in particular at $g = 4$, the sixfold's own base genus; at $g \geq 6$ a window $\frac{g+2}{2} \leq e \leq \frac{2g}{3}$ remains, localized in the two-series incidence framework of [52]. Sharpness is constructed, not conjectured: on any curve with a vanishing thetanull ($\omega = 2N$) the failing bundle exists — two of three preimages over each point of a trigonal fiber, kernel $h^0(N) = 2$ — and the program's own genus-4 tower curve *is* such a cone: the instrument confirmed the predicted kernel jump on trigonal fibers (5/5; controls 5/5; the bundles literally depth-one in 10/12 configurations), which retroactively explains the depth-one battery's 14/14: random sections are collision-free, and collision-free never fails. The arithmetic chain is Lean (`MuThreePrymPetri.lean`: `zone_empty`, `dimension_comparison`, `g4_kill`; standard footprint), including the honesty marker `g6_boundary_not_killed`: an initial "$g \leq 6$" coverage claim was corrected to "$g \leq 5$ plus boundary" when formalization exposed the equality case — the falsification machinery running on our own arithmetic. One caution kept in the record: the general-series hypothesis in Ungureanu's non-existence theorem [51] is essential (composed series violate the unrestricted reading), so it is never invoked here against special pencils.

The rung consumed: the depth-one threefold and the μ_3 parity. The Petri theorem is the deformation control the semiregularity route consumes, and its geometric payoff is now proven and measured. In the torsor $Y = \mathrm{Nm}^{-1}(\omega)$ (tangent space the anti-invariant H^1, dimension $2g-2$) the effective locus $V^1 = \{h^0 \geq 1\}$ has dimension $g - 1$ by a finite parametrization over $|\omega_C|$, and at a depth-one point its tangent space is the annihilator of the Prym–Petri image — so the μ_3 theorem above is *equivalent* to smoothness of V^1 along the depth-one locus, and the normal bundle is canonically the dual Petri image bundle: the Abel–Prym cycle's deformation theory and the Petri map are one object seen three ways. On a certified non-thetanull specimen — a bielliptic tower whose canonical quadric is rank four in closed form; the trinomial towers are *structurally* thetanull ($\mathrm{div}(F_u) = R - 2{\cdot}O$-fiber forces $\omega = 2N$, a design law found the hard way) — the battery lands all 54 depth-one configurations as smooth points of V^1 (rank 3 exactly, Euler checks exact), the Gauss map at a generic depth-one point has full rank three (second-fundamental blocks $(2, 3, 3)$: theta-like, the favorable regime for the semiregularity pairing), and the bielliptic locus's structural bad configurations hide at $h^0 = 3$, outside the stratum. The census then delivered a discovery by refuting the program's own first draft, within the hour: the $\mathbb{F}_{49}$ count $341552/49^3 = 2.903$ tracks $3q^3$, not q^3 — V^1 has *three* irreducible components — and the corrected monodromy names the invariant: the cycle traced by $\mathrm{div}(\eta)$ under an $|\omega|$-loop bounds the swept chain, hence carries zero ε-holonomy, so the total sheet-shift $\Sigma \bmod 3$ is conserved, the monodromy is the index-three wreath subgroup, and Σ is the μ_3 **Prym parity** — the $\mathbb{Z}/3$ twin of Mumford's parity for classical Pryms [48], with no footprint in the literature sweeps. Its distribution law is proven (`MuThreePrymParity.lean`: `uniformity`, `concentration`; standard footprint) and confirmed pointwise, 48/48 over $\mathbb{F}_7$: rational lifts spread uniformly over the three components iff some closed-point degree is prime to three, and concentrate in a single component iff every degree is divisible by three. The Weil-cycle bookkeeping accordingly carries *one* parity component — an irreducible, generically smooth, Gauss-nondegenerate threefold. The semiregularity route is thereby specced to coordinates: on the abelian sixfold the target $H^4(X, \Omega^2)$ trivializes, the pairing columns are integrals of constant forms over the component, and $H^1(N)$ assembles from the same section spaces through the parametrization — the computation executed in the next paragraph; the identification of the component as a translate of W_3 remains the named open question beside it.

The pairing executed: the obstruction calibration and the twist law. The computation just specced has now run, end to end in exact arithmetic over $\mathbb{Q}(\zeta_6)$. The assembly lands as specced: $H^1(N)$ is presented by the sixty-dimensional $H^1(\mathcal{O}_Z) \otimes W$ modulo the twenty-seven-dimensional image of the tangent coboundary δ, the cokernel's thirty-three syzygies graded $(9, 12, 12)$; generic rigidity of the depth-one threefold is *certified* by degeneration — a $(3, 3)$-model with prescribed nodes realizes the specimen and its Kodaira–Spencer block attains full rank twenty-seven, so rigidity propagates to the generic member by semicontinuity; and the Bloch pairing against the constant-form columns (the $\Lambda^4 W^*$ slice of the trivialized $H^4(X, \Omega^2)$) descends to $\operatorname{coker} \delta$ *exactly* once the Abel–Prym scalars $(1 - \zeta^{-j})$ are carried in both the coboundary and the restriction, with rank fifteen of thirty-three. The calibration then came from a contradiction the geometry itself raised: the cycle deforms with the nine-dimensional Weil family, so the family directions must be obstruction-free — their restrictions must lie *inside* $\operatorname{im} \delta$ — yet the pairing measured full rank on the naively symmetrized mixed rows while provably killing $\operatorname{im} \delta$. The resolution is a law, not a patch: the Kodaira–Spencer embedding carries a second Abel–Prym twist on the $H^1(\mathcal{O}_X)$-side of the restriction, and geometry selects it uniquely — of all candidate twist ratios, exactly one places all nine equivariant rows inside $\operatorname{im} \delta$ (a nine-rows-into-codimension-thirty-three containment, an algebraic identity in exact arithmetic), namely $r = 3 = (1 - \zeta_3)(1 - \bar{\zeta}_3) = \operatorname{Nm}(1 - \zeta_3)$, the norm of the Abel–Prym scalar — the two twists compose to the rational norm. The statement is intrinsic, the ratio merely its coordinates: $\operatorname{im} \delta$ meets the mixed Kodaira–Spencer span in dimension exactly nine, so *the Weil tangent is* $\operatorname{im} \delta \cap (\text{mixed block})$, while the naive symmetric frame meets $\operatorname{im} \delta$ in dimension zero — the old frame contained not one obstruction-free direction, which is precisely why it contradicted the geometry. With the frame corrected the true polarized Kodaira–Spencer image decomposes as the twisted mixed block — inside $\operatorname{im} \delta$: the Weil tangent, obstruction-free — plus the twelve pure-grade directions, every one outside $\operatorname{im} \delta$: **the first-order deformation locus of the Abel–Prym cycle in polarized moduli is exactly the Weil family**, certified directly by cokernel membership in exact arithmetic. The pairing then delivered a second law by refuting this paragraph's own first draft within the day (its forward pointer promised "grade-shifted slices"; none exist): the two hundred twenty-five constant-form columns already exhaust the Serre-dual target $H^2(X, \Omega^4)$, and on the full twenty-one-dimensional polarized image the pairing vanishes *identically*, while on the fifteen polarization-breaking directions it is full rank. Far from a defect, this is the program's own universal rigidity law acting as oracle: the middle Abel–Prym class is $\lambda \Theta^3$, the theta class stays $(1, 1)$ under every polarized deformation, so the class-variation that Bloch's map computes must freeze on the polarized locus and move exactly where the polarization breaks — both measured on the nose (0 of 21; 15 of 15), a cross-instrument agreement between the cohomological rigidity law and a deformation pairing built from independent combinatorics. The consequence is the sharpest statement of the rung: the twelve transverse obstructions are *cohomologically invisible* — off the Weil family the class $\lambda\Theta^3$ survives, algebraic via theta intersections, while the cycle itself cannot follow — the deformation-theoretic shadow of Griffiths-group content, and the reason the Abel–Prym cycle's arithmetic lives modulo algebraic equivalence. Semiregular detection of those twelve, the original spec of this rung, is thereby falsified as specced and replaced by the stronger direct certificate; the refuted pointer stays in the record.

The sixfold program: the locus ledger, the calibrated projector, and the explicit base. The frontier now has a worked program, recorded here at its exact registers. The organizing reframe: a divisor is the focal locus of a harmonic value, and the exponential sequence is the grade-one *locus ledger* — integral winding of a harmonic section carves a geometric locus, automatically. The classical absence of this mechanism in codimension ≥ 2 is chart-native ($\mathcal{O}^*$ exists only at GL(1)); the carrier keeps rank-one phasors and a tower-stable winding lattice at every grade, so the honest open object is the grade-p locus ledger, and the named target (the *Locus Ledger Law*, a target, not a claim) is: some refined theta-geometric focal locus of the collective bank has expected codimension and nonzero Weil coefficient. Its first entry is a proven NULL: the plain Abel–Prym locus books $\lambda\Theta^3$ — the rigidity law — so plain theta geometry is Lefschetz-blind on the collective channel. A one-line lemma sharpens the search: the $\mathbb{Z}[\zeta_3]$-action rotates the rational Weil plane with order three and no invariants, so *σ-invariant cycle classes have zero Weil projection* — candidates must be σ-asymmetric (this subsumes the null: deck translation fixes homology classes, so the AP locus never had a chance).

The projector, calibrated (measured register). Frobenius eigen-frames from plain point counts with functional-equation-completed degree-eight Weil polynomials: at split primes the Tate block fuses at one address (multiplicity 18 — reproducing the occupancy law by an independent computation); at generic inert primes it SPLITS: the Lefschetz block at address $+1$ (multiplicity $10 = \mathrm{Sym}^2$ of an NS of rank four) against the Weil-type block at address -1 (multiplicity 8) — the $(-1)^\delta\chi_3$ twist of the freeze determinant law, seen directly in the spectrum. The Weil projection of a candidate class is thereby READABLE: the (-1)-at-generic-inert eigencomponent. Truth-gates ran hot: an impossible null (a spectrum with no Θ^2 line) exposed an invalid character-rail reading at inert primes, corrected to the plain-count route — the instrument record keeps both versions.

The explicit base (instrument-certified). The sixfold's base curve now has candidate equations: in the family $w^3+Aw+B(u,v)=0$ over $E_0 : v^2 = u^3+1$, five genus-four instances pass the hardened gates (unique infinity-fiber factorization pattern per prime; the $k=5$ count PREDICTED by the polynomial fixed at $k \leq 4$; smoothness at two primes; non-Galois by the discriminant gate). Involution triage then excludes the $B_0 = 0$ instances — they carry $(v,w) \mapsto (-v,-w)$, the fatal-involution shape that forces decomposable Pryms, told by their vanishing odd inert traces — leaving the involution-free primary candidates

$$C_0 : \ w^3+2w+(u+v)=0 \qquad \text{and} \qquad C_0' : \ w^3+2w+(u+v+2)=0$$

over E_0, whose rational 3-torsion $(0,\pm1)$ feeds the étale $\mathbb{Z}/3$-cover as the fiber product against the 3-isogeny — counting by splitting weights, no new machinery.

The deliverable and its two open arrows. Target theorem: every rational Hodge class on the explicit Prym sixfold B is algebraic. Chain: certified base $\to$ cover and genus-10 gate $\to$ the census/projector suite ported verbatim $\to$ the refined-locus m-census with both outcomes preregistered (a surviving candidate $\to$ the exact-nonvanishing proof; all-null $\to$ the theta-blindness no-go, itself a theorem) $\to$ the exhaustion $\mathrm{Hdg}^3(B) = \mathbb{Q}\Theta^3 \oplus W$ by a Mumford–Tate computation $\to$ the theorem. Everything in that chain is instrument work or conventional mathematics except one node, named without disguise: the EXISTENCE of a Weil-sensitive algebraic locus on the simple sixfold, within reach of the searchable families. The point counter is the instrument that finds and certifies the missing class; it is not the deliverable.

The downstream implication is now itself a machine-checked theorem (`KSweepCompletion.lean`; standard footprint): given the decomposition with L algebraic, $\dim_K W = 1$, and the K-action algebraic, ONE algebraic cycle with nonzero W-component makes every class in $L \oplus W$ algebraic — subtract the Lefschetz part, sweep the K-line (`completion`, `completion_implication`). Two structural simplifications fall out and are recorded exactly: hypothesis three becomes FREE once the constructed B's deck action is certified (a step-0 gate: the ζ_3-multiplication is then an automorphism whose graph is algebraic by construction), and *no algebraicity of the projector* π_W *is required anywhere* — its only role is the nonvanishing certificate, so the motivated-presentation / standard-conjecture pricing of the collective wall is BYPASSED for this B. Ticket status, stated so nothing is silently assumed: the closure lemma is fixed; the B-specific premises (the decomposition, $\dim_K W = 1$, the certified deck action) await the constructed B; the Weil-sensitive cycle — $m(\mathrm{C1}) \neq 0$ — is OPEN, and it is the search's conclusion, never its assumption. Hodge for B is not proven; what is proven is that it hangs on exactly the named nodes.

Universal reconstruction, extracted: the mechanism is Prym-free. The sixfold proves the mechanism can succeed once; the universal target needs the mechanism without the scaffolding. `UniversalReconstruction.lean` (standard footprint) extracts it. The *seed-transfer theorem* (`seed_transfer`) is stated for an arbitrary SET of algebraicity-preserving operators — the algebraic correspondences of X, whatever they are; no field, no dimension bound, no abelian hypothesis, no Prym, no CM, no split discriminant, no citation of Schoen or Markman: any constituent W of the Hodge space that is *orbit-irreducible* under those operators (every nonzero element reaches every element) becomes wholly algebraic the moment ONE algebraic class has nonzero W-component — subtract the Lefschetz part, transport along the orbit. `exhaustion_of_seeds` iterates: with finitely many constituents, the universal quantifier over classes collapses to one seed per constituent. `kline_seed_transfer` recovers the sixfold's K-sweep as the one-line instance (a K-line is orbit-irreducible under the K-scalar operators): the Prym geometry was one specimen's certificate, never the mechanism. The socket is wired — `sourced_of_sweep` fills the program's single remaining field `sourced_of_pairing`, and `sourceExhaustion_of_sweep` composes with proven retention into the terminus. The price of the universal quantifier is thereby exact, and it is two legs per constituent of $\mathrm{Hdg}^p(X)$: the STRUCTURE leg — $\mathrm{Hdg}^p(X) = L + \sum_i W_i$ with L in the divisor algebra and each W_i orbit-irreducible under the actual correspondences of X — and the SEEDING leg — one fired algebraic class per constituent. Where both legs are theorems today is the low-rung ledger already recorded ($p = 1$; products of elliptic curves and abelian surfaces; abelian threefolds; abelian fourfolds including Weil fourfolds; fivefolds; split Weil sixfolds — and, per the Patel–Zhang audit already in this record, the Weil classes of Pryms of ALL étale abelian covers, in every dimension); the first open constituents live beyond the abelian-cover playbook entirely: non-Prym Weil $2n$-folds and the rigid sixfolds of the cited literature, for which no specimen is constructible by any cover in this program's toolkit. The necessity claim is itself machine-checked:

`seed_without_irreducibility_fails` exhibits a model satisfying every hypothesis EXCEPT orbit-irreducibility in which the seeded constituent is NOT algebraic — without irreducibility a seed certifies only itself, so the reduction's entire content lives in the structure leg, and the register says so. Scope: none of this assumes or proves the Hodge conjecture; it prices it, exactly, and the Lean enforces the price.

The compression theorem: seeds are free, transitivity is proven, the wall is one field. At *full Hodge-correspondence strength* both legs close unconditionally, and `HodgeCompression.lean` (standard footprint) proves it. The structure leg: polarizable rational Hodge structures form a semisimple category (classical — polarization positivity), so $H^{2p}(X, \mathbb{Q})$ splits as $T \oplus N$ with T the Tate-isotypic block, whose underlying rational space is exactly the span of the Hodge classes: $\mathrm{Hdg}^p(X) = T$. Every linear endomorphism of T extended by zero on N is a morphism of Hodge structures ($\mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(-p)^m, \mathbb{Q}(-p)^m) = M_m(\mathbb{Q})$), hence — Künneth plus Poincaré duality — is induced by a HODGE class on $X \times X$; the block is therefore orbit-irreducible under these transports, and that transitivity is now machine-checked linear algebra (`transport_exists`, `hodge_block_irreducible`). The seeding leg: the intrinsic seed is the polarization power h^p — algebraic, Hodge, nonzero for $p \leq \dim X$ by hard Lefschetz; no Prym, no CM, no discovery search. The compression (`hodge_compression`): with the splitting and the free seed, the Hodge conjecture for (X, p) IS the single field `halg` — *the Tate-block transport operators preserve algebraicity.* The self-reference is stated without disguise: a rank-one transport carrying the seed to a target β is built from β itself, so the field is the conjecture one degree up on $X \times X$; the compression is a reorganization of the wall, not an escape. Its value is coordinates, and they are operative: the TRADE-OFF LAW (`block_closed_of_algebraic_transport`) says that restricting to proven-algebraic correspondences makes stability a theorem while irreducibility and seeding go open (the sixfold's position), while enlarging to full Hodge transport makes irreducibility and seeding theorems while stability goes open (the compression's position) — the Hodge conjecture is precisely the assertion that the two positions meet, for every (X, p), in a correspondence set that is simultaneously algebraic and transitive. This coincidence deserves the emphasis of a display, because it is the exact diagnosis rather than a weakness: *permit every Hodge correspondence and transitivity is a theorem; restrict to known algebraic correspondences and algebraicity is a theorem; the Hodge conjecture is exactly the statement that the two correspondence algebras coincide on the Tate-block transport operators.* And the coincidence is itself machine-checked as an equivalence, in both directions (`hodge_iff_positions_meet`): given the splitting and the free seed, the block is algebraic IF AND ONLY IF some algebraicity-preserving correspondence set acts transitively on it — the conjecture is not merely *reduced to* the coincidence; it *is* the coincidence. Earlier reductions that seemed to "move the difficulty around" were not failing; they were moving between the two coordinate charts of this one statement.

The instruments point at this field at the correct register — and the register matters. The census's collective projector is a FROBENIUS spectral projector cutting the Weil channel; its measured p-independent signature $(D_p, \mathrm{mult}_W) = (13, 18)$ across nine primes is *consistent with* — probes, but does not yet measure — the characteristic-zero transport operator `halg` names. The identification requires one more theorem, and it is the program's existing glue field in a new shape: a specialization/comparison bridge carrying Frobenius spectral projectors to absolute-Hodge transport operators. Until that bridge is proven, the census is evidence about the field, not a reading of it — the same wall the universal harmonic clock states as "the clock's graph exists in characteristic zero," and the descent interface types as `glue`. Scope, unchanged: nothing here assumes or proves the Hodge conjecture; what is proven is that the conjecture, whole, is one named coincidence — and the bridge that would let the instruments read it directly is itself a named, typed target.

The abelian discharge: on the specimen class the cable narrows to the André gap. The ladder from projector data to the conjecture — spectral data $\Rightarrow$ absolute-Hodge/motivated projector $\Rightarrow$ algebraic correspondence $\Rightarrow$ transitivity $\Rightarrow$ Hodge — carries one dangerous step: "motivated/absolute-Hodge $\Rightarrow$ algebraic" is not known universally, and asserting it would simply rename the conjecture. But every specimen in this program — the fourfold, the sixfold, the eightfold — is an abelian variety, and *on abelian varieties the dangerous step splits, and both closable halves close by cited classical theorems.* Hodge $\Rightarrow$ absolutely Hodge on abelian varieties is Deligne's theorem; our transports are Hodge classes on $X \times X$ (proven above), $X \times X$ is abelian, so the transports are absolutely Hodge for free. And motivated-with-abelian-auxiliaries $\Rightarrow$ algebraic on abelian varieties is a THEOREM: standard conjecture B holds for abelian varieties (Lieberman; via the Fourier–Mukai transform, Beauville), products of abelian varieties are abelian, so the Lefschetz-involution factor in any motivated presentation over abelian auxiliaries is itself algebraic. `AbelianTransportLadder.lean` (standard footprint) formalizes the discharge: `MotivatedPresentation` (an algebraic sandwich around one Lefschetz factor), `motivated_stab` (given the B-discharge, motivated operators preserve algebraicity), `hodge_of_motivated_glue` (motivated transitivity plus the free seed close the block), and `abelian_cable_iff` — the residue as an equivalence. What remains on the abelian class is therefore strictly narrower than "manufacture an algebraic cycle": manufacture a MOTIVATED presentation of the Tate-block transports from the glue data — absolutely Hodge $\Rightarrow$ motivated, for one operator class. That is the André gap, not the whole conjecture: the algebraicity of the Lefschetz factor is already banked, and this is the one rung of the ladder with known nontrivial successful instances (the Kuga–Satake correspondence and CM classes are motivated — André). The cable is located, typed, and narrowed; it is not fixed, and the register says exactly that.

The non-abelian cable: separator plus mixer. The matrix-unit demand was too strong, and the repair is non-abelian: two noncommuting algebraic correspondences with separator-plus-mixer behavior generate the needed transports automatically as OUTPUT — they are never required as input. `NonabelianCable.lean` (standard footprint) proves both halves. The repair theorem (`block_closed_of_irreducible`): if the Tate block is an irreducible module under a composition-closed set of algebraic correspondences, the orbit span of the free seed h^p is a nonzero invariant subspace, hence everything — irreducibility $\Rightarrow$ cyclicity $\Rightarrow$ Hodge for (X, p). The operational form (`hodge_of_nonabelian_connectivity`): a SEPARATOR A with simple rational spectrum on the block — entering through its Lagrange spectral projectors e_i, polynomials in A and hence algebraic, exactly the clock-projector machinery already in the tree — plus a MIXER B whose transition graph $i \to j \iff e_j B e_i \neq 0$ between the one-dimensional eigenchannels is strongly connected, together close the block. The Lean proof is constructive in exactly the sense the instruments need: the seed has a nonzero component in some channel, each edge is a nonzero map between one-dimensional spaces, and path composition transports that component into a nonzero algebraic spanning vector of EVERY channel — connectivity *manufactures* the per-channel seeds that the abelian route could only demand. Noncommutativity is load-bearing, not incidental: a mixer commuting with the separator commutes with its projectors, every off-diagonal block vanishes, and the graph has no edges. The Hodge conjecture for (X, p) is thereby reduced to a finite, operational existence statement: *one algebraic separator with simple rational spectrum on the block, one algebraic mixer with strongly connected transitions.* Register, exact: the pair's existence is the new open target, consumed never asserted; recovering the separator from the stable Frobenius spectral data still passes through the bridge theorem above (with the motivated-to-algebraic step banked on abelian specimens via B); the mixer is a genuinely new instrument job — a geometric correspondence OUTSIDE the separator's commutant; and the connectivity check, unlike every previous formulation of the wall, is computable mod p from point counts. The census's job list is now three items long, and each item is falsifiable.

The rigidity program: the overdetermined transport system and the two-sided squeeze. Three genuinely-new-shaped programs present themselves beyond the organizational work: RECONSTRUCTION (prove the transport algebra uniquely reconstructible from local/Frobenius/deformation data — partially embodied in the freeze-decoder queue), COMPRESSION to a finite generating mechanism (already delivered above in its sharpest form: the transport basis has size two — one separator, one mixer), and RIGIDITY: prove that any transport system satisfying enough independent compatibilities is forced to be unique, and that the unique realization is algebraic. Rigidity is the bet, and `TransportRigidity.lean` (standard footprint) banks its skeleton. Let C be the space of transports satisfying the compatibility stack — Hodge (morphism of structures), Frobenius (Galois-equivariance of the ℓ-adic realization), polarization — and A_{alg} the span of algebraic transports. Three registers: $A_{\mathrm{alg}} \subseteq C$ (theorem-backed inclusions); $\dim C$ bounded ABOVE by finite computation — C embeds in the Frobenius-fixed part of the transport Künneth block of $X \times X$, the commutant of finitely many Frobenii is computed exactly from census eigenvalue data, is monotone decreasing in the prime set, and converges to the true Galois commutant by Chebotarev; this uses the comparison bridge in its EASY, theorem-backed direction (characteristic zero into characteristic p as a constraint), not the hard construction direction that gates the glue — and the house occupancy machinery already computes exactly these bounds; $\dim A_{\mathrm{alg}}$ bounded BELOW by exhibiting correspondences. The squeeze (`rigidity_squeeze`): if the bounds MEET, $C = A_{\mathrm{alg}}$ — every compatible transport is algebraic. The terminus (`hodge_of_transport_rigidity`): the meet, plus the Deligne register on abelian specimens (Hodge transports are absolutely Hodge, hence Galois-invariant, hence in C), plus the free seed, close the block — *a specimen's Hodge conjecture closes by finite computation the moment the census upper bound touches the known-correspondence lower bound; the closing step constructs no cycle.* The falsifier register is pre-committed and named (`phantomCount`, `meet_iff_phantomCount_eq_zero`): if the bounds stabilize apart, the gap is the measured integer count of phantom compatible transports — the conjecture's residue displayed as a number, per specimen; on any étale-Prym sixfold the expected count is 0 once the Schoen/Patel–Zhang classes enter the lower bound (a calibration reading), and a genuine discovery readout awaits a beyond-playbook specimen. Honesty note, stated in the file: per specimen the meet is equivalent to a Tate-block algebraicity statement on $X \times X$ — rigidity is not a free lunch; what distinguishes it from "assume every projector is algebraic" is epistemic access: the hypothesis is a NUMBER MEET between two convergent, independently computable estimators, checkable prime by prime, whose failure mode is itself a measurement. Protocol, pre-registered: run the squeeze instrument first on the decomposable fourfold, where the meet MUST occur (everything algebraic is known there — our construction plus Markman); a non-meet there falsifies the instrument, not the mathematics; then the sixfold as the second calibration rung (étale-Prym, hence theorem-covered — see the reclassification below).

The bounded-complexity lift: the conductor is uniform complexity. There is now a route where the final implication is genuine algebraic geometry rather than the conjecture renamed — verified in dialogue, with two repairs, before formalization. The lifting theorem: let X be smooth projective over a number field with integral model $\mathcal{X} \to S$, and β a rational Hodge class. If a SINGLE Hilbert polynomial P admits, at infinitely many good places v, a codimension-p cycle $Z_v \subseteq \mathcal{X}_v$ with $\mathrm{Hilb}(Z_v) = P$ and $\mathrm{cl}_\ell(Z_v) = \mathrm{sp}_v(\beta_\ell)$, then (after a finite extension, up to a Galois conjugation) a characteristic-zero cycle Z on X has $\mathrm{cl}_B(Z) = \beta$. The proof is classical spreading-out: the relative Hilbert scheme $\mathrm{Hilb}_P(\mathcal{X}/S)$ is projective with finitely many irreducible components; witnesses at infinitely many places force a component carrying them over infinitely many fibers (the pigeonhole — Lean-proven, `exists_component_with_infinite_places`); a constructible image in the one-dimensional S that is infinite dominates, so the generic fiber is nonempty; the universal family's relative cycle class is a section of a lisse sheaf, rigid on an irreducible base, so agreement with β_ℓ at one witness point propagates to the generic point; Betti–étale comparison and injectivity of scalar extension finish. The two repairs from verification, recorded: take the component inside the Zariski closure of the witness points themselves, so witnesses are dense and the generic transfer of open conditions is lisse rigidity (equal-at-one-point $\Rightarrow$ equal-at-generic), not a discard-a-closed-subset argument that the witnesses might evade; and the cospecialization identification returns a Galois conjugate of β_ℓ — repaired by conjugating the cycle. Bounded degree suffices in place of a fixed polynomial: bounded-degree subvarieties realize finitely many Hilbert polynomials, and the same pigeonhole applies a second time.

Applied to the separator–mixer package rather than to single cycles, the lift becomes the program's terminus in its strongest honest form (`BoundedLift.lean`, standard footprint): `SeparatorMixerData` packages the two-operator criterion, `BoundedLiftPackage` types the ticket, and `hodge_of_bounded_complexity` proves: witnesses at infinitely many places with fixed Hilbert data, plus the lift field (the classical assembly above, consumed as cited geometry), close the block — *the closing argument never asserts "Tate implies algebraic"; the characteristic-zero correspondence EXISTS by geometry once complexity is bounded.* The named ticket is therefore

> `uniform_cycle_complexity`: *every compatible Frobenius-defined Tate transport admits algebraic realizations in the reductions whose degrees are bounded independently of the prime,*

and the serious remaining problem is the uniform degree bound — a quantitative geometric problem, not a bare existential conjecture. The census upgrade this demands: record not only eigenvalues but the realizing cycles' degrees and Hilbert data at every prime — the projector-realizing correspondence, the mixer, the transition minors — and watch the one experimental question that now matters: *do the required cycles recur with bounded degree as $p \to \infty$?* An infinite hypothesis can never be verified outright; it can be falsified (degrees grow without bound — itself a structural discovery about why the conjecture is hard) or supported (a stable finite list of Hilbert data recurring), and pigeonhole converts support into the lifting theorem's exact hypothesis.

First light: both instruments on the fourfold. Built and run (`tmp/cable_instruments.sage`, `tmp/cable_lower.sage`; split primes $7, 31, 37, 43$, inert $5, 11, 17, 23$). *Squeeze census.* The formal split-torus model derived from the proven relations ($a\bar{a} = b\bar{b} = p$, multiplicity two) predicts the H^4 profile $\{18, 8^{\times 4}, 4^{\times 4}, 1^{\times 4}\}$, $D = 13$, commutant 648, Tate 18 — and all four split primes match it EXACTLY, explaining the census's measured $D = 13/\mathrm{mult}_W = 18$ from first principles. The inert primes carry the constraints the split primes cannot see: at $p = 11, 17, 23$ the

H^1 commutant bound drops to $\mathbf{8}$ — and at the H^1 level the true value is pinned by Faltings ($\dim \mathrm{End}^0(B)$), so the squeeze mechanism has closed its first meet at a level where the answer is a theorem: the instrument's built-in truth-gate, passed. At H^4 the over-$\mathbb{Q}$ bound is 328; the 8-dimensional Weil-type cluster moves to -1 at inert primes (the $\mathbb{Q}(\sqrt{-3})$-twist, reproducing the earlier $(1,10)+(2,8)$ inert splitting by an independent computation), so the over-K bound is 648 — the instrument caught the field-of- definition subtlety in the transports' Galois equivariance. The $p=5$ reading is anomalous (extra coincidences, 946) and harmless (minima govern). *Separator/mixer live test.* On the 18-dimensional Tate block: simple-spectrum separators exist in the torus-block algebra (18 distinct eigenvalues, every trial), and the seed orbit of Θ^2 fills the block (18/18, deficit 0) — with one calibration flag stated at register: the sampled block algebra is the TORUS-compatible one (dim 16 on H^1), which the H^1 squeeze itself just measured to be twice the algebraic algebra (dim 8), so deficit-0 is an upper-side reading and the algebraic-orbit rerun with the true End^0-generators is the named v2 item. The mixer finding is strengthened, not weakened, by the overshoot: strong connectivity FAILED in all trials — structural zero blocks — and failure in the larger algebra implies failure in every subalgebra, so *no mixer from the Frobenius-block algebra can strongly connect the fourfold's Tate block*: the block algebra acts cyclically but reducibly, the two-operator criterion needs a genuinely non-torus correspondence for its mixer, and the operative closing route on real specimens is the multi-constituent one the Lean already carries (`exhaustion_of_seeds`: the single polarization seed seeds every constituent through its projections). Theory and measurement met exactly where the theorems said they would. *Degree ledger.* On the fourfold all realizers descend from characteristic zero, so bounded recurrence holds by descent — the tautological gate, stated as such; the sixfold rung (below) is, per the reclassification that follows it, ALSO descent-covered in principle, so the genuinely non-tautological run needs a beyond-playbook object.

Step-0 executed: the simple sixfold exists. (`tmp/sixfold_step0.sage`.) The primary certified base $C_0\colon w^3+2w+(u+v)=0$ over $E_0\colon v^2=u^3+1$, the 3-torsion $\eta=(0,1)$, the Vélu 3-isogeny $E_0\to E_1$, and the fiber-product cover $\widetilde{C}=C_0\times_{E_0}E_1$ were point-counted through $\mathbb{F}_{7^6}$, and every step-0 gate passed at $p=7$: the degree-12 Prym factor is integral, Weil-pure, and functional-equation-consistent (a disconnected cover cannot produce this, so $f^*\eta\neq 0$ and $\widetilde{C}$ is connected of genus 10 — gate G1 closed by one prime); $P_B=\mathrm{Nm}_{\mathbb{Q}(\zeta_3)/\mathbb{Q}}(\Pi)$ with Π of degree 6 (the $\mathbb{Z}[\zeta_3]$-action signature); the freeze scalar is EXACTLY 1 (the rail-freeze law holds on the sixfold); and — the verdict the construction was designed for — P_B is IRREDUCIBLE over $\mathbb{Q}$, so B is **simple** (a decomposable B would factor at every prime): the non-Galois degree-3 base kills the bielliptic involution exactly as intended. The Frobenius field is a degree-12 CM field whose unique quadratic subfield is $\mathbb{Q}(\sqrt{-3})$: the full Weil-type $(3,3)$ signature. The sixfold squeeze's first reading: Λ^6 Tate multiplicity 22 at $p=7$, resolving as 20 pairing monomials ($\binom{6}{3}$, the Lefschetz-type inflation) plus $\mathbf{2}$ — the collective Weil pair, visible in first light; H^1 eigenvalues all distinct (per-prime commutant 12). Register notes: the $j=0$ base makes E_0 supersingular at inert primes, so inert readings carry supersingular inflation ($p=5$: Tate 44, and the infinity-pattern fit is ambiguous there — the $p=7$ fit is unique); $p=11,13$ runs in progress.

Reclassification — the audit already in this record applies to the new specimen, and the session's framing is corrected. The construction above is an étale $\mathbb{Z}/3$ (abelian) Prym. Two results already in this paper's record therefore govern it: the étale-cyclic no-go (uniform-rescale polarization type $\Rightarrow$ det $= -3^b \Rightarrow$ disc $\equiv -1$: SPLIT type, always), and the Patel–Zhang audit (their Thm 1.2, read at source: the Weil classes of the Prym of ANY étale abelian cover are algebraic, all dimensions, unconditionally; the cyclic primitive-character case is Schoen 1988). So the new B's Weil classes were never open, and several formulations used in the preceding paragraphs of this section — "the non-split sixfold," "the first open constituent," "the discovery node $m(\mathrm{C1}) \neq 0$," "expected phantom count 2" — were MISCALIBRATED against this paper's own audit; they are corrected in place above and retracted here for the record. The honest ledger: the new B is the program's second certified-simple Weil sixfold and its correct role is the DIM-6 CALIBRATION RUNG — the squeeze must meet, the phantom count must read 0 once the Schoen/Patel–Zhang classes enter the lower bound, bounded recurrence holds by descent — one dimension above the fourfold rung, one register below the frontier. An explicit locus with $m \neq 0$ on B remains worthwhile mathematics at the NEW-CONSTRUCTION register (Patel–Zhang existence is not an explicit cycle), never at the new-existence register. The open frontier stands exactly where the audit put it: beyond the abelian-cover playbook — non-Prym Weil $2n$-folds, the rigid sixfolds of the cited literature, the semiregularity/variational wall — objects this program cannot yet construct, which is the frontier's content, not a failure of the instruments.

The calibration rung, completed — and the frontier spec. The two-prime certification playbook was then run on the new B (subfield audit + power stability of both Frobenius fields): $\mathbb{Q}(\pi_7)$ and $\mathbb{Q}(\pi_{13})$ each have proper subfields only in degrees $\{2, 6\}$, the degree-6 subfields DIFFER, the unique common proper subfield is $\mathbb{Q}(\sqrt{-3})$, and both fields are power-stable at $n \in \{2, 3, 4, 6, 12\}$ — so $\mathrm{End}^0(B_{\bar{k}}) = \mathbb{Q}(\zeta_3)$ exactly and geometrically, two maximal power-stable Frobenius tori force $\mathrm{MT}(B) = \mathrm{GU}(3, 3)$ full (Serre, cited), and the accounting closes: $\mathrm{Hdg}(B) = \theta\text{-algebra} \oplus W$ with $\dim_K W = 1$. Every premise of `KSweepCompletion` is thereby certified on the actual B; with Schoen/Patel–Zhang supplying the Weil classes, *the Hodge conjecture for the new specimen closes in every degree at the cited-theorem register* — the dim-6 calibration rung is DONE, and the instruments now stand calibrated one register below the frontier. The frontier spec itself (`tmp/beyond_playbook_spec.md`), three directions: the RIGID/CM sixfolds of Mostaed (arXiv 2603.20268, abstract-level read — full-text verification pending before any build): McMullen's curve meets the Weil locus in finitely many CM points with $M = \mathbb{Q}(\zeta_{42})$, whose Hodge–Weil classes are absolutely Hodge yet inaccessible to every existing theorem, and the $d \in \{3, 7\}$ case reduces to 2816 explicit equations at $\ell = 43$ via Hecke correspondences — CM-explicit and character-sum countable, a computational attack surface shaped like house territory; NON-ABELIAN étale covers — the Patel–Zhang boundary is abelian, so isotypic Prym pieces of étale S_3-covers are outside their theorem and constructible by house fiber-product methods (the uniform-rescale/split argument breaks exactly at 2-dimensional irreps: the opening the no-go leaves); and the LATTICE spec (odd inert-prime glue), which is the non-abelian direction in lattice language. Priority: the full-text read gates direction one; the S_3 design mathematics is the genuinely new territory.

Both frontier directions executed: the Mostaed read, and the non-abelian glue verdict. Direction one, read at full text (ar5iv): Mostaed's structure is three named open problems — O1, a FINITE ONE-VARIABLE SEARCH (2816 Hecke fixed-point conditions on McMullen's curve at $\ell = 43$, with norm equations $a^2 + 3b^2 = 43$ and $a^2 + 7b^2 = 43$, both visibly solvable: $(4, 3)$ and $(6, 1)$); O2, CM-type verification (20 of the oriented CM types are Weil-compatible);

O3, the algebraicity itself, blocked by three independent obstructions (CM isolation kills deformation/semiregularity arguments; no K-secant structure kills Markman's route; uncontrolled discriminant kills every discriminant-specific theorem). The paper gives NO explicit period lattices or countable models — the specimen search (O1+O2) is genuinely open, finite, and house-shaped: CM Frobenius data comes from Hecke characters, so the census instruments would run at the EXACT register on any specimen found. Direction two, executed: the cut-glue integral model generalized to arbitrary finite groups (`tmp/integral_model_s3.sage`), TRIPLE-GATED before reading the verdict (cyclic $\mathbb{Z}/3$ reproduces Lange–Ortega at two genera; the $\mathbb{Z}/6$ primitive packet reproduces the house μ_6 verdict $(2,2,2,6,6,6)$ exactly). The $S_3 \times \mathbb{Z}/3$ reading, $\rho \otimes \omega$ packet, transposition-invariant copy: base genus 2 (Weil fourfold, sig $(2,2)$): type $(2,6,6,18)$, $\mathrm{Pf} = 2^4 3^4$; base genus 3 (Weil eightfold, sig $(4,4)$): type $(2,2,6,6,6,6,18,18)$, $\mathrm{Pf} = 2^8 3^8$. Two findings, opposite signs. The NULL: the 2-valuations are EVEN, so $|\det| \in \mathrm{Nm}(K^\times)$ (the free-$\mathbb{Z}[\omega]$-lattice class argument makes the parity basis-independent) and the split verdict EXTENDS to the first non-abelian design — published per the falsifiability register. The DISCOVERY: the type is NOT a uniform rescale — $(2,6,6,18) = 2 \cdot (1,3,3,9)$ carries a 9-scale, the first ever observed (the cyclic averaging argument provably forbids 9s), so the 2-dim irrep genuinely BREAKS the uniform-rescale mechanism even though the split conclusion survives through even 2-powers. The non-abelian route is therefore not closed: the named next search is designs whose Pfaffian lands ODD at an inert prime — quaternionic irreps (Schur index 2: Q_8-covers, $K = \mathbb{Q}(i)$), mixed involution-invariant lattices — now a cheap scan over $(G, \mathrm{packet}, \mathrm{involution})$ in the triple-validated generic model.

The scan ran the same night, ten designs, and produced two laws and one live lead. THE BLOCK-PARITY LAW (new, structural): the handle blocks of the cut-glue model contribute orthogonally, so $v(\mathrm{Pf}) = (g-1) \cdot v_{\mathrm{block}}$ — non-split REQUIRES odd base-genus-minus-one together with odd per-block valuation at a K-inert prime (this retroactively explains the eightfold's $\mathrm{Pf} = (2^4 3^4)^2$: even by block-squaring, always). The dimension menu at one block is $\dim B = s \cdot d$ (d = irrep dimension, s = Galois-orbit size), which selects the designs worth running. The sharpest: $\mathrm{PSL}(2,7)$ — two 3-dimensional irreps with character field EXACTLY $\mathbb{Q}(\sqrt{-7})$ (where 2 splits and $3, 5$ are inert), so an étale $\mathrm{PSL}(2,7)$-cover of a genus-2 curve (existence via Ore: every element of a simple group is a commutator) yields a $\mathbb{Q}(\sqrt{-7})$-Weil-type SIXFOLD from a SIMPLE group — outside Patel–Zhang by the widest possible margin. The model confirmed the design's ranks exactly ($36 = H^1(B^3)$ isotypic, $12 = H^1(B)$ on the $\mathbb{Z}/3$-invariant slice) and returned the verdict: type $(24, 24, 24, 168, 168, 168) = 24 \cdot (1^3, 7^3)$ — the Lange–Ortega shape resurfacing through the Sylow-7 — with $v_7 = 3$ odd but 7 RAMIFIED (norm-safe), $v_3 = 6$, $v_5 = 0$: SPLIT AGAIN, the third null of the night, published. The live lead: odd inert-parity DOES occur in the landscape ($\mathbb{Z}/3 \rtimes \mathbb{Z}/8$: a slice with $v_3 = 1$, and two of that group's four 2-dimensional irreps have character field $\mathbb{Q}(i)$, where 3 is inert) — the packet-field disambiguation is the named next computation; even where the resulting fourfold is Markman-covered, the block-parity law scales the class to a candidate non-split $\mathbb{Q}(i)$-Weil TWELVEFOLD at three blocks. The frontier hunt is now a lattice-arithmetic search with validated instruments, explicit selection laws, and nulls accumulating exactly where the theory says they must.

The eightfold, constructed. Dimension eight is where the literature's construction stops: the secant-sheaf candidates on abelian eightfolds await a semiregularity that has, by its author's statement, not yet been addressed [44]; the route above never invokes it. The genus-five base is governed by the index formula $g' = 3K - I + 1$ (K the pole renormalization, I the order index of the plane model), which prices the two available routes, and both prices are now laws: the *flex* construction (tangency at the 3-torsion flex of the base curve) dies of a correspondence leak — the flex tangent is itself a second magic function, and the rail instrument diagnosed the resulting elliptic invaders inside V_ω as CM-lines with eigenvalues $\mp(3\omega+1)$ — while the *tuned-pair* construction pays only in nodes: its split-index points are nodes of the plane model, a per-level bookkeeping correction that the genus-five functional equation then confirms exactly (predicted and actual $\#C'(\mathbb{F}_{7^6})$ and $\#C'(\mathbb{F}_{13^6})$ agree on the nose). The surviving specimen — $C'\colon u^3-3x^2u+(xy+3y+2x+1)=0$ over the conductor-19 curve, index pair at $x^2+x+1=0$, non-torsion by design — is a **simple Weil-type abelian eightfold**: its node-corrected degree-16 Frobenius polynomial at the split prime 7 is irreducible with exact purity (the correction fit unique, ten orders beyond its rivals), its Frobenius field a degree-16 CM field whose only quadratic subfield is $\mathbb{Q}(\sqrt{-3})$, power-stable through $n=16$. It inherits $m=4$ and the battery from the model, so $\Xi=W_4-\Theta^4/1944$ is nonzero, algebraic, and primitive by the universal law. Register, exact — and corrected by the program's own literature audit: for *Prym-type* Weil classes, algebraicity at every dimension is covered by Schoen's cycles, recently recast and generalized to all abelian covers by Patel–Zhang [55]; so this specimen's Weil classes were never open, and what the Ξ-route adds on the Prym locus is an independent elementary proof with the cycle exhibited (the same register the sixfold holds against Markman). The genuinely open frontier — Markman's semiregularity-blocked candidates [44] — is the *general* Weil eightfold, and the audit yields a no-go sharper than the split law: abelian étale glue lands in the covered locus *by construction*, so no cover in this playbook reaches it; the beyond-boundary hunt requires non-abelian structure. The accounting ($\mathrm{End}^0=K$ and full $GU(4,4)$, second-prime compute-gated) completes our proof, not the existence. The falsification record — the flex specimen's death by its own rails, and this paragraph's own first draft, corrected within the hour by the audit — is part of the claim.

The harmonic turn: the μ_6 tower. Changing the harmonic — base curve with rational $\mathbb{Z}/6$ torsion, magic function the Miller product $f_3^2\,(x-x(3P_6))$ with divisor $6[P_6]-6[O]$, top cover $t^6=h$ — yields three rail systems in one object: a χ-primitive K-Weil sixfold B_6, a μ_3 control sixfold B_3, and a $\mathbb{Z}/2$-Prym B_2. The first roll certified B_6 *and* B_3 simple (irreducible degree-12 at the split prime, exact purity, the six twisted-count conventions self-validated, the descent character calibrated at the torsion support by a leading-coefficient regularization computed by hand and confirmed by the instrument), with order-one freezes on all three rails and the $\mathbb{Z}/2$ content quarantined exactly where the inert-prime traces predicted it. The prize question — whether B_6's 2-adic glue breaks the split class — was computed exactly on the μ_6 integral model and answered *no*: the induced type is $2\cdot(1,1,1,3,3,3)$, the 2-glue enters as a uniform scale, and the discriminant class is -1 again. The split no-go thereby *extends* from μ_3 to μ_6 towers — published here as the null it is, per the falsifiability register — and is then itself subsumed by the literature audit: Patel–Zhang's theorem [55] covers the Weil classes of *every* abelian-cover Prym regardless of discriminant, so the entire cyclic playbook lands in covered territory twice over. The beyond-boundary specification is now exact: non-abelian covers, or constructions that are not Pryms at all.

Conclusion

What stands, and at what strength. The depth dictionary is a machine-checked theorem about a non-circular, machine-fixed filtration, transport-invariant, and unique among filtrations graded by its tower — coinciding with Bloch–Beilinson only under the (unverified) hypothesis that the latter is graded by the same tower. No-silent-layer is unconditional on every finite normalized extension stack: semisimple values and all retained jet layers are separated layerwise. In the first-order model drift is value-blind, jet-visible, and never silent, with positive-definite response energy — the model form of Beilinson–Bloch anisotropy — and the measured law $k^* =$ drift dimension holds across the census with machine-zero silence. Depth one is measured at machine precision with all three coordinates independently resolved and the obstruction landing on exact squares; depth two fired on schedule; depth three is measured on the Laga–Shnidman family with an exact isotropy boundary and a discovered torsion-translation symmetry, and has a machine-checked typed landing from the standard height certificate — whose L-side is now on record: the Klein quartic's Ceresa channel carries forced sign -1 and central derivative $0.8299 \neq 0$, matching the proven non-torsion of the cycle (Theorem 9.1; the decimal remains in the measured register). Recognition has an explicit higher-grade source constructor and exact landing interface, while grade one is gate-certified on house instruments. Grade four — the first unnamed rung — is mapped to its three walls (non-critical center, parity ladder, split no-go) and crossed: the homeless degree-16 fiber is assembled on the carrier, its degenerate case factoring through the lower harmonics exactly, its primitive case certified irreducible, its first datum on record — and its fourth wall, the information wall, is measured on three faces and dismantled: the functional-equation sheet is pinned from the twenty integers with the root number proven $+1$, the strip-interior values exist, the center is read non-vanishing in the framework register, the unread number is identified as a tropical-dominated regulator volume, and the fiber's formal home is a Lean theorem. The instruments that got inside — the harmonic lattice, the conjugate-rail pairing, the portal, the μ_6-address locator — obey one measured admissibility criterion, harmonic self-compatibility, and they scale: grades five and six run with no new method, the Catalan spine and the parity law confirm across the ladder, and the residual obstruction at every grade is one named theorem-target, the value-registration law. The terminus architecture is kernel-fixed with its two hypotheses named and asserted by no one. The end goal is the Hodge conjecture as source exhaustion, and the route to it is the same route the main paper walked for functoriality: build the object, keep the ledger, induce every layer to appear, and let the recognition theorem finish. The collective campaign (§15–§16) is that route walked to its current end: detection of the hardest classes is a theorem with a proven mechanism; their censuses are kernel-checked identities; the first collective recognition loop is closed on exact integers; and what remains is one located coordinate — the curve, invisible to the projection, retained by the radius, reachable by Torelli — plus the genuinely open sixfold frontier, under attack with pre-registered outcomes and no assertion beyond what closes.

The arrow ledger. The terminus chain — actual rational Hodge class $\to$ faithful carrier state $\to$ firing coordinate $\to$ constructed algebraic cycle — carries an exact per-arrow account. The middle arrow, separation, is proven in general, at every finite extension order (`generalExtensionTower_exhaustive`). The composition is proven in general, and it is exact and irredundant: given the detector, recognition is *equivalent* to source exhaustion (`sourceExhaustion_iff_recog`
neither factor alone suffices, and the two failure modes are exhibited as consistent models — the pre-registered falsifiers (`factorization_exact_and_independent`). The first arrow, realization,

is typed in general with its retention transport proven (`FaithfulRealization.retention`); it is proven, not projected, on the executed rungs — the group algebra and the Artin motive of every cyclic Galois number field — where the third arrow, recognition, is also proven (diagonal recognition; Artin descent), so the complete chain runs end to end, unconditionally, on those objects (`cyclic_sourceExhaustion`, `galois_sourceExhaustion`). At depth one the first arrow is typed against the library curve with kernel exactly the torsion from the two cited classical inputs (`realization_eq_zero_iff`), and grade-one recognition is classical (Gross–Zagier–Kolyvagin). What remains open is exactly the first and third arrows *above* the executed rungs — new realization terms and new recognition theorems, one grade at a time; the rungs already executed are theorems, not projections.

The bottom-out, agreed and typed. The audit form of the obstruction — geometric rational (p, p) class $\to$ faithful finite/controlled carrier extension state $\to$ recognized algebraic source — is verbatim the artifact's typed frontier. The finiteness half of "finite/controlled" is not where the load sits: the geometric sides at issue are finite-dimensional by their own theorems (the cohomology of a smooth projective variety; Mordell–Weil by the cited classical input), so finite support and distinct frequencies are the bank's structural fields, and the load is faithfulness — fields (iii)–(v) of `FaithfulRealization`, proven on the executed rungs, cited-conditional at depth one, open above. And "recognition is essentially the algebraicity theorem" is priced by the exactness theorem: modulo the proven detector it is *exactly* the algebraicity theorem (`sourceExhaustion_iff_recognition`), neither more nor less — which is what a lossless reduction means. The audit's bottom-out and the program's named frontier are the same object; it is typed, its transport is proven, and it is climbed rung by rung.

The recognition mechanism, staked. The program's candidate for *why* completeness should force a source is stated here at its exact register. Three legs. *First* (proven and measured): the reading harmonic changes with the rank — depth-d data registers at harmonic d and not before (the drift law $k^* = \dim$, the jet ladder; classically, Griffiths transversality) — so a frozen class faces one cancellation test *per harmonic*, an a priori infinite tower of conditions. *Second* (confirmed where tested): the harmonics are not independent — they are compatible, generated by one structure. On the tower this is the Catalan spine, exact through C_{20}: every even-channel moment is a moment of the *single* semicircle pairing — one-pairing generation of all harmonics, machine precision; at rank one it is the matching law, the leading jet generated place-by-place by the one height pairing; classically it is Schmid's SL(2)-orbit theorem locally (all levels the weight ladder of a single $\mathfrak{sl}_2$) and Saito's *admissibility* globally — the classical condition bearing, independently, the same name as the house condition it mirrors. *Third* (the staked step): compatibility converts the infinite tower of tests into finite generation — a class passing every harmonic test with compatible harmonics is presented by a finite lattice with one Hermitian pairing whose exterior powers reproduce every jet — and that finite generator is the source candidate: a cycle lattice with its height pairing is exactly such an object. Recognition then reads: *realize the generator geometrically.* The classical bridge naming this step is Green–Griffiths–Kerr: the Hodge conjecture is equivalent to the existence of singularities of the associated (admissible) normal functions — in house coordinates, the harmonic-1 drift transport of the frozen class breaking over the family, exactly where the source materializes. The orphan this mechanism excludes is sharply shaped: an infinite harmonic tower, compatible at every level, with no finite geometric generator. Falsifier, pre-registered: the drift-singularity probe on the cyclic-cover family, calibrated on members where algebraicity is a theorem (Markman's fourfolds, the CM members) — there the singularity is theorem-bound to exist, and an instrument silent on those members is dead on arrival; an instrument firing there and silent on an open sixfold member would be the mechanism's first genuine hit against the program.

A Reproducibility

All instruments are oracle-free — no L-function call inside a locator; published values enter only final verification columns. Lean footprints are uniformly {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`, audited by `#print axioms`.

- **The filtration dictionary**, `RequestProject/HodgeLedgerFiltration.lean`: `carrierFiltration_antito` `ledger_kernel`, `mem_carrierFiltration_succ`, `no_early_detection`, `grade_visibility`, `firstVisibleDepth_eq_grade`, `isFirstVisible_unique`, `isFirstVisible_map_iff`, `carrierFiltration_` `blochBeilinson_coincides`, `exhaustive_of_radical_trivial`, `grade_visible_of_nondegenerate`, `radical_of_factorsThrough`, `not_mem_radical_of_detectedPast`.

- **The finite-extension retention theorem**, `RequestProject/CarrierTowerSeparation.lean` and `RequestProject/GeneralExtensionRetention.lean`: `radical_trivial_of_tower_injective`, `momentTower_detects`, `momentTower_radical_trivial`, `momentTower_exhaustive`, `generalExtensionTow` `generalExtensionTower_exhaustive`, `generalExtension_retention`.

- **The realization bridge and the contentful model dial**, `RequestProject/HodgeRealizationBridge.le` `modelHodgeDial`, `modelHodge_retention`, `modelDC_iff_angularFixed`, `FaithfulRealization`, `FaithfulRealization.retention`, `FaithfulRealization.isFirstVisible_iff`, `FaithfulRealization.s`

- **The cyclic instance**, `RequestProject/CyclicRealizationInstance.lean`: `windingTransform_injectiv` `winding_dc`, `winding_rational`, `cyclicRealization`, `cyclic_retention`, `cyclic_recognition`, `cyclic_sourceExhaustion`, `cyclic_diagonal_law`.

- **The Artin-motive instance**, `RequestProject/GaloisRealizationInstance.lean`: `finDFT_detects`, `galoisTransform_injective`, `galois_dc`, `galois_recognition_of_frozen`, `galoisRealization`, `galois_sourceExhaustion`, `galois_descent_law`, `galois_sourceExhaustion_of_isCyclic`.

- **The depth-one literature-hypothesis rung**, `RequestProject/EllipticDepthOneRung.lean` (conditional on the cited bundle fields, which are classical): `DepthOneHeightData`, `pairing_torsion_right`, `faithful_mod_torsion`, `no_silent_point`, `coordinateTower_detects_nonTorsion`, `realization_eq_zer` `elliptic_no_silent_point`.

- **Architecture exactness and independence**, `RequestProject/ArchitectureExactness.lean`: `sourceExhaustion_iff_recognition`, `phantomDial`, `sourcelessDial`, `factorization_exact_and_indepe` `detector_not_constructor`.

- **The depth-one census**, `tmp/jet_census.py` and `tmp/sha_hinge.py`: full BSD readings at ranks 0–3; $|Ш|$ landed as exact squares $1, 4, 9$ (margins $\leq 3.6 \times 10^{-15}$ at rank zero, $\leq 1.6 \times 10^{-4}$ through rank three); $L^{(r)}/r!$ via incomplete-Gamma Taylor; heights by x-only 4^{-n} doubling; Tate c_p and group-law torsion in-house; Sage-oracle validated once, then run oracle-free.

- **The layer calibration**, `tmp/hodge_layer_calib.py`: order/regulator/obstruction resolved as independent coordinates across the census.

- **The delayed-signature test**, `tmp/tower_exhaustion_test.py`: point-counting tower; CM self-correspondence silent at depth one, first firing at depth two; first nonzero slot matches expected filtration depth for every tested class $(1, 1, 1, 2, 3)$; no class silent at every level.

- **The Griffiths probe**, `tmp/hodge_griffiths_probe.py`: value channel fires for every homologically-trivial cycle in the census; the recursive tower shows classes absent at level one appearing at level ≥ 2.

- **The ledger-failure search, v1**, `tmp/hodge_fast_tests.py`: the $11a$ isogeny class. Identical a_p at every good $p < 400$, certified by point counts — so the 1D readout provably cannot separate the members (Faltings) — while the retained channels separate every pair (Ω, T, c_p, each computed from the curve, never from L), and the Cassels combination $\Omega \prod c_p/|T|^2$ is invariant across the class to 2.2×10^{-16}; $|\text{Ш}|$ landed $= 1$ on every member. No ledger failure found. The v2 escalation — a pair with equal *full* ledger and different Ш or, in higher dimension, equal zeta and different Chow data — is the genuine clause-(c) falsifier hunt.

- **The drift model and terminus spine**, `RequestProject/JordanDriftTower.lean` (drift value-blind / jet-visible / never silent; `driftPairing_posDef`; `anisotropic_of_definite_factor`), `RequestProject/GeneralExtensionRetention.lean` (all finite jet layers retained by the paired layer/moment tower), `RequestProject/HigherGradeRecognition.lean` (explicit certified source, exact higher-grade realization, and grade-split recognition), `RequestProject/CeresaGrossS` (height nonvanishing, depth-three first visibility, analytic-to-cycle source landing, and the certificate anatomy: `CeresaIdentification.fires_iff`, `landing_iff_identification` — the depth-three firing equivalent to the central-derivative nonvanishing, established not supplied), `RequestProject/HodgeSourceExhaustionAssembly.lean` (source exhaustion over all states of the model finite-extension dial, DC and Rational total by construction), `RequestProject/HodgeDial.lean` (Retention, Recognition, `hodge_of_retention_recognition` `model_sourceExhaustion_of_recognition`), `RequestProject/Rung4SplitNoGo.lean` (`split_unit_pairi` All standard footprint, no `sorry`.

- **The harmonic frame, calibrated**, `tmp/hodge_clock_demo.py`: $E_1 \times E_2$ Picard numbers read two ways — period-clock alignments (Hecke scan certified against the exact $j = c_4^3/\Delta$) and count-side moments — 6/6 cases matching Lefschetz $(1, 1)$ truth; CM discs $-3, -4$ recovered; the two-CM-fields control correctly refuses to align.

- **The drift ladder**, `tmp/drift_observability.py`: jet ladders $k = 0..4$ for eight curves, root numbers certified in-house (split-vs-direct $\Lambda(3)$); $k^* =$ drift dimension throughout; responses $= \mathrm{Reg}\cdot|\text{Ш}|$; three same-placement falsifier curves (certified by prime discriminant and exhibited integral points) separate cleanly.

- **The dial**, `tmp/twistor_hunt.py`: DC-locking on the product-family twistor dial is discrete and exactly algebraic; irrational controls at $O(1)$; locks are local minima (the base point is a critical point of j — quadratic shoulders).

- **The mis-clocking battery**, `tmp/harmonic_falsifier.py` and `tmp/mu_ladder.py`: wrong root / wrong harmonic / wrong rate each break their readings (separations 10^3–10^7); rungs μ_3–μ_{12} reconstruct integer counts exactly with their own roots (purity at 10^{-15}, DC slots as predicted, sphere-lock at inert primes) and miss at the integer level with wrong ones.

- **Recognition at grade one**, `tmp/heegner_recognition.py`: point counts $\to$ modular image $\to$ Heegner alignment ($D = -7$) $\to$ self-certified lattice (both half-period identities exact) $\to$ $P = (1, 0)$ recognized and verified in exact arithmetic; $\hat{h}(P)/\hat{h}(\text{gen}) = 4.0006$.

- **Depth-three anisotropy**, `tmp/ceresa_anisotropy.py`: $\hat{h}(Q_t)$ over $\mathbb{Q}(\sqrt[3]{t^2-1})$ from scratch; exact torsion boundary at $t = 0, \pm 3$; census non-isotropic 0.59–1.16; the torsion-translation symmetry discovered and group-law-verified in-instrument.

- **Rung four**, `tmp/quadruple_rung.py` (occupancies 0/1/Catalan 2/3 with parity controls), `tmp/weil_detection.py` (the Weil-assembly excess, $\mathbb{Q}$-count 0 vs $\overline{\mathbb{Q}}$ occupancy, orthogonality controls), `tmp/farside_quadruple.py` (the degree-16 fiber: degenerate channel split at 3.3×10^{-16}, coefficient factorization $\zeta^2 * \mathrm{Sym}^{2*3} * \mathrm{Sym}^4$ to 3.6×10^{-14}, Schur irreducibility $0.930 \to 1$ vs $13.703 \to 14$, window growth exponent 0.577).

- **The certified evaluator and value chain**, `tmp/center_reading.py`, `tmp/deg6_center.py`, `tmp/theta_cache.py`: one-contour-pass split identity with two-point split-vs-direct self-certification; gates $\zeta(\frac{1}{2})$, $L(11a1, \frac{1}{2})$; oracle validation PARI `lfunsympow` (degrees 3–5) and Dokchitser (degree 4); the new degree-4 and degree-6 centers; $O(1)$ θ_p caches (148,933 primes to 2×10^6).

- **Wall four, three faces**, `tmp/farside_center.py`, `tmp/carrier_center.py`, `tmp/adapted_center.py`: windowed protocol soundness ($A = 1.5315$ vs analytic 1.5323); the information bound measured (primitive transient $X^{0.19}$, no plateau at any reachable height); solved warps close cells but fail both value gates — summability, not values.

- **The dismantling**, `tmp/wall_dissolve.py`: the sheet from twenty integers (forced 3/3 split), first strip-interior values, mirror zeros, center order zero in the framework register, the tropical layer ($\tau(C_n) = n/12$, $\sum\tau = 2/3$, C_5 moments $m_1 = 0, m_2 = 5/36$).

- **The root number, proven**, `tmp/eps_quadruple.py` and `tmp/eps_quadruple_notes.md`: the Deligne/Tate local computation, nine-for-nine postdiction battery (reduction types recomputed via Tate's algorithm, non-circular), conductor-exponent consistency, Sage oracle cross-checks.

- **The lattice, the rails, the portal, the door**, `tmp/harmonic_lattice.py` (exact lane identities, incommensurate falsifier), `tmp/helix_pairing.py` (single rail diverges / conjugate pair converges to the certified value / both mis-pairs diverge), `tmp/hodge_portal.py` (channel moments, zero-variance degenerate constancy, interior Gram 17/32–1/4, 144σ falsifier), `tmp/spectral_door.py` and `tmp/spectral_door2.py` (fixed-head μ_6-rail locator, 12/12 oracle gate, Rayleigh-limit pair split, quadruple clean null with hostile checks); `tmp/helix_tower.py` spectral half: the growth-window rerun of the gate (11/12 bare rails; the diagonal schedule starves at tensor conductor, as diagnosed) and of the quadruple (pre-registered null; the recurrent $t = 2.68$ dip killed a second time by Z-wander and both scrambles); the arrow-scale falsifier shown to be gauge in fixed-head scans (sign-scramble is the honest falsifier there, 0/12); and the growth-vs-flat resolution comparison — a bounded null: at reachable head the grown and equal-magnitude reads resolve the close pair at the same data volume; `tmp/helix_toll.py` and `tmp/helix_spectral.py`: the $g = 6$ follow-up with two pre-registered predictions falsified and one first-run claim retracted (the transient growth exponent tracks *degree* through phase cancellation, not conductor — a third independent confirmation that cheap descriptors carry no value information; the carrier-representation identity holds at both grades), and the Sym^4 escalation: 8/19 oracle zeros located at degree five, the detected subset head-stable and dying under both scrambles — the locator's first degree-five focal events.

- **The Ceresa L-side**, `tmp/ceresa_lside.py`: from-scratch Hecke L-series over $\mathbb{Q}(\sqrt{-7})$; integer-exact match to the weight-4 level-49 CM form; two-point sign certificates; central derivative 0.8298893 (PARI agreement 3.6×10^{-6}); the model subtlety (naive Klein model is a cubic twist) caught by point counts.

- **The scaling test**, `tmp/grade56_scaling.py`, `tmp/theta_79a1.npy`, `tmp/theta_83a1.npy`: pre-registered exact rationals at grades five and six, all landing $\leq 1.2\sigma$; Catalan spine $C_3 = 5$ and Schur companions C_4, C_5, C_6; parity law (no DC channel at grade five); falsifiers 132–139σ, degenerate-scramble 437σ; design-change log: the odd-weight Γ branch only.

- **The clause-(c) hunt off the diagonal**, `tmp/k3_ledger.py` and `tmp/k3_nondiag.py`: the diagonal-quartic reader (Fermat class $\rho - 1 = 19$ at split primes, exact locks on inert progressions) and the non-diagonal escalation (Dwork pencil and block-separable generics; two independent exact counters validated against brute enumeration; forty-four pairs all separated, forced-isomorphic control identical; exact-discriminant screening catching a bad prime at 229 the small-p screen missed).

- **The tower atlas**, `tmp/tower_atlas.py`: grades 2–20 on the first twenty prime-conductor curves; exact $O(g^2)$ convolution portal (brute-checked through $g = 10$); the $\varepsilon(g)$ law with the Kummer derivation; Catalan spine to C_{20}; three-tier resolution law with theoretical (not empirical) error bands; three-version falsifier sweep with the support-dilution curve; `tmp/harmonic_compat.py`: the compatibility audit (conjugate closure at machine zero, the affine-in-$c(w)$ channel-moment law, rational-vs-incommensurate discrimination by lattice membership); `tmp/helix_tower.py`: the height run (Sym^3 pairing converges on the carrier to the certified value, growth -0.067, single-rail and mis-pair diverge; the degree-16 $G * \bar{G}$ identity at 2.4×10^{-13}; the radial-head toll measured; the unit-scale window falsifier).

- **Recognition at grade four**, `tmp/recognition_loop_g4.py`: the four-arm loop on three quadruples (degenerate / two isogeny pairs / scrambled), truth consulted only at grading; the live-direction law; independent counting- and period-side alignment certificates; isogeny degrees found with 10^{-15}-scale j-residual landings; occupancy and channel-profile recognition certificates; scrambled null and false-alignment rejection.

- **The constancy mechanism**, `tmp/constancy_mechanism.py` and `tmp/constancy_mechanism_notes.md`: the annihilator theorem with citations; variance spectroscopy across generic/degenerate/isogenous/CM regimes (new exact rationals 22/10/1; the CM ladder law); the blind inversion demo (four cases); the reading-scale falsifier table (truth only on μ_6; over-split/collapse/irrational off-grid); `RequestProject/ChannelConstancy.lean` (ten theorems, standard footprint) —

the finite-abelian model, the diagonal case matching `EvenWeightDC`, the double-annihilator inversion, and the freeze scale-invariance.

- **The grade-four assembly**, `tmp/matching_law_g4.py`: the tropical invariants derived from the admissible Green's function (exact rationals, ledger-matching); the frozen ledger determinant; the from-scratch $\sim$26-digit constituent recomputation (refining the degenerate center within the stated oracle floor); the two-precision PSLQ battery with its decisive nulls and the honestly-flagged under-determined edge-ratio tests (extended, after the grade-two translation, with the lane-count primes $\{7, 13\}$ — the Catalan spine factors through 7 — and an explicit π-power scan $k \in [-30, 30]$, all null: the geometric-integral factor is irreducible); the pre-registered primitive identity with its four falsifiers, stated volume-not-height on the grade-two template.

- **Recognition at grade two (even rung)**, `tmp/bf_loop_g2.py` and `tmp/bf_loop_g2_notes.md`: the trivial-zero/FE bridge derived and confirmed; the certified Rankin–Selberg values at the Beilinson point; the Rogers–Zudilin corner closed from scratch (Mahler integral vs evaluator, 1.55×10^{-5}, rational landing 24); the genuine-product construction gap documented as the literature's own state; the carrier translation of the even-rung construction arm.

- **The value-registration census**, `tmp/value_registration.py`: frozen forcible-closure protocol (faithfulness anchor reproducing the published mis-registrations exactly); 24 rows over the six certified truths; leave-one-object-out cross-validation killing the closure-improvement and degree/conductor candidates and confirming off-lattice distance as the governing regressor; the graded-family monotone degradation ($0.165 \to 0.200 \to 0.228 \to 0.236$ as off-lattice freedom is added); the finite-difference Jacobian requirement (real banks are first-order-stationary for the closure objective) documented in-source.

- **The grade-one pairing-law calibration**, `tmp/matching_law_g1.py`: canonical heights decomposed place-by-place (archimedean theta/sigma via complex AGM — the true lattice $\tau = \omega_2/\omega_1$, not j-inversion's fundamental-domain representative — plus Bernoulli Green's functions on the reduction graphs); one calibrated constant (the classical factor two), then frozen and blind; jet comparison non-circular; wrong-graph falsifier missing by the exact predicted rational.

- **The rung-four Lean bricks**, `RequestProject/QuadrupleFiber.lean` (thirteen theorems: the degree-16 fiber's determinant-one ledger, $(-X)^{16}$ per-place functional equation, completed reflection, strand exchange) and `RequestProject/EvenWeightDC.lean` (ten theorems: `dc_channel_iff_even_weight` and the grade-4/grade-5 corollaries). Standard footprint on all twenty-three, no `sorry`.

- **The collective-class campaign**, `tmp/weil_scout.py` (family portfolio, CM-type censuses incl. composite fields, the S1–S6 exact-angle families), `tmp/weil_excess.py` (the trace-level detector, ten-configuration calibration), `tmp/weil_rails.py` (the rail-native primary: freeze scalar, no-orphan decomposition, order law, reading-scale and mis-pair falsifiers), `tmp/weil_sixfold_count.py` (the non-CM counter, ∞-ramification law), `tmp/weil_composite_cens` `tmp/freeze_mechanism.py` + `_notes.md` (the mechanism theorem, link-by-link), `tmp/schoen_explicit.py` (the explicit member, fiber-product counter, the rail-native $6 + 2$ reading).

- **The recognition bridge**, `tmp/construction_arm.py` + `_notes.md` (the cycle pinned, eight landings, the generalization memo), `tmp/carrier_inversion.py` + `_notes.md` (blind inversion 4/4, the three-tier Faltings boundary), `tmp/k3_isogeny.py` (the per-channel retention result), `tmp/g9_verification.py` and `tmp/g9_rail_pairing.py` (the Sym^9 forced zero and derivative; the certificate-collapse curve; the general m-identity), `tmp/transient_form.py`, `tmp/transient_pairing.py`, and `tmp/registration_theorem.py` + `_notes.md` (the derived mirror, the product anomaly, the dichotomy theorem).

- **The law bricks**, `RequestProject/RootNumberLaw.lean` (the tower's root-number law, unconditional, Kummer step included), `RequestProject/CMTypeCensus.lean` (both Weil-type censuses as kernel-checked enumerations, *empty* axiom list), `RequestProject/MomentLaw.lean` (the channel-moment closed form as finite algebra), `RequestProject/LiveDirection.lean` (the live-direction law with the pointer property), `RequestProject/NoOrphan.lean` (occupancy = divisor + collective with zero remainder, regrounded on the instrument's exact frozen-rail model — the #eval table reproduces the measured decompositions bit-for-bit), `RequestProject/BalancedCount.lean` (the $\binom{2n}{n}$ identity as a general theorem; `uniform_twelve`; `obstruction_via_identity` = "$6-6=0$"), `RequestProject/CatalanMoment.lean` (the Sato–Tate/Catalan moment integrals, discharging `MomentLaw`'s hypotheses), `RequestProject/ResCube.` (the mechanism theorem's integer heart: the resultant-cube identity, the cubic-character kill, and the multiplicity-one counterfactual), and `RequestProject/RecognitionBridge.lean` (the bridge architecture; axiom-free glue; the `boundary_necessity` impossibility). Standard footprint or below throughout.

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